

STAT 423 Winter 2026 - Final Project

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Data Processing - Krish Doshi

```
orig_data <- read.csv("billboard.csv")

cleaned_data <- data.frame(artist = orig_data$artist, rating = orig_data$overall_rating, age=orig_data$age,
                           gender = orig_data$gender, race = orig_data$race, genre = orig_data$genre, bpm=orig_data$bpm,
                           energy=orig_data$energy, danceability = orig_data$danceability, has_winds = orig_data$has_winds,
                           has_brass = orig_data$has_brass, has_percussion = orig_data$has_percussion, has_vocal_or_whistle = orig_data$has_vocal_or_whistle)

cleaned_data$gender <- ifelse(orig_data$artist_male == 1, "Male", "Not Male")

cleaned_data$race <- ifelse(orig_data$artist_white == 1, "White", "Not White")

cleaned_data$genre <- ifelse(is.na(orig_data$cdr_genre), "Genre Not Specified", orig_data$cdr_genre)

cleaned_data$genre <- ifelse(grepl("Pop", orig_data$cdr_genre), "Pop",
                             ifelse(grepl("Country", orig_data$cdr_genre), "Country",
                                     ifelse(grepl("Hip-Hop", orig_data$cdr_genre), "Hip-Hop",
                                             ifelse(grepl("Rock", orig_data$cdr_genre), "Rock",
                                                     ifelse(grepl("Jazz", orig_data$cdr_genre), "Jazz", "Other")))))

cleaned_data <- cbind(cleaned_data, bpm=orig_data$bpm, energy=orig_data$energy, danceability = orig_data$danceability)

#Wind
cleaned_data$has_winds <- ifelse(orig_data$clarinet, "True",
                                ifelse(orig_data$flute_piccolo, "True",
                                        ifelse(orig_data$harmonica, "True",
                                              ifelse(orig_data$kazoo, "True",
                                                      ifelse(orig_data$ocarina, "True",
                                                                ifelse(orig_data$saxophone, "True",
                                                                      ifelse(orig_data$accordion, "True", "False"))))))))

#Brass
cleaned_data$has_brass <- ifelse(orig_data$trumpet, "True", "False")

#Percussion
cleaned_data$has_percussion <- ifelse(orig_data$piano_keyboard_based, "True",
                                      ifelse(orig_data$bongos, "True",
                                              ifelse(orig_data$cowbell, "True",
                                                      ifelse(orig_data$handclaps_snaps, "True", "False")))))

#Vocal or Whistle
cleaned_data$has_vocal_or_whistle <- ifelse(orig_data$vocally_based, "True",
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        ifelse(orig_data$falsetto_vocal, "True",
              ifelse(orig_data$human_whistling, "True", "False")))

#String
cleaned_data$has_strings <- ifelse(orig_data$bass_based, "True",
                                ifelse(orig_data$guitar_based, "True",
                                      ifelse(orig_data$orchestral_strings, "True",
                                            ifelse(orig_data$banjo, "True",
                                                  ifelse(orig_data$mandolin, "True",
                                                        ifelse(orig_data$pedal_lap_steel, "True",
                                                                ifelse(orig_data$sitar, "True",
                                                                    ifelse(orig_data$ukulele, "True",
                                                                        ifelse(orig_data$violin, "True", "False")))))))

cleaned_data$year <- as.numeric(substr(orig_data$date, 1, unlist(gregexpr("-", orig_data$date))[1] - 1))

cleaned_data <- cleaned_data[rowSums(is.na(cleaned_data)) == 0 , ]

cutoff <- round(min(cleaned_data$year) + ((max(cleaned_data$year) - min(cleaned_data$year)) / 2))

first_subpop <- cleaned_data[cleaned_data$year < cutoff, ]
second_subpop <- cleaned_data[cleaned_data$year >= cutoff, ]

model_results <- list()

for(i in seq(1960, 2030, 10)){
  decade_label <- paste0(i - 10, "s")

  subpop <- subpop_60s <- cleaned_data[cleaned_data$year >= i - 10 & cleaned_data$year < i, ]

  # --- SAFETY CHECK 1: Ensure enough rows exist ---
  # If you have < 50 rows, the interaction model will likely fail.
  if(nrow(subpop) < 50) {
    message(paste("Skipping", decade_label, "- Not enough data points."))
    next
  }

  factor_counts <- sapply(subpop[, c("genre", "gender", "race")], function(x) length(unique(na.omit(x))))

  if(any(factor_counts < 2)) {
    message(paste("Skipping", decade_label, "due to single-level factor:",
                  names(factor_counts)[factor_counts < 2]))
    next
  }

  first_model <- lm(data = subpop, rating ~ age + race + gender + genre + energy + danceability + happi

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second_model <- lm(data = subpop, rating ~ age + gender + race + age*genre + age*energy + age*danceab

# --- Plotting (Wrapped in 'try' so one bad plot doesn't stop the whole loop) ---
try({
  plot(first_model, which = 1:2)
  boxcox(first_model, plotit=TRUE)
  plot(second_model, which = 1:2)
  boxcox(second_model, plotit=TRUE)
})
optimal_model <- step(second_model, direction="backward")

model_results[[decade_label]] <- list(
  "first" = first_model,
  "second" = second_model,
  "optimal" = optimal_model
)

print(decade_label)

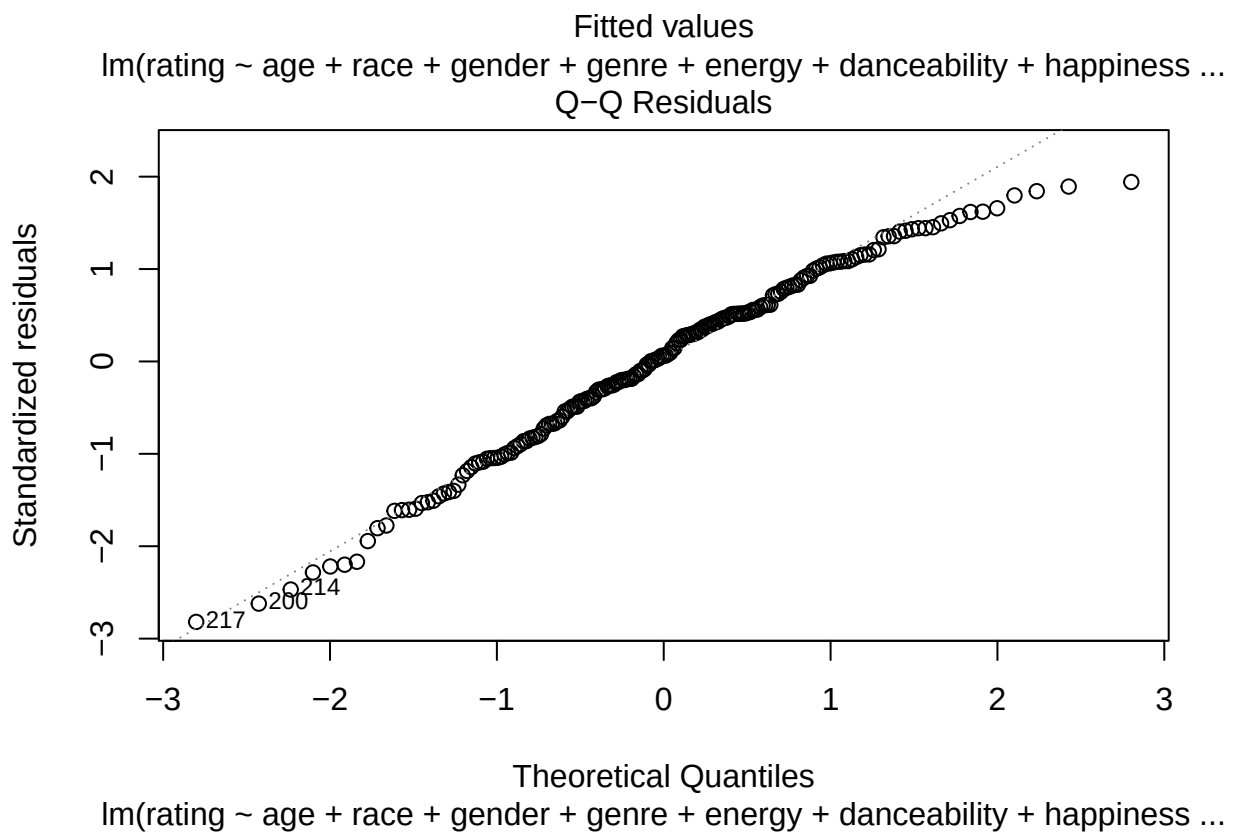
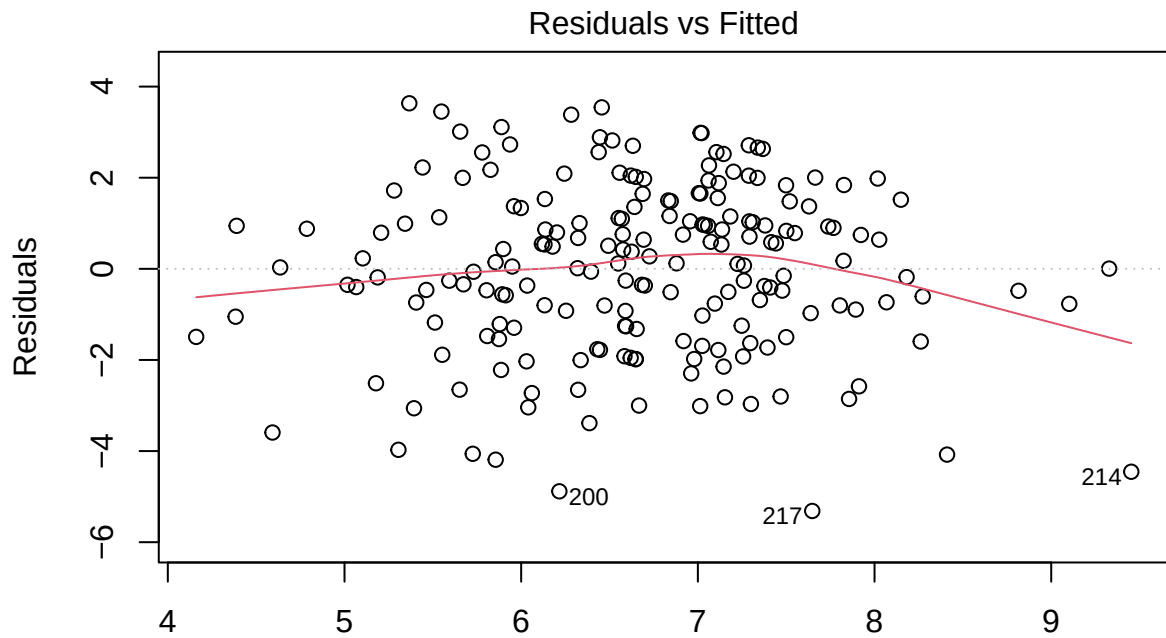
summary(first_model)

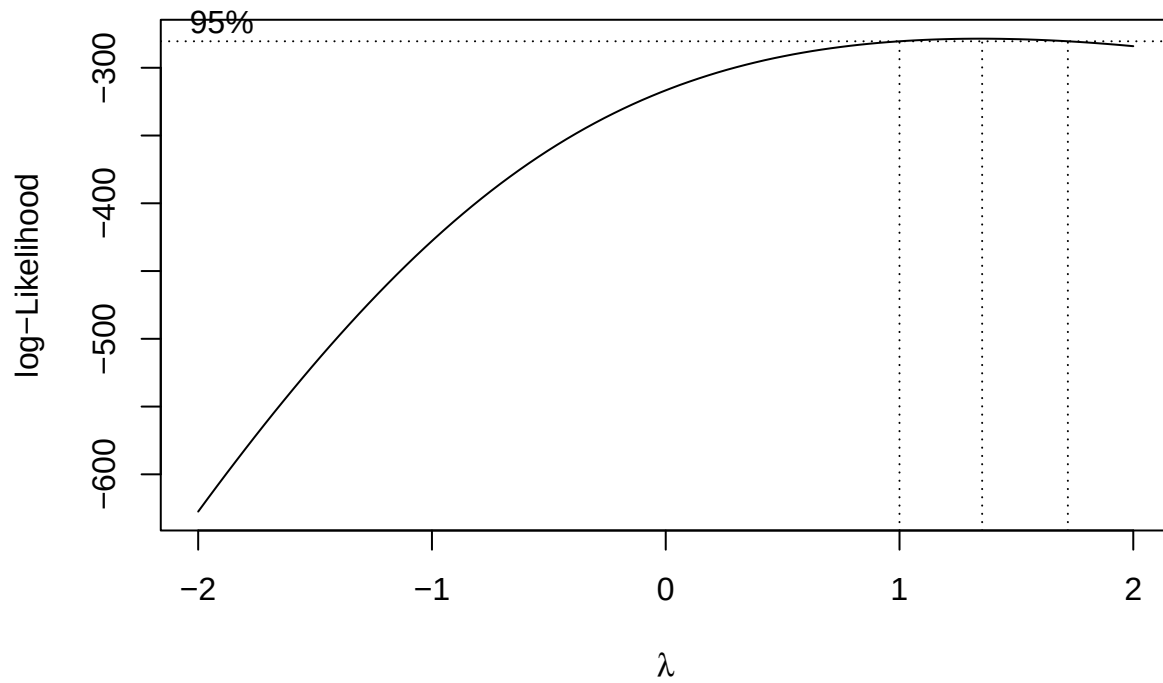
summary(second_model)

summary(optimal_model)
}

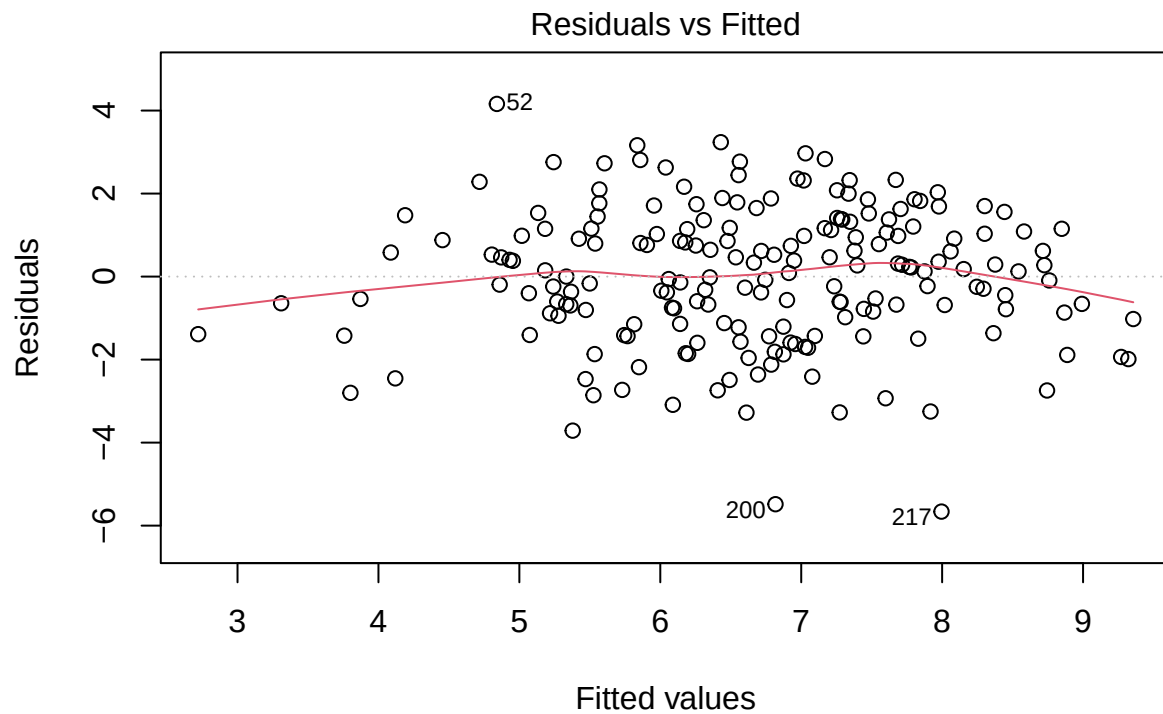
## Skipping 1950s - Not enough data points.

```

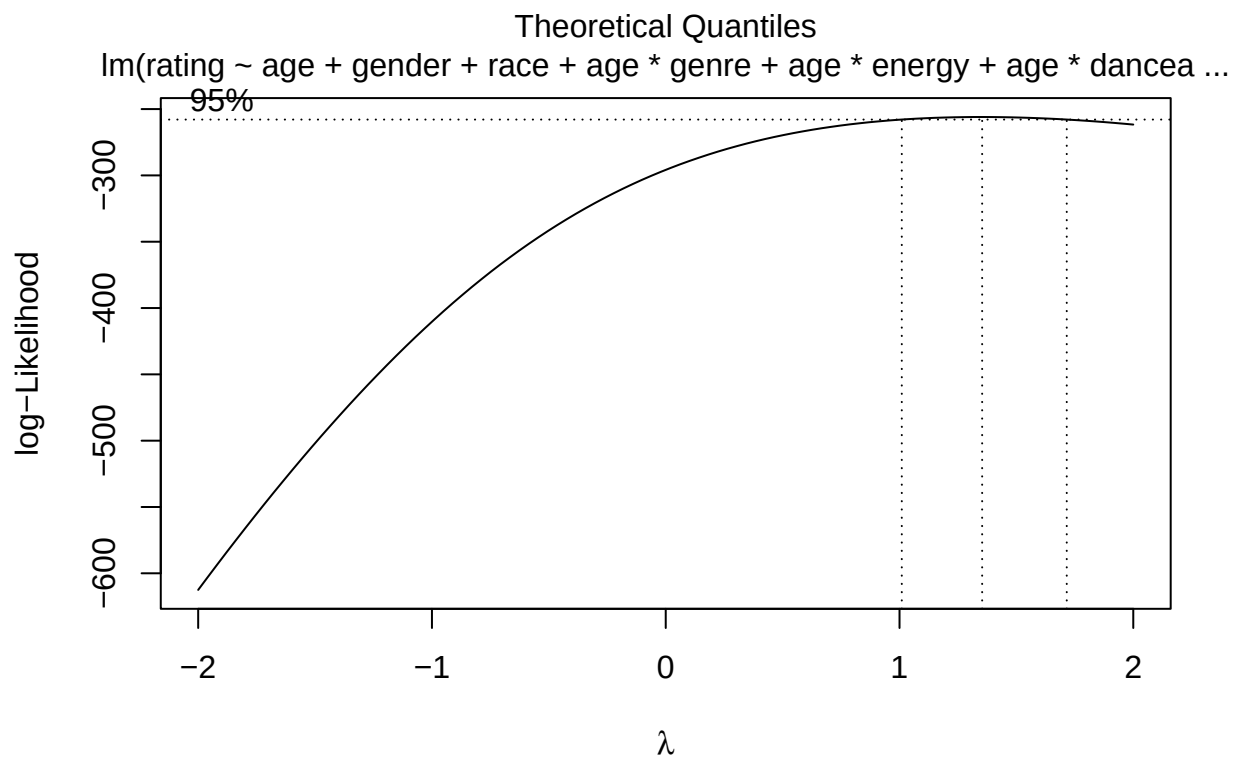
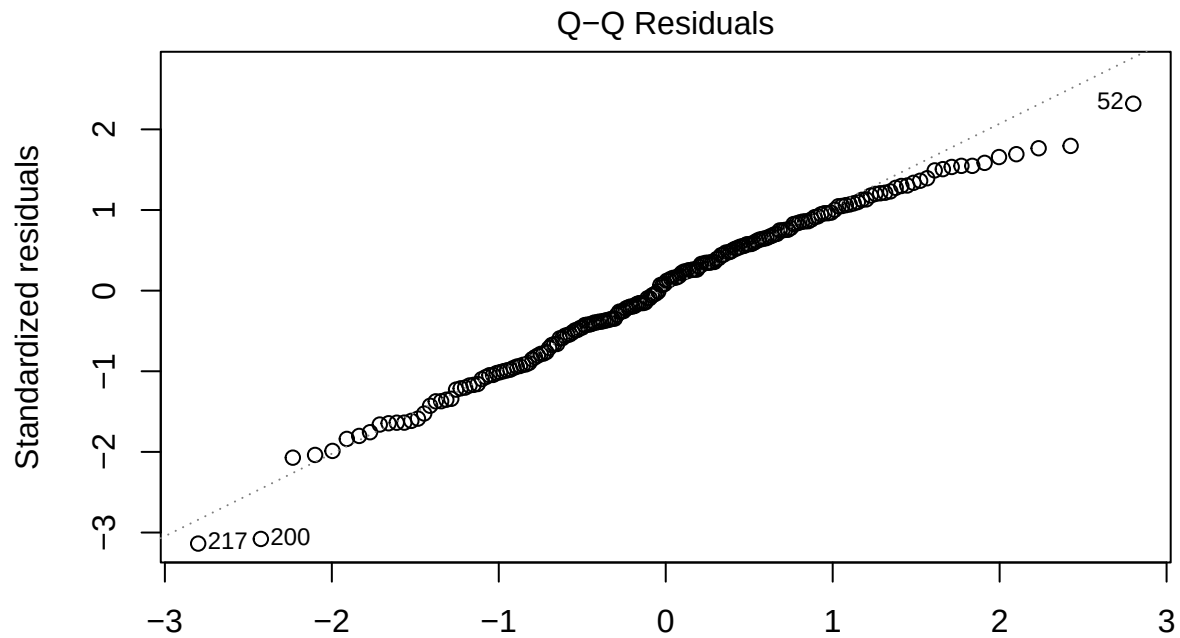




```
## Warning: not plotting observations with leverage one:
## 48
```



lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...



```
## Start: AIC=281.18
## rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + +gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##
```

```

##              Df Sum of Sq    RSS    AIC
## - age:genre      4      6.3839 542.36 275.51
## - gender:genre    3      0.9873 536.97 275.54
## - age:energy      1      0.1576 536.14 279.23
## - age:length_sec  1      0.2498 536.23 279.27
## - race:bpm        1      0.3150 536.30 279.29
## - age:bpm         1      0.3777 536.36 279.31
## - age:happiness   1      0.4214 536.40 279.33
## - gender:bpm      1      0.8413 536.82 279.49
## - race:happiness   1      0.9928 536.97 279.54
## - race:loudness_db 1      1.1628 537.14 279.60
## - age:danceability 1      1.7691 537.75 279.82
## - gender:length_sec 1      2.3411 538.32 280.04
## - age:loudness_db  1      2.7934 538.77 280.20
## - race:energy      1      4.0209 540.00 280.65
## <none>              535.98 281.18
## - race:length_sec  1      6.7873 542.77 281.65
## - gender:energy     1      7.2507 543.23 281.82
## - race:danceability 1      8.3951 544.38 282.24
## - race:genre        3     19.8442 555.82 282.34
## - gender:loudness_db 1      8.9645 544.94 282.44
## - gender:happiness  1      9.1272 545.11 282.50
## - gender:danceability 1     30.1379 566.12 289.95
##
## Step:  AIC=275.51
## rating ~ age + gender + race + genre + energy + danceability +
##          happiness + bpm + loudness_db + length_sec + age:energy +
##          age:danceability + age:happiness + age:bpm + age:loudness_db +
##          age:length_sec + race:genre + race:energy + race:danceability +
##          race:happiness + race:bpm + race:loudness_db + race:length_sec +
##          gender:genre + gender:energy + gender:danceability + gender:happiness +
##          gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:genre      3      2.648 545.01 270.47
## - age:energy         1      0.023 542.39 273.52
## - age:length_sec     1      0.040 542.40 273.52
## - age:bpm            1      0.091 542.45 273.54
## - age:happiness      1      0.318 542.68 273.62
## - race:bpm           1      0.667 543.03 273.75
## - race:happiness      1      0.669 543.03 273.75
## - race:loudness_db    1      1.144 543.51 273.92
## - age:loudness_db     1      1.414 543.78 274.02
## - gender:length_sec   1      1.771 544.14 274.15
## - gender:bpm          1      1.934 544.30 274.21
## - age:danceability    1      2.999 545.36 274.60
## - race:energy         1      3.834 546.20 274.90
## <none>              542.36 275.51
## - race:length_sec    1      6.161 548.53 275.73
## - gender:energy       1      7.395 549.76 276.18
## - race:danceability   1      8.293 550.66 276.50
## - gender:happiness    1      8.431 550.79 276.55
## - gender:loudness_db  1      8.748 551.11 276.66
## - race:genre         3     28.215 570.58 279.50

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```

## - gender:danceability 1 32.747 575.11 285.06
##
## Step: AIC=270.47
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:happiness + age:bpm + age:loudness_db +
## age:length_sec + race:genre + race:energy + race:danceability +
## race:happiness + race:bpm + race:loudness_db + race:length_sec +
## gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
##
## Df Sum of Sq RSS AIC
## - age:bpm 1 0.018 545.03 268.48
## - age:energy 1 0.022 545.03 268.48
## - age:length_sec 1 0.127 545.14 268.51
## - age:happiness 1 0.321 545.33 268.58
## - race:loudness_db 1 0.556 545.57 268.67
## - race:bpm 1 0.661 545.67 268.71
## - race:happiness 1 1.298 546.31 268.94
## - gender:bpm 1 1.387 546.40 268.97
## - age:loudness_db 1 1.579 546.59 269.04
## - gender:length_sec 1 1.847 546.86 269.13
## - age:danceability 1 2.747 547.76 269.46
## - race:energy 1 3.197 548.21 269.62
## <none> 545.01 270.47
## - gender:energy 1 6.580 551.59 270.83
## - race:length_sec 1 6.863 551.88 270.93
## - gender:happiness 1 8.063 553.07 271.36
## - gender:loudness_db 1 8.252 553.26 271.43
## - race:danceability 1 9.685 554.70 271.94
## - race:genre 3 30.197 575.21 275.09
## - gender:danceability 1 32.042 577.05 279.72
##
## Step: AIC=268.47
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:happiness + age:loudness_db + age:length_sec +
## race:genre + race:energy + race:danceability + race:happiness +
## race:bpm + race:loudness_db + race:length_sec + gender:energy +
## gender:danceability + gender:happiness + gender:bpm + gender:loudness_db +
## gender:length_sec
##
##
## Df Sum of Sq RSS AIC
## - age:energy 1 0.019 545.05 266.48
## - age:length_sec 1 0.133 545.16 266.52
## - age:happiness 1 0.304 545.33 266.58
## - race:loudness_db 1 0.543 545.57 266.67
## - race:bpm 1 0.683 545.71 266.72
## - race:happiness 1 1.318 546.35 266.95
## - gender:bpm 1 1.375 546.41 266.97
## - age:loudness_db 1 1.616 546.65 267.06
## - gender:length_sec 1 1.829 546.86 267.13
## - race:energy 1 3.191 548.22 267.62
## - age:danceability 1 3.256 548.29 267.65

```



```

## <none> 545.03 268.48
## - gender:energy 1 6.586 551.62 268.84
## - race:length_sec 1 7.035 552.07 269.00
## - gender:happiness 1 8.050 553.08 269.36
## - gender:loudness_db 1 8.236 553.27 269.43
## - race:danceability 1 9.688 554.72 269.95
## - race:genre 3 30.248 575.28 273.12
## - gender:danceability 1 32.457 577.49 277.87
##
## Step: AIC=266.48
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:danceability +
## age:happiness + age:loudness_db + age:length_sec + race:genre +
## race:energy + race:danceability + race:happiness + race:bpm +
## race:loudness_db + race:length_sec + gender:energy + gender:danceability +
## gender:happiness + gender:bpm + gender:loudness_db + gender:length_sec
##
## Df Sum of Sq RSS AIC
## - age:length_sec 1 0.122 545.17 264.53
## - age:happiness 1 0.354 545.40 264.61
## - race:loudness_db 1 0.553 545.60 264.68
## - race:bpm 1 0.669 545.72 264.72
## - race:happiness 1 1.320 546.37 264.96
## - gender:bpm 1 1.397 546.45 264.99
## - gender:length_sec 1 1.813 546.86 265.14
## - age:loudness_db 1 2.605 547.65 265.42
## - age:danceability 1 3.281 548.33 265.66
## - race:energy 1 3.431 548.48 265.72
## <none> 545.05 266.48
## - gender:energy 1 6.572 551.62 266.84
## - race:length_sec 1 7.064 552.11 267.02
## - gender:happiness 1 8.032 553.08 267.36
## - gender:loudness_db 1 8.265 553.31 267.45
## - race:danceability 1 9.700 554.75 267.96
## - race:genre 3 30.254 575.30 271.12
## - gender:danceability 1 32.678 577.73 275.95
##
## Step: AIC=264.53
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:danceability +
## age:happiness + age:loudness_db + race:genre + race:energy +
## race:danceability + race:happiness + race:bpm + race:loudness_db +
## race:length_sec + gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
## Df Sum of Sq RSS AIC
## - age:happiness 1 0.271 545.44 262.62
## - race:loudness_db 1 0.590 545.76 262.74
## - race:bpm 1 0.796 545.97 262.81
## - race:happiness 1 1.332 546.50 263.01
## - gender:bpm 1 1.422 546.59 263.04
## - gender:length_sec 1 1.953 547.12 263.23
## - age:loudness_db 1 2.507 547.68 263.43
## - age:danceability 1 3.162 548.33 263.67

```

```

## - race:energy          1      3.593 548.76 263.82
## <none>                  545.17 264.53
## - gender:energy        1      6.527 551.70 264.87
## - race:length_sec      1      7.135 552.31 265.09
## - gender:happiness     1      7.931 553.10 265.37
## - gender:loudness_db   1      8.277 553.45 265.49
## - race:danceability    1      9.791 554.96 266.03
## - race:genre           3     30.745 575.92 269.33
## - gender:danceability  1     32.799 577.97 274.04
##
## Step: AIC=262.62
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:danceability +
## age:loudness_db + race:genre + race:energy + race:danceability +
## race:happiness + race:bpm + race:loudness_db + race:length_sec +
## gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:loudness_db      1      0.609 546.05 260.84
## - race:bpm              1      0.786 546.23 260.91
## - race:happiness        1      1.233 546.67 261.07
## - gender:bpm            1      1.457 546.90 261.15
## - gender:length_sec     1      2.090 547.53 261.38
## - age:danceability      1      3.012 548.45 261.71
## - age:loudness_db       1      3.366 548.81 261.83
## - race:energy           1      3.726 549.17 261.96
## <none>                   545.44 262.62
## - gender:energy        1      6.614 552.06 263.00
## - race:length_sec      1      7.221 552.66 263.21
## - gender:happiness     1      7.671 553.11 263.38
## - gender:loudness_db   1      8.356 553.80 263.62
## - race:danceability    1      9.685 555.13 264.09
## - race:genre           3     30.717 576.16 267.42
## - gender:danceability  1     32.532 577.97 272.04
##
## Step: AIC=260.84
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:danceability +
## age:loudness_db + race:genre + race:energy + race:danceability +
## race:happiness + race:bpm + race:length_sec + gender:energy +
## gender:danceability + gender:happiness + gender:bpm + gender:loudness_db +
## gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:bpm              1      0.758 546.81 259.12
## - race:happiness        1      1.140 547.19 259.25
## - gender:bpm            1      1.556 547.61 259.40
## - gender:length_sec     1      1.836 547.89 259.50
## - age:danceability      1      3.084 549.13 259.95
## - race:energy           1      3.350 549.40 260.05
## - age:loudness_db       1      3.601 549.65 260.14
## <none>                   546.05 260.84
## - gender:energy        1      6.219 552.27 261.07

```

```

## - race:length_sec      1      6.678 552.73 261.24
## - gender:happiness     1      7.681 553.73 261.60
## - gender:loudness_db   1      7.792 553.84 261.63
## - race:danceability    1      9.970 556.02 262.41
## - race:genre           3     30.684 576.73 265.61
## - gender:danceability  1     32.813 578.86 270.34
##
## Step: AIC=259.12
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:danceability +
## age:loudness_db + race:genre + race:energy + race:danceability +
## race:happiness + race:length_sec + gender:energy + gender:danceability +
## gender:happiness + gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:happiness      1      0.864 547.67 257.43
## - gender:bpm           1      1.059 547.87 257.50
## - gender:length_sec    1      1.782 548.59 257.76
## - race:energy          1      2.937 549.75 258.17
## - age:loudness_db      1      3.349 550.16 258.32
## - age:danceability     1      3.376 550.18 258.33
## <none>                  546.81 259.12
## - gender:energy        1      6.306 553.11 259.38
## - race:length_sec      1      6.687 553.50 259.51
## - gender:loudness_db   1      8.286 555.09 260.08
## - gender:happiness     1      8.709 555.52 260.23
## - race:danceability    1      9.911 556.72 260.65
## - race:genre           3     31.011 577.82 263.98
## - gender:danceability  1     33.009 579.82 268.66
##
## Step: AIC=257.43
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:danceability +
## age:loudness_db + race:genre + race:energy + race:danceability +
## race:length_sec + gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:bpm           1      0.865 548.54 255.74
## - gender:length_sec     1      1.573 549.25 255.99
## - age:danceability      1      3.165 550.84 256.56
## - age:loudness_db       1      3.329 551.00 256.62
## <none>                  547.67 257.43
## - race:energy          1      5.919 553.59 257.55
## - race:length_sec       1      6.071 553.74 257.60
## - gender:energy         1      6.920 554.59 257.90
## - gender:loudness_db    1      8.084 555.76 258.31
## - race:danceability     1      9.447 557.12 258.80
## - gender:happiness      1     11.330 559.00 259.46
## - race:genre           3     32.597 580.27 262.82
## - gender:danceability   1     34.647 582.32 267.51
##
## Step: AIC=255.74
## rating ~ age + gender + race + genre + energy + danceability +

```

```

##      happiness + bpm + loudness_db + length_sec + age:danceability +
##      age:loudness_db + race:genre + race:energy + race:danceability +
##      race:length_sec + gender:energy + gender:danceability + gender:happiness +
##      gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:length_sec    1      1.365 549.90 254.23
## - bpm                  1      1.607 550.14 254.31
## - age:danceability     1      3.185 551.72 254.88
## - age:loudness_db      1      3.620 552.16 255.03
## <none>                  548.54 255.74
## - race:energy          1      5.613 554.15 255.74
## - race:length_sec      1      6.189 554.73 255.95
## - gender:energy        1      6.192 554.73 255.95
## - gender:loudness_db   1      7.441 555.98 256.39
## - race:danceability    1      9.572 558.11 257.15
## - gender:happiness     1     12.437 560.97 258.15
## - race:genre           3     33.928 582.47 261.56
## - gender:danceability  1     34.121 582.66 265.63
##
## Step: AIC=254.23
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:danceability +
##      age:loudness_db + race:genre + race:energy + race:danceability +
##      race:length_sec + gender:energy + gender:danceability + gender:happiness +
##      gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - bpm                  1      1.740 551.64 252.85
## - age:danceability     1      3.052 552.95 253.32
## - age:loudness_db      1      3.350 553.25 253.42
## - race:length_sec      1      4.840 554.74 253.95
## - race:energy          1      4.922 554.82 253.98
## <none>                  549.90 254.23
## - gender:energy        1      5.917 555.82 254.34
## - gender:loudness_db   1      7.215 557.12 254.80
## - race:danceability    1      8.651 558.55 255.30
## - gender:happiness     1     15.057 564.96 257.55
## - race:genre           3     33.710 583.61 259.95
## - gender:danceability  1     35.155 585.06 264.44
##
## Step: AIC=252.85
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + age:danceability +
##      age:loudness_db + race:genre + race:energy + race:danceability +
##      race:length_sec + gender:energy + gender:danceability + gender:happiness +
##      gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - age:loudness_db      1      3.113 554.76 251.96
## - age:danceability     1      3.743 555.39 252.18
## - race:energy          1      4.950 556.59 252.61
## - race:length_sec      1      4.981 556.62 252.62
## <none>                  551.64 252.85

```

```

## - gender:energy      1      6.148 557.79 253.03
## - gender:loudness_db 1      8.025 559.67 253.70
## - race:danceability  1      8.473 560.12 253.85
## - gender:happiness   1     14.433 566.08 255.94
## - race:genre         3     34.698 586.34 258.87
## - gender:danceability 1     34.474 586.12 262.79
##
## Step: AIC=251.96
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + age:danceability +
##      race:genre + race:energy + race:danceability + race:length_sec +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - age:danceability  1      4.140 558.90 251.42
## - race:energy      1      4.411 559.17 251.52
## <none>                                554.76 251.96
## - gender:energy    1      5.774 560.53 252.00
## - race:length_sec  1      5.824 560.58 252.02
## - gender:loudness_db 1      7.492 562.25 252.60
## - race:danceability 1      8.520 563.28 252.96
## - gender:happiness  1     13.672 568.43 254.75
## - race:genre       3     35.753 590.51 258.26
## - gender:danceability 1     34.021 588.78 261.68
##
## Step: AIC=251.42
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + race:genre + race:energy +
##      race:danceability + race:length_sec + gender:energy + gender:danceability +
##      gender:happiness + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - gender:energy    1      5.246 564.14 251.26
## - race:energy      1      5.291 564.19 251.28
## <none>                                558.90 251.42
## - gender:loudness_db 1      6.576 565.47 251.73
## - race:length_sec  1      6.646 565.54 251.75
## - race:danceability 1      8.657 567.55 252.45
## - age              1     11.424 570.32 253.41
## - gender:happiness  1     14.543 573.44 254.49
## - race:genre       3     35.352 594.25 257.51
## - gender:danceability 1     31.578 590.47 260.25
##
## Step: AIC=251.26
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + race:genre + race:energy +
##      race:danceability + race:length_sec + gender:danceability +
##      gender:happiness + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - gender:loudness_db 1      1.810 565.95 249.90
## - race:energy        1      3.897 568.04 250.62
## - race:length_sec    1      4.870 569.01 250.96

```

```

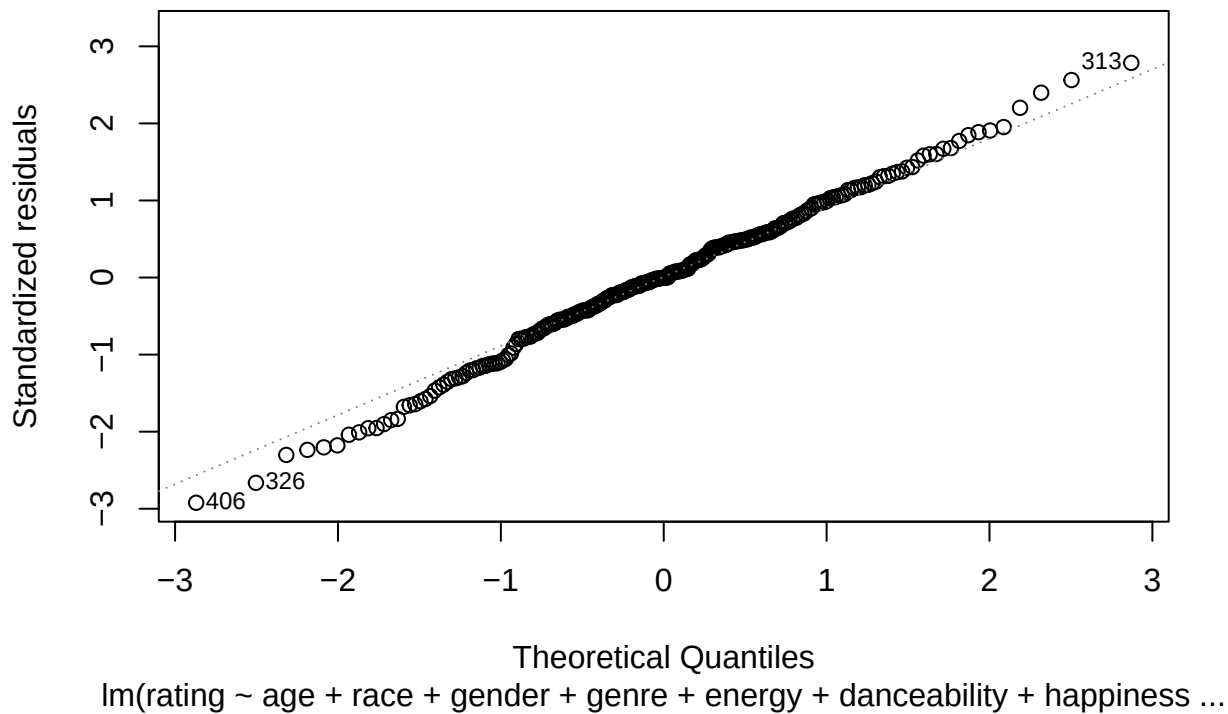
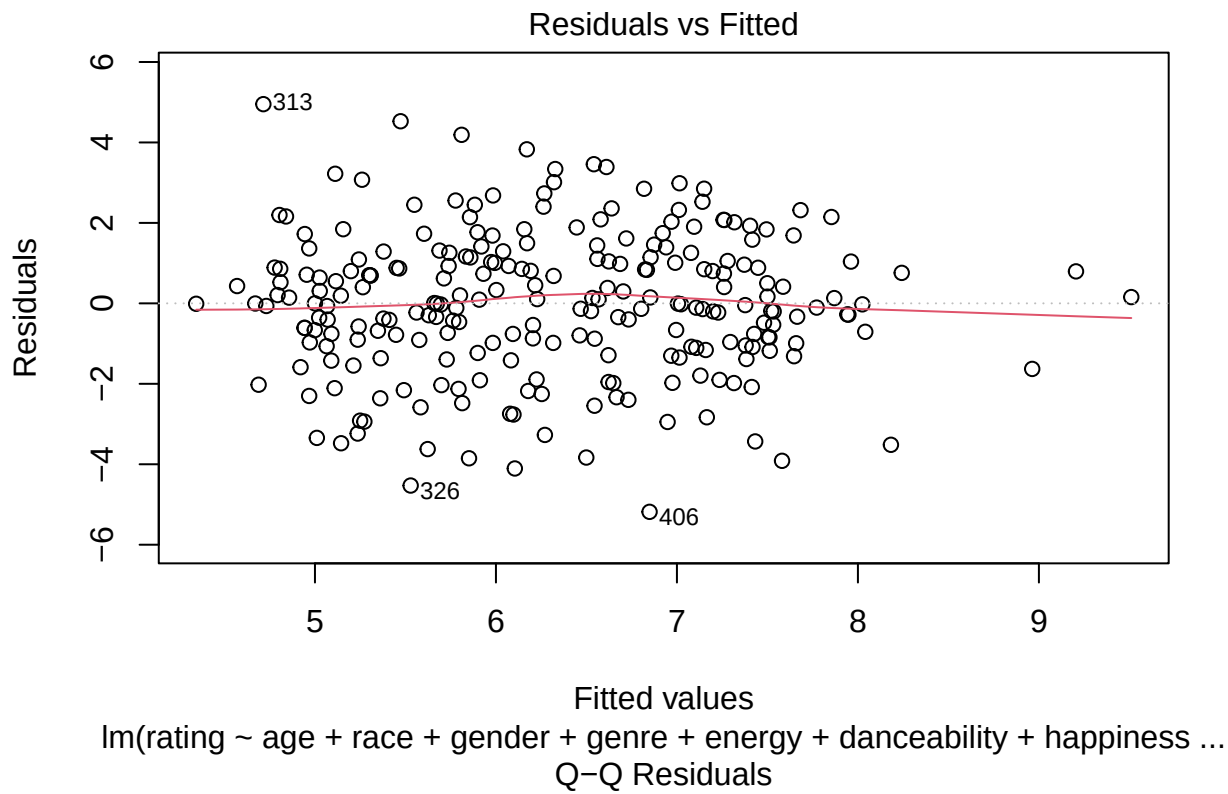
## <none>                                564.14 251.26
## - race:danceability    1      8.107 572.25 252.07
## - age                  1      9.432 573.57 252.53
## - gender:happiness     1      9.738 573.88 252.64
## - race:genre           3     35.039 599.18 257.13
## - gender:danceability  1     30.692 594.83 259.70
##
## Step: AIC=249.9
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + race:genre + race:energy +
##      race:danceability + race:length_sec + gender:danceability +
##      gender:happiness
##
##              Df Sum of Sq    RSS    AIC
## - race:length_sec    1      4.800 570.75 249.56
## - loudness_db        1      4.839 570.79 249.57
## <none>                                565.95 249.90
## - race:energy        1      6.645 572.60 250.19
## - race:danceability  1      9.038 574.99 251.02
## - age                1     10.136 576.09 251.39
## - gender:happiness   1     12.208 578.16 252.10
## - race:genre         3     33.271 599.22 255.15
## - gender:danceability 1     30.208 596.16 258.14
##
## Step: AIC=249.56
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + race:genre + race:energy +
##      race:danceability + gender:danceability + gender:happiness
##
##              Df Sum of Sq    RSS    AIC
## - loudness_db        1      4.255 575.01 249.02
## <none>                                570.75 249.56
## - race:energy        1      6.452 577.20 249.77
## - age                1     10.174 580.93 251.04
## - gender:happiness   1     11.277 582.03 251.41
## - race:danceability  1     12.736 583.49 251.91
## - race:genre         3     28.783 599.54 253.25
## - length_sec         1     17.833 588.58 253.62
## - gender:danceability 1     33.114 603.87 258.67
##
## Step: AIC=249.02
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + length_sec + race:genre + race:energy + race:danceability +
##      gender:danceability + gender:happiness
##
##              Df Sum of Sq    RSS    AIC
## <none>                                575.01 249.02
## - race:energy        1      6.391 581.40 249.20
## - age                1      9.812 584.82 250.36
## - gender:happiness   1     11.732 586.74 251.00
## - race:danceability  1     14.130 589.14 251.81
## - race:genre         3     28.484 603.49 252.55
## - length_sec         1     18.065 593.07 253.12
## - gender:danceability 1     34.156 609.16 258.39

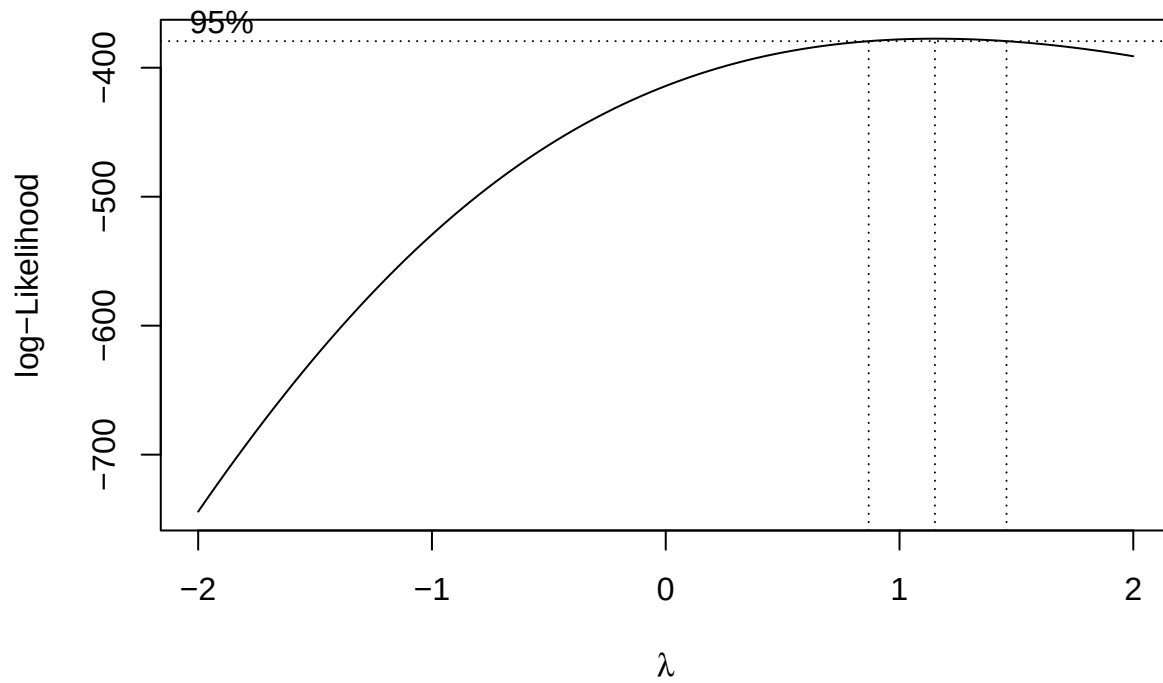
```

```
## [1] "1960s"
```

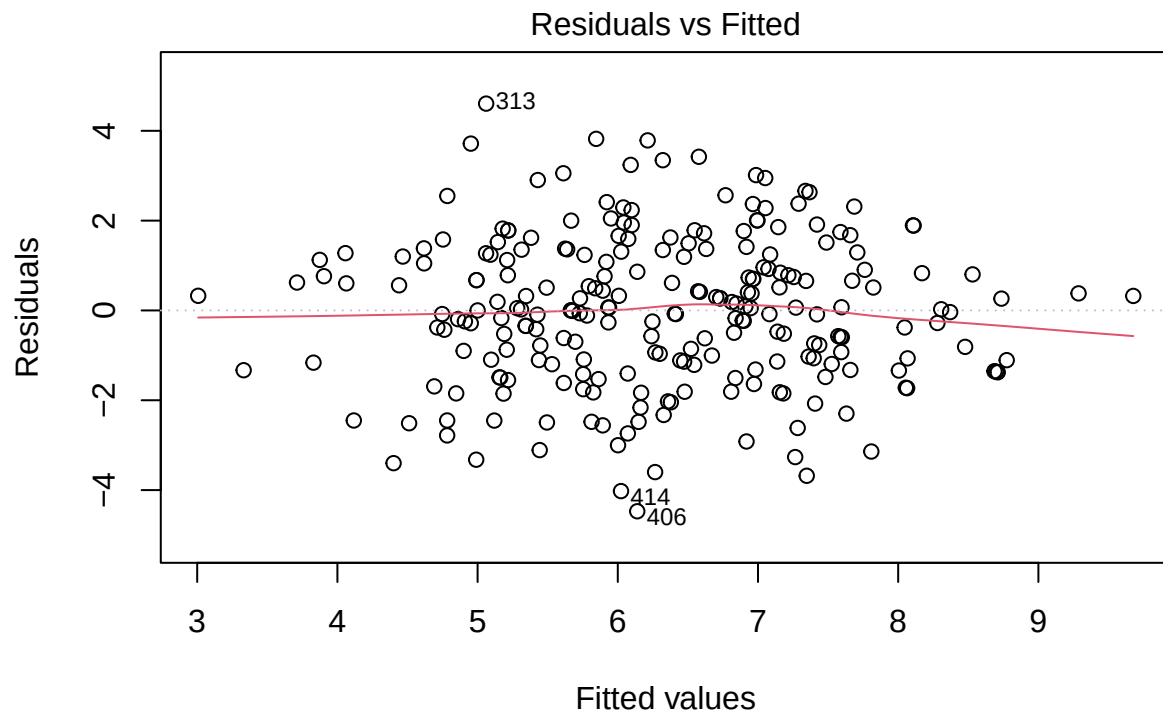
```
## Warning: not plotting observations with leverage one:
```

```
## 239
```

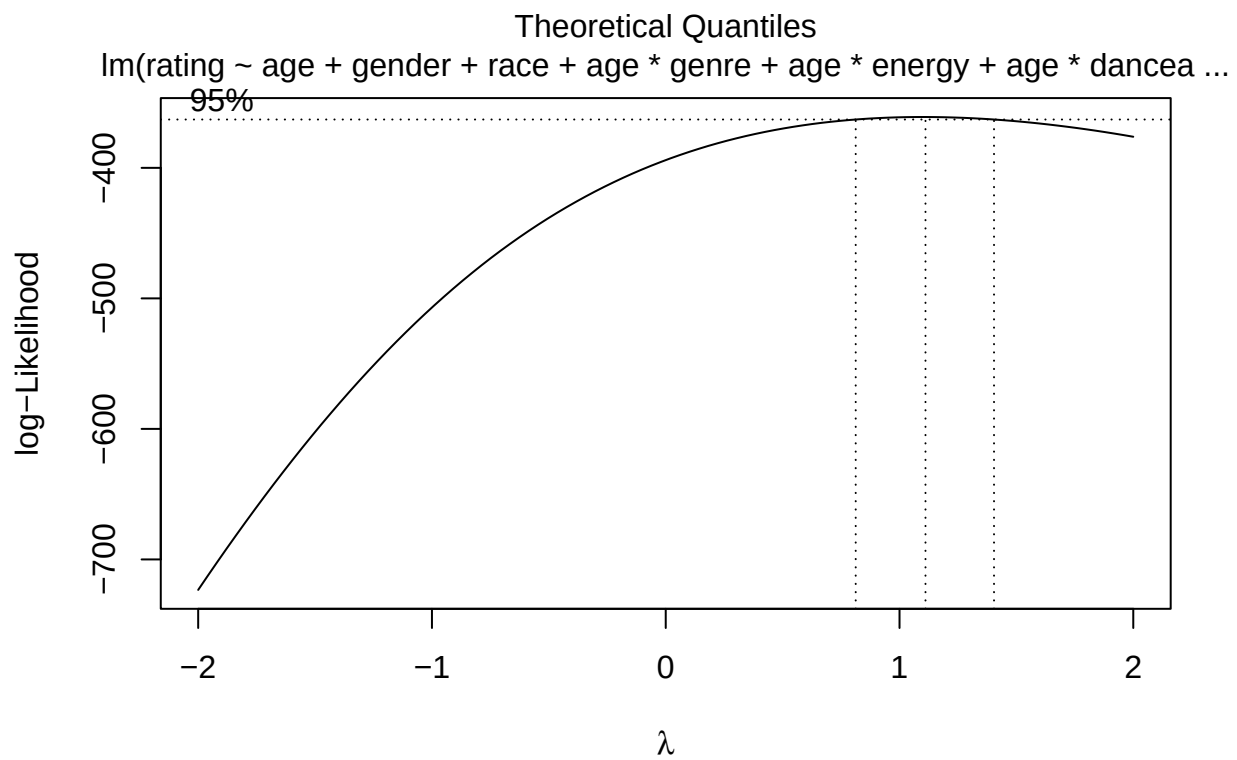
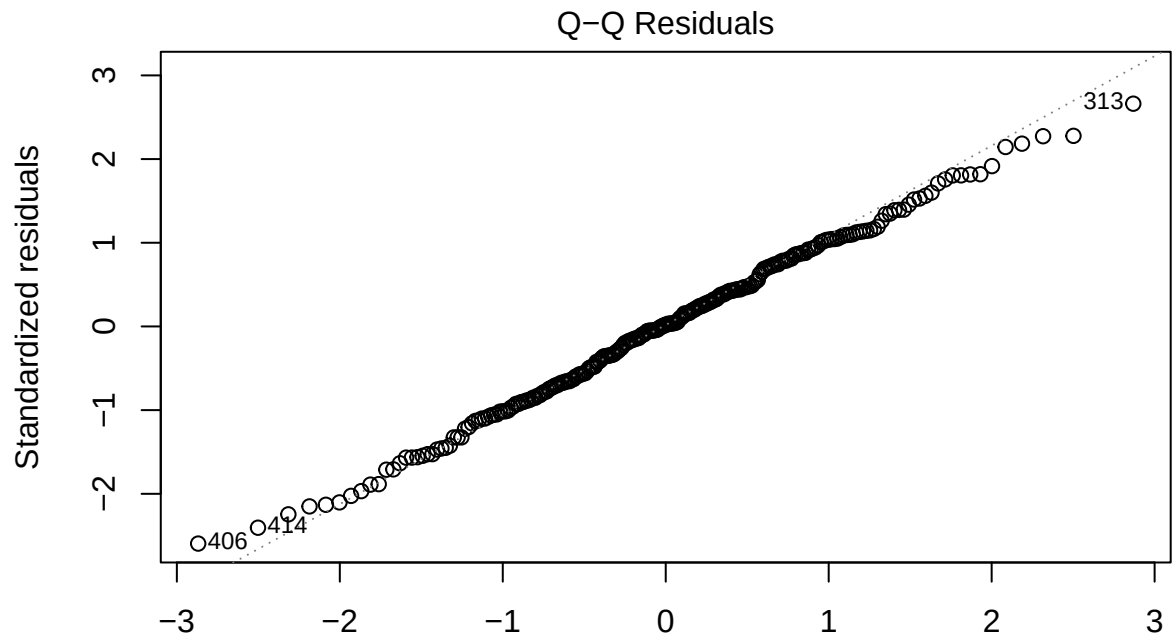




```
## Warning: not plotting observations with leverage one:
## 38, 239
```



lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...



```
## Start: AIC=327.31
## rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + +gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##
```

```

##          Df Sum of Sq    RSS    AIC
## - gender:genre      3    4.5148 671.33 322.96
## - race:happiness     1    0.0509 666.87 325.33
## - gender:bpm         1    0.0575 666.87 325.33
## - race:loudness_db   1    0.0637 666.88 325.33
## - gender:happiness   1    0.0693 666.88 325.33
## - gender:danceability 1    0.0693 666.88 325.33
## - race:danceability  1    0.5466 667.36 325.51
## - age:loudness_db    1    0.6594 667.47 325.55
## - age:danceability   1    1.2483 668.06 325.77
## - gender:loudness_db 1    1.7006 668.52 325.93
## - gender:energy      1    1.9573 668.77 326.03
## - race:length_sec    1    2.1587 668.97 326.10
## - race:genre         3   14.9204 681.74 326.73
## - age:energy         1    3.9706 670.79 326.76
## <none>                666.81 327.31
## - age:happiness      1    5.9102 672.73 327.47
## - race:bpm           1    6.7182 673.53 327.76
## - race:energy        1    7.1011 673.92 327.90
## - age:bpm            1   10.1186 676.93 329.00
## - gender:length_sec  1   12.1319 678.95 329.72
## - age:genre          3   23.7323 690.55 329.88
## - age:length_sec     1   14.2379 681.05 330.48
##
## Step:  AIC=322.96
## rating ~ age + gender + race + genre + energy + danceability +
##          happiness + bpm + loudness_db + length_sec + age:genre +
##          age:energy + age:danceability + age:happiness + age:bpm +
##          age:loudness_db + age:length_sec + race:genre + race:energy +
##          race:danceability + race:happiness + race:bpm + race:loudness_db +
##          race:length_sec + gender:energy + gender:danceability + gender:happiness +
##          gender:bpm + gender:loudness_db + gender:length_sec
##
##          Df Sum of Sq    RSS    AIC
## - race:loudness_db   1    0.0027 671.33 320.96
## - gender:danceability 1    0.0064 671.34 320.96
## - gender:happiness    1    0.0151 671.34 320.97
## - race:happiness      1    0.0240 671.35 320.97
## - gender:bpm          1    0.3552 671.69 321.09
## - race:danceability   1    0.4579 671.79 321.13
## - age:loudness_db     1    0.4797 671.81 321.14
## - gender:loudness_db  1    0.6231 671.95 321.19
## - gender:energy       1    0.7602 672.09 321.24
## - age:danceability    1    1.0366 672.37 321.34
## - race:length_sec     1    2.5747 673.90 321.90
## - race:genre          3   14.1348 685.46 322.07
## - age:energy          1    3.4393 674.77 322.21
## <none>                671.33 322.96
## - age:happiness      1    6.3026 677.63 323.25
## - race:energy         1    6.7154 678.05 323.40
## - race:bpm           1    7.2589 678.59 323.60
## - age:bpm            1   11.3985 682.73 325.09
## - age:genre          3   24.5416 695.87 325.76
## - age:length_sec     1   14.3055 685.64 326.13

```

```

## - gender:length_sec      1    16.1867 687.52 326.80
##
## Step: AIC=320.96
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:energy +
##      race:danceability + race:happiness + race:bpm + race:length_sec +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:bpm + gender:loudness_db + gender:length_sec
##
##
##      Df Sum of Sq    RSS    AIC
## - gender:danceability  1      0.0071 671.34 318.96
## - gender:happiness     1      0.0159 671.35 318.97
## - race:happiness       1      0.0312 671.36 318.97
## - gender:bpm           1      0.3731 671.71 319.10
## - race:danceability    1      0.4623 671.79 319.13
## - age:loudness_db      1      0.4824 671.81 319.14
## - gender:loudness_db   1      0.6332 671.97 319.19
## - gender:energy        1      0.7591 672.09 319.24
## - age:danceability     1      1.0353 672.37 319.34
## - race:length_sec      1      2.5770 673.91 319.90
## - race:genre           3     14.3567 685.69 320.15
## - age:energy           1      3.4369 674.77 320.21
## <none>                  671.33 320.96
## - age:happiness        1      6.3995 677.73 321.29
## - race:bpm             1      7.4374 678.77 321.66
## - age:bpm              1     11.4625 682.79 323.11
## - race:energy          1     11.7473 683.08 323.21
## - age:genre            3     25.2477 696.58 324.01
## - age:length_sec       1     14.4572 685.79 324.18
## - gender:length_sec    1     16.1849 687.52 324.80
##
## Step: AIC=318.96
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:energy +
##      race:danceability + race:happiness + race:bpm + race:length_sec +
##      gender:energy + gender:happiness + gender:bpm + gender:loudness_db +
##      gender:length_sec
##
##
##      Df Sum of Sq    RSS    AIC
## - gender:happiness     1      0.0100 671.35 316.97
## - race:happiness       1      0.0299 671.37 316.98
## - gender:bpm           1      0.3868 671.73 317.11
## - race:danceability    1      0.4569 671.80 317.13
## - age:loudness_db      1      0.4817 671.82 317.14
## - gender:loudness_db   1      0.6269 671.97 317.19
## - gender:energy        1      0.7663 672.11 317.24
## - age:danceability     1      1.0643 672.40 317.35
## - race:length_sec      1      2.5720 673.91 317.90
## - race:genre           3     14.3666 685.71 318.15
## - age:energy           1      3.4407 674.78 318.22

```

```

## <none>                                671.34 318.96
## - age:happiness          1      6.4179 677.76 319.30
## - race:bpm               1      7.4310 678.77 319.66
## - age:bpm                1     11.4599 682.80 321.11
## - race:energy            1     11.7443 683.08 321.21
## - age:genre              3     25.2747 696.61 322.02
## - age:length_sec         1     14.5147 685.85 322.20
## - gender:length_sec      1     16.4894 687.83 322.91
##
## Step:  AIC=316.97
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:energy +
##      race:danceability + race:happiness + race:bpm + race:length_sec +
##      gender:energy + gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:happiness      1      0.0225 671.37 314.98
## - gender:bpm           1      0.3845 671.73 315.11
## - race:danceability    1      0.4543 671.80 315.13
## - age:loudness_db      1      0.4860 671.84 315.15
## - gender:loudness_db   1      0.7785 672.13 315.25
## - age:danceability     1      1.0593 672.41 315.35
## - gender:energy        1      1.4978 672.85 315.51
## - race:length_sec      1      2.6242 673.97 315.92
## - race:genre           3     14.4707 685.82 316.19
## - age:energy           1      3.4312 674.78 316.22
## <none>                                671.35 316.97
## - age:happiness        1      6.4119 677.76 317.30
## - race:bpm             1      7.4527 678.80 317.67
## - age:bpm              1     11.4618 682.81 319.12
## - race:energy          1     12.8095 684.16 319.60
## - age:genre            3     25.3765 696.73 320.06
## - age:length_sec       1     14.8398 686.19 320.32
## - gender:length_sec    1     16.7887 688.14 321.02
##
## Step:  AIC=314.98
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:energy +
##      race:danceability + race:bpm + race:length_sec + gender:energy +
##      gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:bpm           1      0.3886 671.76 313.12
## - age:loudness_db      1      0.4659 671.84 313.15
## - race:danceability    1      0.4709 671.84 313.15
## - gender:loudness_db   1      0.7622 672.13 313.25
## - age:danceability     1      1.1477 672.52 313.39
## - gender:energy        1      1.5000 672.87 313.52
## - race:length_sec      1      2.6029 673.98 313.92
## - race:genre           3     14.5057 685.88 314.21

```

```

## - age:energy          1      3.4088 674.78 314.22
## <none>                  671.37 314.98
## - age:happiness       1      6.6747 678.05 315.40
## - race:bpm            1      7.5442 678.92 315.71
## - age:bpm             1     11.4466 682.82 317.12
## - age:genre           3     25.7596 697.13 318.20
## - age:length_sec      1     14.8393 686.21 318.33
## - race:energy         1     15.8498 687.22 318.69
## - gender:length_sec   1     17.0128 688.38 319.11
##
## Step:  AIC=313.12
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:energy +
##      race:danceability + race:bpm + race:length_sec + gender:energy +
##      gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:danceability  1      0.4025 672.16 311.26
## - age:loudness_db    1      0.5193 672.28 311.31
## - gender:loudness_db 1      0.8559 672.62 311.43
## - age:danceability   1      1.1403 672.90 311.53
## - gender:energy      1      1.8979 673.66 311.81
## - race:length_sec    1      2.5654 674.33 312.05
## - race:genre         3     14.4815 686.24 312.34
## - age:energy         1      3.5062 675.27 312.39
## <none>                671.76 313.12
## - age:happiness      1      6.6135 678.37 313.52
## - race:bpm           1      7.8870 679.65 313.98
## - age:bpm            1     11.5684 683.33 315.30
## - age:genre          3     25.4697 697.23 316.24
## - age:length_sec     1     14.8834 686.64 316.49
## - race:energy        1     15.7902 687.55 316.81
## - gender:length_sec  1     17.0058 688.77 317.24
##
## Step:  AIC=311.26
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:energy +
##      race:bpm + race:length_sec + gender:energy + gender:loudness_db +
##      gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - age:loudness_db    1      0.5933 672.76 309.48
## - age:danceability    1      1.1692 673.33 309.69
## - gender:loudness_db  1      1.1994 673.36 309.70
## - gender:energy      1      2.2366 674.40 310.08
## - race:length_sec    1      2.4276 674.59 310.15
## - age:energy         1      3.5750 675.74 310.56
## - race:genre         3     16.4217 688.59 311.18
## <none>                672.16 311.26
## - age:happiness      1      6.8066 678.97 311.73

```

```

## - race:bpm          1      7.6990 679.86 312.06
## - age:bpm           1     11.6400 683.80 313.47
## - age:genre         3     25.2709 697.43 314.31
## - age:length_sec    1     15.1451 687.31 314.72
## - race:energy       1     16.3393 688.50 315.15
## - gender:length_sec  1     16.7323 688.90 315.29
##
## Step:  AIC=309.48
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:length_sec + race:genre + race:energy + race:bpm + race:length_sec +
##      gender:energy + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq   RSS   AIC
## - gender:loudness_db  1      0.9620 673.72 307.83
## - age:danceability    1      1.1381 673.89 307.89
## - gender:energy       1      2.0913 674.85 308.24
## - race:length_sec     1      2.3781 675.13 308.35
## - age:energy          1      3.5224 676.28 308.76
## - race:genre          3     16.1447 688.90 309.29
## <none>                672.76 309.48
## - race:bpm           1      7.5503 680.31 310.21
## - age:happiness       1     10.0098 682.77 311.10
## - age:bpm            1     11.5872 684.34 311.67
## - age:genre           3     25.0042 697.76 312.42
## - race:energy         1     15.7540 688.51 313.15
## - age:length_sec      1     16.6943 689.45 313.49
## - gender:length_sec   1     17.5417 690.30 313.79
##
## Step:  AIC=307.83
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:length_sec + race:genre + race:energy + race:bpm + race:length_sec +
##      gender:energy + gender:length_sec
##
##              Df Sum of Sq   RSS   AIC
## - age:danceability    1      1.0841 674.80 306.23
## - gender:energy       1      1.1310 674.85 306.24
## - race:length_sec     1      2.5843 676.30 306.77
## - age:energy          1      3.8324 677.55 307.22
## <none>                673.72 307.83
## - race:genre          3     17.8821 691.60 308.25
## - race:bpm           1      7.2729 680.99 308.46
## - loudness_db         1      7.5347 681.25 308.56
## - age:happiness       1      9.8237 683.54 309.38
## - age:bpm            1     11.4953 685.21 309.98
## - age:genre           3     25.1786 698.90 310.82
## - race:energy         1     16.9276 690.65 311.91
## - age:length_sec      1     17.0886 690.81 311.97
## - gender:length_sec   1     18.2172 691.94 312.37
##
## Step:  AIC=306.22

```

```

## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:genre +
##   age:energy + age:happiness + age:bpm + age:length_sec + race:genre +
##   race:energy + race:bpm + race:length_sec + gender:energy +
##   gender:length_sec
##
##           Df Sum of Sq   RSS   AIC
## - danceability      1    0.6482 675.45 304.46
## - gender:energy      1    1.0456 675.85 304.60
## - race:length_sec    1    2.6561 677.46 305.19
## - age:energy         1    3.3828 678.19 305.45
## <none>                674.80 306.23
## - race:genre        3   18.0765 692.88 306.70
## - race:bpm          1    7.0787 681.88 306.78
## - loudness_db       1    7.4797 682.28 306.93
## - age:happiness     1    9.8679 684.67 307.78
## - age:bpm           1   12.2391 687.04 308.63
## - age:genre         3   25.6748 700.48 309.37
## - age:length_sec    1   16.0048 690.81 309.97
## - race:energy       1   16.9075 691.71 310.29
## - gender:length_sec  1   17.8339 692.64 310.62
##
## Step:  AIC=304.46
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##   loudness_db + length_sec + age:genre + age:energy + age:happiness +
##   age:bpm + age:length_sec + race:genre + race:energy + race:bpm +
##   race:length_sec + gender:energy + gender:length_sec
##
##           Df Sum of Sq   RSS   AIC
## - gender:energy      1    1.0840 676.53 302.85
## - race:length_sec    1    2.4834 677.93 303.36
## - age:energy         1    3.3649 678.82 303.68
## <none>                675.45 304.46
## - race:genre        3   17.6009 693.05 304.76
## - race:bpm          1    7.2391 682.69 305.07
## - loudness_db       1    7.3376 682.79 305.11
## - age:happiness     1    9.7081 685.16 305.96
## - age:bpm           1   12.3161 687.77 306.89
## - age:genre         3   25.2517 700.70 307.45
## - age:length_sec    1   15.9393 691.39 308.17
## - race:energy       1   16.7780 692.23 308.47
## - gender:length_sec  1   18.1069 693.56 308.94
##
## Step:  AIC=302.85
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##   loudness_db + length_sec + age:genre + age:energy + age:happiness +
##   age:bpm + age:length_sec + race:genre + race:energy + race:bpm +
##   race:length_sec + gender:length_sec
##
##           Df Sum of Sq   RSS   AIC
## - race:length_sec    1    2.5400 679.07 301.77
## - age:energy         1    3.2509 679.79 302.03
## <none>                676.53 302.85
## - race:genre        3   18.5325 695.07 303.47

```

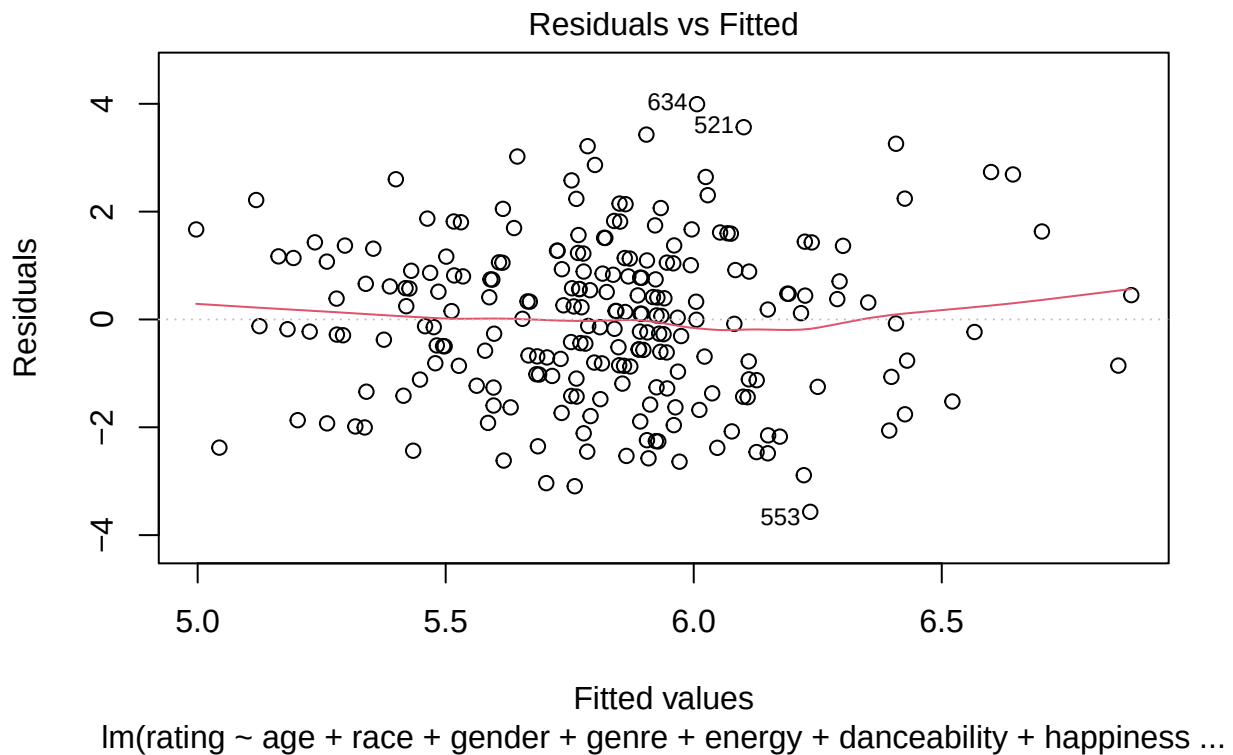
```

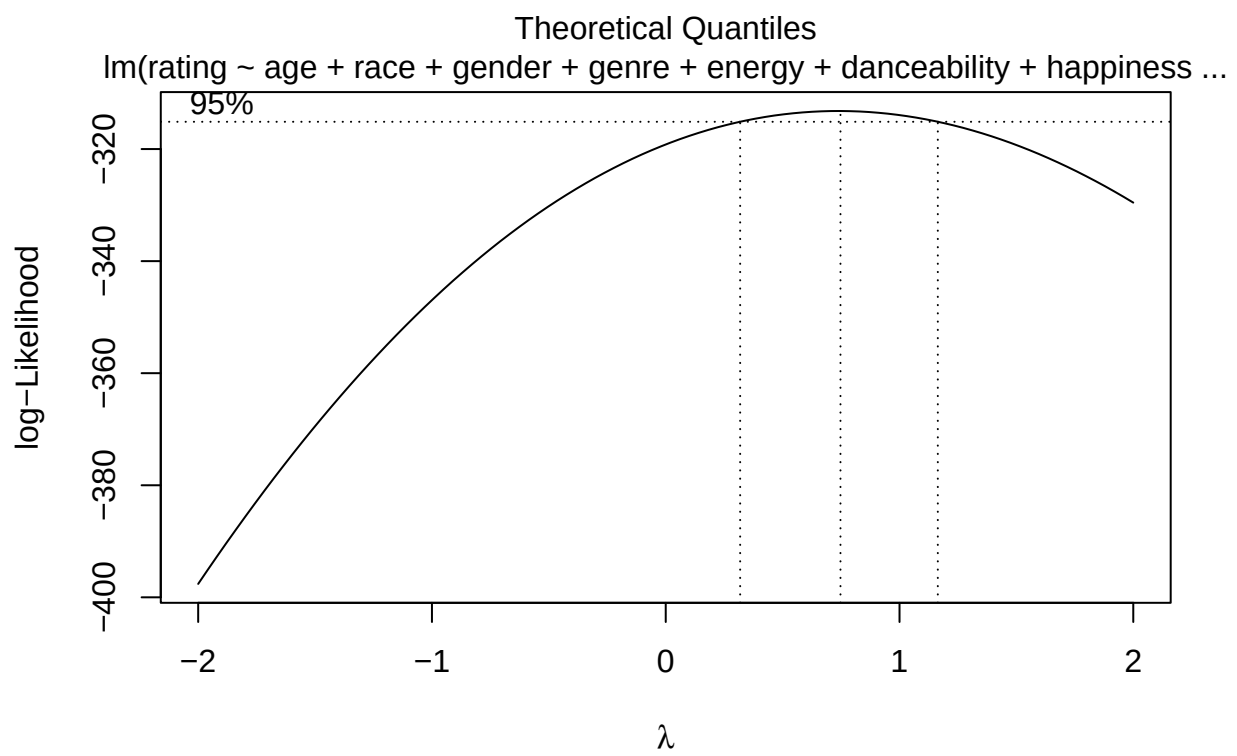
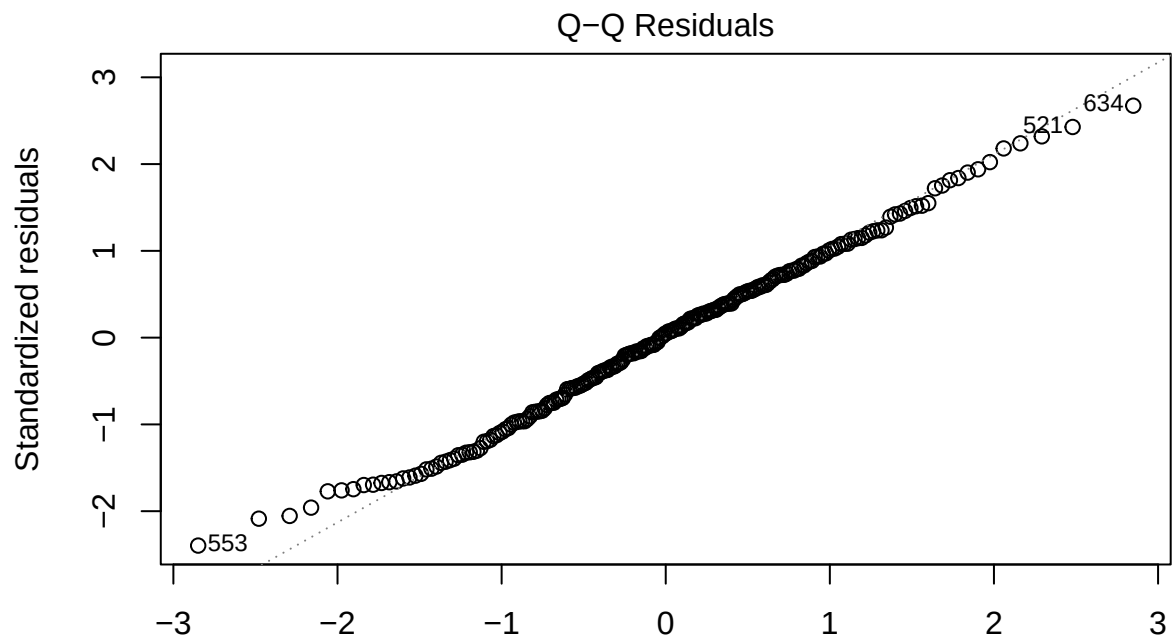
## - race:bpm          1    7.7735 684.31 303.65
## - loudness_db       1    8.4575 684.99 303.90
## - age:happiness     1    9.4000 685.93 304.23
## - age:bpm           1   12.2406 688.78 305.25
## - age:genre         3   24.6657 701.20 305.63
## - age:length_sec    1   15.6285 692.16 306.45
## - race:energy       1   16.3123 692.85 306.69
## - gender:length_sec 1   18.0731 694.61 307.31
##
## Step:  AIC=301.77
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##      loudness_db + length_sec + age:genre + age:energy + age:happiness +
##      age:bpm + age:length_sec + race:genre + race:energy + race:bpm +
##      gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - age:energy    1    3.6200 682.69 301.07
## - race:genre    3   16.7212 695.80 301.73
## <none>              679.07 301.77
## - race:bpm      1    7.1994 686.27 302.36
## - age:happiness  1    8.3054 687.38 302.75
## - loudness_db   1    8.6956 687.77 302.89
## - age:bpm       1   12.2490 691.32 304.15
## - age:genre     3   24.0867 703.16 304.31
## - race:energy   1   14.9064 693.98 305.09
## - age:length_sec 1   16.2469 695.32 305.56
## - gender:length_sec 1  16.4467 695.52 305.63
##
## Step:  AIC=301.07
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##      loudness_db + length_sec + age:genre + age:happiness + age:bpm +
##      age:length_sec + race:genre + race:energy + race:bpm + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:genre    3   16.5383 699.23 300.94
## <none>              682.69 301.07
## - race:bpm      1    6.1217 688.82 301.26
## - loudness_db   1    7.8042 690.50 301.86
## - age:genre     3   21.4854 704.18 302.67
## - age:bpm       1   11.7041 694.40 303.24
## - race:energy   1   14.0155 696.71 304.05
## - gender:length_sec 1  16.6269 699.32 304.97
## - age:length_sec 1   17.3916 700.09 305.24
## - age:happiness  1   23.6421 706.34 307.41
##
## Step:  AIC=300.94
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##      loudness_db + length_sec + age:genre + age:happiness + age:bpm +
##      age:length_sec + race:energy + race:bpm + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:bpm      1    3.8475 703.08 300.28
## <none>              699.23 300.94
## - age:bpm       1    6.7730 706.01 301.30

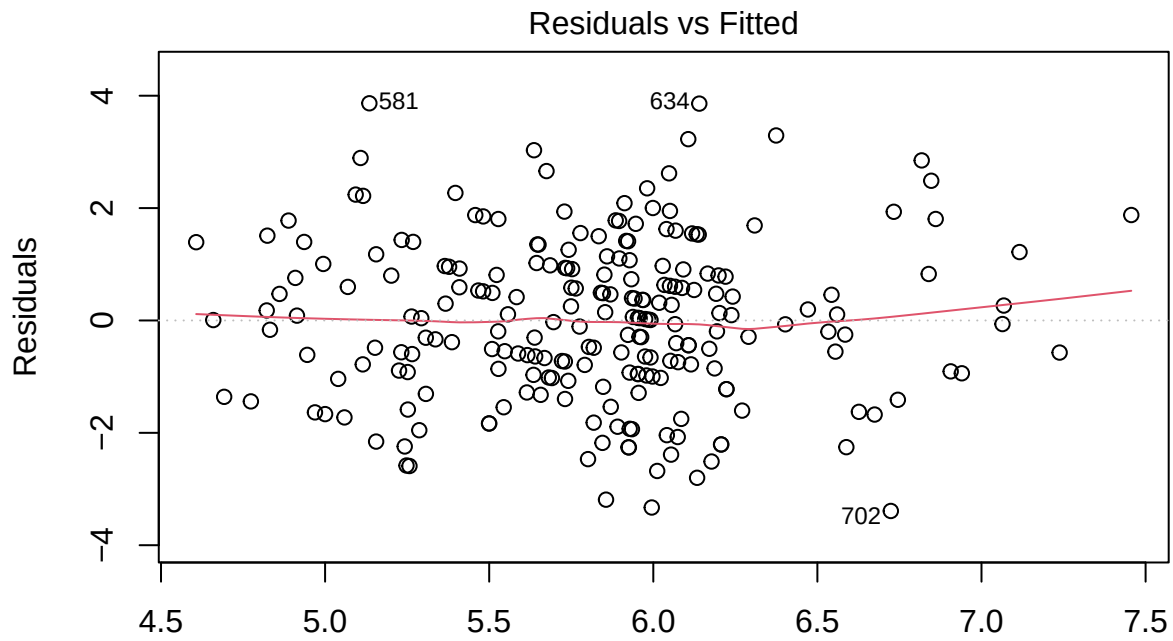
```



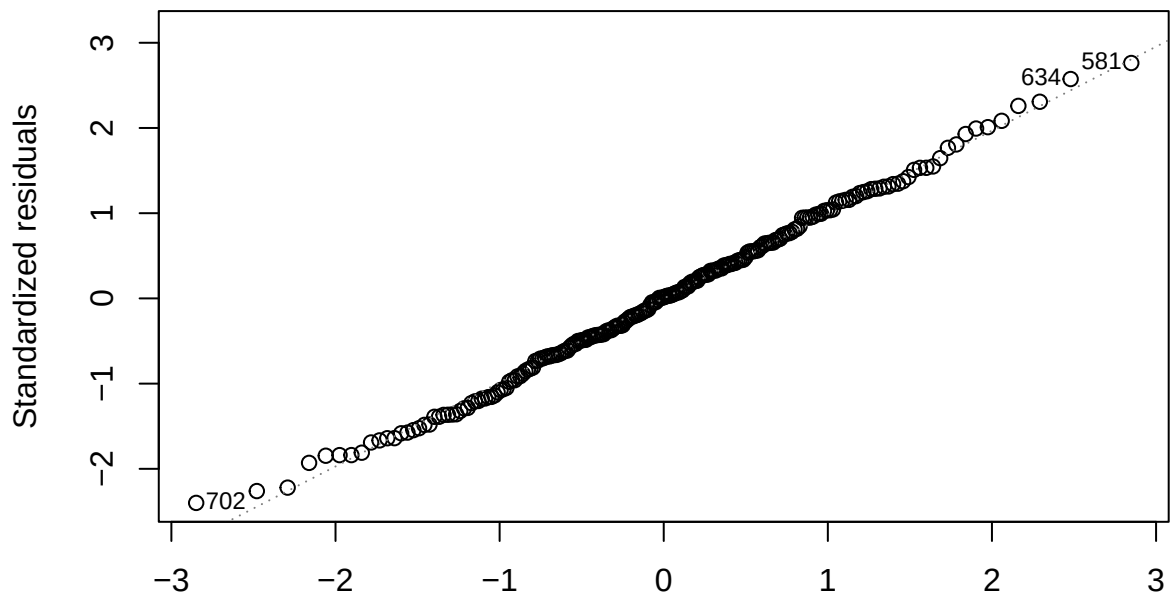
```
## - loudness_db      1      7.1708 706.40 301.44
## - race:energy      1      8.1624 707.40 301.78
## - age:genre        3     20.3701 719.60 301.97
## - age:happiness    1     19.1919 718.43 305.57
## - gender:length_sec 1     19.3127 718.55 305.61
## - age:length_sec   1     21.0050 720.24 306.19
##
## Step:  AIC=300.28
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##      loudness_db + length_sec + age:genre + age:happiness + age:bpm +
##      age:length_sec + race:energy + gender:length_sec
##
##              Df Sum of Sq   RSS   AIC
## <none>                703.08 300.28
## - loudness_db        1    5.8449 708.93 300.31
## - age:bpm            1    7.7319 710.81 300.96
## - race:energy        1    7.8422 710.92 301.00
## - age:genre          3   19.6097 722.69 301.02
## - age:happiness      1   17.9609 721.04 304.46
## - gender:length_sec  1   19.5122 722.59 304.99
## - age:length_sec     1   21.1947 724.28 305.56
## [1] "1970s"
```



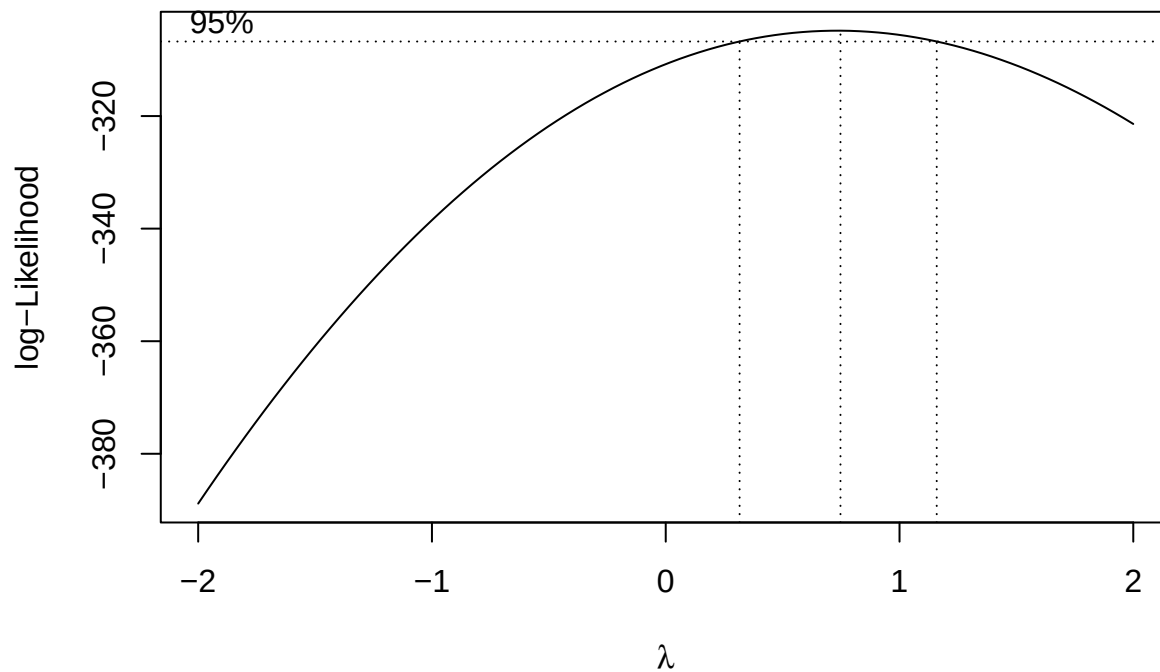




lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...
Q-Q Residuals



lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...



```
## Start: AIC=231.98
## rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + +gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##
##
##           Df Sum of Sq  RSS   AIC
## - age:genre      2    1.9499 461.86 228.95
## - age:energy      1     0.0012 459.91 229.98
## - age:bpm         1     0.0300 459.94 230.00
## - gender:length_sec 1     0.0874 460.00 230.03
## - age:length_sec   1     0.1084 460.02 230.04
## - age:danceability 1     0.1513 460.06 230.06
## - gender:bpm       1     0.1817 460.09 230.07
## - race:energy       1     0.2867 460.19 230.12
## - race:loudness_db  1     0.2893 460.20 230.13
## - gender:danceability 1     0.3624 460.27 230.16
## - gender:happiness  1     0.3944 460.30 230.18
## - age:loudness_db   1     0.4693 460.38 230.22
## - race:happiness    1     0.5862 460.49 230.27
## - race:danceability 1     0.6606 460.57 230.31
## - gender:loudness_db 1     0.9713 460.88 230.46
## - gender:genre      2     5.2307 465.14 230.56
## - gender:energy      1     1.5799 461.49 230.76
## - age:happiness      1     2.2112 462.12 231.08
## - race:length_sec    1     2.6993 462.61 231.32
## <none>                                459.91 231.98
## - race:bpm          1     4.0558 463.96 231.99
## - race:genre         2     9.6085 469.52 232.70
```

```

##
## Step: AIC=228.95
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:happiness + age:bpm + age:loudness_db +
## age:length_sec + race:genre + race:energy + race:danceability +
## race:happiness + race:bpm + race:loudness_db + race:length_sec +
## gender:genre + gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
##
## Df Sum of Sq RSS AIC
## - age:bpm 1 0.0045 461.86 226.95
## - age:energy 1 0.0295 461.89 226.96
## - gender:length_sec 1 0.0540 461.91 226.97
## - age:danceability 1 0.1190 461.98 227.01
## - age:length_sec 1 0.1455 462.00 227.02
## - race:loudness_db 1 0.2041 462.06 227.05
## - race:energy 1 0.2552 462.11 227.07
## - gender:bpm 1 0.2679 462.13 227.08
## - gender:danceability 1 0.4433 462.30 227.17
## - race:happiness 1 0.5201 462.38 227.20
## - race:danceability 1 0.5212 462.38 227.21
## - gender:genre 2 4.6735 466.53 227.24
## - gender:happiness 1 0.6296 462.49 227.26
## - age:loudness_db 1 0.6516 462.51 227.27
## - gender:loudness_db 1 1.2440 463.10 227.56
## - gender:energy 1 1.6035 463.46 227.74
## - age:happiness 1 2.3741 464.23 228.12
## - race:length_sec 1 2.5741 464.43 228.22
## <none> 461.86 228.95
## - race:bpm 1 4.3388 466.20 229.08
## - race:genre 2 10.2311 472.09 229.94
##
## Step: AIC=226.95
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:happiness + age:loudness_db + age:length_sec +
## race:genre + race:energy + race:danceability + race:happiness +
## race:bpm + race:loudness_db + race:length_sec + gender:genre +
## gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
##
## Df Sum of Sq RSS AIC
## - age:energy 1 0.0261 461.89 224.96
## - gender:length_sec 1 0.0523 461.91 224.98
## - age:danceability 1 0.1262 461.99 225.01
## - age:length_sec 1 0.1489 462.01 225.02
## - race:loudness_db 1 0.2017 462.06 225.05
## - race:energy 1 0.2510 462.11 225.07
## - gender:bpm 1 0.2874 462.15 225.09
## - gender:danceability 1 0.4557 462.32 225.18
## - race:danceability 1 0.5182 462.38 225.21
## - race:happiness 1 0.5209 462.38 225.21
## - gender:genre 2 4.6699 466.53 225.24

```

```

## - gender:happiness      1      0.6337 462.50 225.26
## - age:loudness_db       1      0.6645 462.53 225.28
## - gender:loudness_db    1      1.2642 463.13 225.57
## - gender:energy         1      1.5993 463.46 225.74
## - age:happiness         1      2.3730 464.24 226.12
## - race:length_sec       1      2.5699 464.43 226.22
## <none>                  461.86 226.95
## - race:bpm              1      4.4004 466.26 227.11
## - race:genre            2     10.3181 472.18 227.99
##
## Step:  AIC=224.96
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:danceability +
##      age:happiness + age:loudness_db + age:length_sec + race:genre +
##      race:energy + race:danceability + race:happiness + race:bpm +
##      race:loudness_db + race:length_sec + gender:genre + gender:energy +
##      gender:danceability + gender:happiness + gender:bpm + gender:loudness_db +
##      gender:length_sec
##
##              Df Sum of Sq   RSS   AIC
## - gender:length_sec      1    0.0538 461.94 222.99
## - age:danceability        1    0.1239 462.01 223.02
## - age:length_sec          1    0.1623 462.05 223.04
## - race:loudness_db        1    0.1880 462.08 223.06
## - race:energy             1    0.2289 462.12 223.08
## - gender:bpm              1    0.2800 462.17 223.10
## - gender:danceability     1    0.4581 462.35 223.19
## - race:happiness          1    0.4974 462.39 223.21
## - race:danceability       1    0.5180 462.41 223.22
## - gender:genre            2    4.6438 466.53 223.24
## - gender:happiness        1    0.6203 462.51 223.27
## - gender:loudness_db      1    1.2404 463.13 223.57
## - gender:energy           1    1.5733 463.46 223.74
## - age:loudness_db         1    2.1843 464.07 224.04
## - race:length_sec         1    2.5787 464.47 224.23
## - age:happiness           1    2.8850 464.77 224.38
## <none>                   461.89 224.96
## - race:bpm               1    4.4130 466.30 225.13
## - race:genre              2   10.3089 472.20 226.00
##
## Step:  AIC=222.99
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:danceability +
##      age:happiness + age:loudness_db + age:length_sec + race:genre +
##      race:energy + race:danceability + race:happiness + race:bpm +
##      race:loudness_db + race:length_sec + gender:genre + gender:energy +
##      gender:danceability + gender:happiness + gender:bpm + gender:loudness_db
##
##              Df Sum of Sq   RSS   AIC
## - age:danceability        1    0.1331 462.08 221.06
## - age:length_sec          1    0.1717 462.11 221.07
## - race:loudness_db        1    0.1770 462.12 221.08
## - race:energy             1    0.2288 462.17 221.10
## - gender:bpm              1    0.3042 462.25 221.14

```

```

## - gender:danceability 1 0.4512 462.39 221.21
## - race:happiness 1 0.5282 462.47 221.25
## - race:danceability 1 0.5683 462.51 221.27
## - gender:genre 2 4.6989 466.64 221.30
## - gender:happiness 1 0.7816 462.72 221.38
## - gender:loudness_db 1 1.3202 463.26 221.64
## - gender:energy 1 1.7167 463.66 221.84
## - age:loudness_db 1 2.1634 464.11 222.06
## - race:length_sec 1 2.5613 464.50 222.25
## - age:happiness 1 2.8505 464.79 222.39
## <none> 461.94 222.99
## - race:bpm 1 4.3683 466.31 223.13
## - race:genre 2 10.2581 472.20 224.00
##
## Step: AIC=221.06
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:happiness +
## age:loudness_db + age:length_sec + race:genre + race:energy +
## race:danceability + race:happiness + race:bpm + race:loudness_db +
## race:length_sec + gender:genre + gender:energy + gender:danceability +
## gender:happiness + gender:bpm + gender:loudness_db
##
## Df Sum of Sq RSS AIC
## - race:loudness_db 1 0.1780 462.25 219.14
## - age:length_sec 1 0.1984 462.27 219.15
## - race:energy 1 0.2120 462.29 219.16
## - gender:bpm 1 0.2824 462.36 219.19
## - gender:danceability 1 0.4577 462.53 219.28
## - race:happiness 1 0.5060 462.58 219.31
## - race:danceability 1 0.5069 462.58 219.31
## - gender:happiness 1 0.7686 462.84 219.43
## - gender:genre 2 4.9016 466.98 219.46
## - gender:loudness_db 1 1.2314 463.31 219.66
## - gender:energy 1 1.6534 463.73 219.87
## - age:loudness_db 1 2.0440 464.12 220.06
## - race:length_sec 1 2.6033 464.68 220.34
## - age:happiness 1 3.9750 466.05 221.01
## <none> 462.08 221.06
## - race:bpm 1 4.2954 466.37 221.16
## - race:genre 2 10.2976 472.37 222.08
##
## Step: AIC=219.14
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:happiness +
## age:loudness_db + age:length_sec + race:genre + race:energy +
## race:danceability + race:happiness + race:bpm + race:length_sec +
## gender:genre + gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db
##
## Df Sum of Sq RSS AIC
## - race:energy 1 0.0542 462.31 217.17
## - age:length_sec 1 0.2032 462.46 217.24
## - gender:bpm 1 0.2909 462.54 217.29
## - race:happiness 1 0.3706 462.62 217.33

```

```

## - race:danceability      1      0.4256 462.68 217.35
## - gender:danceability    1      0.5276 462.78 217.40
## - gender:genre           2      4.7716 467.02 217.48
## - gender:happiness       1      0.7445 463.00 217.51
## - gender:loudness_db     1      1.3677 463.62 217.82
## - gender:energy          1      1.7275 463.98 217.99
## - age:loudness_db        1      2.1081 464.36 218.18
## - race:length_sec        1      2.4861 464.74 218.37
## - age:happiness          1      3.9373 466.19 219.08
## <none>                   462.25 219.14
## - race:bpm               1      4.2157 466.47 219.21
## - race:genre             2     10.1449 472.40 220.09
##
## Step:  AIC=217.17
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:happiness +
##         age:loudness_db + age:length_sec + race:genre + race:danceability +
##         race:happiness + race:bpm + race:length_sec + gender:genre +
##         gender:energy + gender:danceability + gender:happiness +
##         gender:bpm + gender:loudness_db
##
##              Df Sum of Sq  RSS    AIC
## - age:length_sec      1    0.1868 462.49 215.26
## - gender:bpm          1    0.2846 462.59 215.31
## - race:happiness       1    0.3169 462.62 215.33
## - race:danceability    1    0.3881 462.70 215.36
## - gender:danceability  1    0.5096 462.82 215.42
## - gender:genre         2    4.9490 467.26 215.60
## - gender:happiness     1    0.8685 463.18 215.60
## - gender:loudness_db   1    1.3731 463.68 215.85
## - gender:energy        1    1.8062 464.11 216.06
## - age:loudness_db      1    2.0596 464.37 216.18
## - race:length_sec      1    2.4341 464.74 216.37
## - age:happiness        1    3.8898 466.20 217.08
## <none>                 462.31 217.17
## - race:bpm            1    4.3531 466.66 217.31
## - race:genre           2   10.5639 472.87 218.32
##
## Step:  AIC=215.26
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:happiness +
##         age:loudness_db + race:genre + race:danceability + race:happiness +
##         race:bpm + race:length_sec + gender:genre + gender:energy +
##         gender:danceability + gender:happiness + gender:bpm + gender:loudness_db
##
##              Df Sum of Sq  RSS    AIC
## - gender:bpm          1    0.2664 462.76 213.39
## - race:happiness       1    0.3265 462.82 213.42
## - race:danceability    1    0.4058 462.90 213.46
## - gender:danceability  1    0.4601 462.95 213.49
## - gender:happiness     1    0.8225 463.32 213.67
## - gender:genre         2    4.9349 467.43 213.68
## - gender:loudness_db   1    1.3852 463.88 213.94
## - gender:energy        1    1.8147 464.31 214.16

```



```

## - age:loudness_db      1      2.0156 464.51 214.25
## - race:length_sec      1      2.7952 465.29 214.64
## <none>                  462.49 215.26
## - race:bpm             1      4.4195 466.91 215.43
## - age:happiness         1      4.4987 466.99 215.47
## - race:genre           2     10.4449 472.94 216.35
##
## Step:  AIC=213.39
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:happiness +
##      age:loudness_db + race:genre + race:danceability + race:happiness +
##      race:bpm + race:length_sec + gender:genre + gender:energy +
##      gender:danceability + gender:happiness + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - race:happiness      1      0.3541 463.11 211.57
## - gender:danceability  1      0.3876 463.15 211.58
## - race:danceability    1      0.4207 463.18 211.60
## - gender:happiness      1      0.8365 463.60 211.81
## - gender:genre         2      5.1465 467.91 211.91
## - gender:loudness_db   1      1.2417 464.00 212.00
## - gender:energy        1      1.6599 464.42 212.21
## - age:loudness_db      1      1.8410 464.60 212.30
## - race:length_sec      1      3.0825 465.84 212.91
## <none>                  462.76 213.39
## - race:bpm             1      4.5066 467.27 213.60
## - age:happiness         1      4.5776 467.34 213.64
## - race:genre           2     10.4590 473.22 214.49
##
## Step:  AIC=211.57
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:happiness +
##      age:loudness_db + race:genre + race:danceability + race:bpm +
##      race:length_sec + gender:genre + gender:energy + gender:danceability +
##      gender:happiness + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - race:danceability    1      0.1513 463.27 209.64
## - gender:danceability  1      0.3694 463.48 209.75
## - gender:happiness      1      0.9129 464.03 210.02
## - gender:genre         2      5.1128 468.23 210.07
## - gender:loudness_db   1      1.1987 464.31 210.16
## - age:loudness_db      1      1.7438 464.86 210.42
## - gender:energy        1      1.7740 464.89 210.44
## - race:length_sec      1      2.8635 465.98 210.97
## <none>                  463.11 211.57
## - age:happiness         1      4.3610 467.48 211.70
## - race:bpm             1      4.9196 468.03 211.98
## - race:genre           2     10.7089 473.82 212.78
##
## Step:  AIC=209.64
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:happiness +
##      age:loudness_db + race:genre + race:bpm + race:length_sec +

```

```

##      gender:genre + gender:energy + gender:danceability + gender:happiness +
##      gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - gender:danceability  1      0.3893 463.66 207.83
## - gender:happiness     1      0.9126 464.18 208.09
## - gender:loudness_db   1      1.1555 464.42 208.21
## - gender:genre         2      5.4341 468.70 208.30
## - gender:energy        1      1.7125 464.98 208.48
## - age:loudness_db      1      1.7312 465.00 208.49
## - race:length_sec      1      3.1413 466.41 209.18
## <none>                  463.27 209.64
## - age:happiness        1      4.5633 467.83 209.88
## - race:bpm             1      4.7694 468.04 209.98
## - race:genre           2     10.9350 474.20 210.96
##
## Step:  AIC=207.83
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:happiness +
##      age:loudness_db + race:genre + race:bpm + race:length_sec +
##      gender:genre + gender:energy + gender:happiness + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - danceability         1      0.7050 464.36 206.18
## - gender:genre          2      5.0799 468.74 206.32
## - gender:loudness_db    1      1.0823 464.74 206.37
## - age:loudness_db       1      1.6314 465.29 206.63
## - gender:energy         1      1.7430 465.40 206.69
## - gender:happiness      1      2.2347 465.89 206.93
## - race:length_sec       1      3.4994 467.15 207.55
## <none>                  463.66 207.83
## - age:happiness         1      4.5415 468.20 208.06
## - race:bpm              1      4.6745 468.33 208.12
## - race:genre            2     11.4150 475.07 209.38
##
## Step:  AIC=206.18
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##      loudness_db + length_sec + age:happiness + age:loudness_db +
##      race:genre + race:bpm + race:length_sec + gender:genre +
##      gender:energy + gender:happiness + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - gender:loudness_db    1      1.3663 465.73 204.85
## - age:loudness_db       1      1.3747 465.74 204.85
## - gender:genre          2      5.5955 469.96 204.91
## - gender:energy         1      2.2257 466.59 205.27
## - gender:happiness      1      2.7064 467.07 205.50
## - race:length_sec       1      3.5065 467.87 205.90
## <none>                  464.36 206.18
## - age:happiness         1      4.4755 468.84 206.37
## - race:bpm              1      5.5505 469.91 206.89
## - race:genre            2     12.6578 477.02 208.31
##
## Step:  AIC=204.85

```

```

## rating ~ age + gender + race + genre + energy + happiness + bpm +
##   loudness_db + length_sec + age:happiness + age:loudness_db +
##   race:genre + race:bpm + race:length_sec + gender:genre +
##   gender:energy + gender:happiness
##
##           Df Sum of Sq   RSS   AIC
## - gender:energy      1    0.9021 466.63 203.29
## - age:loudness_db     1    1.2394 466.97 203.46
## - gender:happiness    1    2.0811 467.81 203.87
## - gender:genre        2    6.5221 472.25 204.02
## - race:length_sec     1    3.9711 469.70 204.79
## <none>                                465.73 204.85
## - age:happiness       1    4.1688 469.90 204.88
## - race:bpm            1    5.5953 471.32 205.57
## - race:genre          2   13.7119 479.44 207.47
##
## Step:   AIC=203.29
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##   loudness_db + length_sec + age:happiness + age:loudness_db +
##   race:genre + race:bpm + race:length_sec + gender:genre +
##   gender:happiness
##
##           Df Sum of Sq   RSS   AIC
## - energy      1    0.6261 467.25 201.60
## - gender:happiness  1    1.2457 467.87 201.90
## - age:loudness_db  1    1.4788 468.11 202.01
## - gender:genre     2    5.9962 472.62 202.20
## - race:length_sec  1    3.4423 470.07 202.97
## - age:happiness    1    3.9814 470.61 203.23
## <none>                                466.63 203.29
## - race:bpm         1    5.2795 471.91 203.86
## - race:genre        2   14.2021 480.83 206.13
##
## Step:   AIC=201.6
## rating ~ age + gender + race + genre + happiness + bpm + loudness_db +
##   length_sec + age:happiness + age:loudness_db + race:genre +
##   race:bpm + race:length_sec + gender:genre + gender:happiness
##
##           Df Sum of Sq   RSS   AIC
## - gender:happiness  1    1.3568 468.61 200.26
## - age:loudness_db   1    1.3626 468.62 200.26
## - gender:genre      2    6.2055 473.46 200.60
## - age:happiness     1    3.8434 471.10 201.46
## - race:length_sec   1    3.8805 471.14 201.48
## <none>                                467.25 201.60
## - race:bpm          1    5.4241 472.68 202.23
## - race:genre        2   13.7123 480.97 204.19
##
## Step:   AIC=200.26
## rating ~ age + gender + race + genre + happiness + bpm + loudness_db +
##   length_sec + age:happiness + age:loudness_db + race:genre +
##   race:bpm + race:length_sec + gender:genre
##
##           Df Sum of Sq   RSS   AIC

```

```

## - age:loudness_db 1 1.4500 470.06 198.96
## - gender:genre 2 5.6214 474.23 198.98
## - race:length_sec 1 3.8041 472.42 200.10
## <none> 468.61 200.26
## - age:happiness 1 5.2600 473.87 200.80
## - race:bpm 1 5.4847 474.10 200.91
## - race:genre 2 13.9050 482.52 202.93
##
## Step: AIC=198.96
## rating ~ age + gender + race + genre + happiness + bpm + loudness_db +
## length_sec + age:happiness + race:genre + race:bpm + race:length_sec +
## gender:genre
##
## Df Sum of Sq RSS AIC
## - loudness_db 1 1.8844 471.95 197.87
## - gender:genre 2 6.1122 476.17 197.91
## <none> 470.06 198.96
## - age:happiness 1 4.1987 474.26 198.99
## - race:length_sec 1 4.4861 474.55 199.13
## - race:bpm 1 5.7202 475.78 199.72
## - race:genre 2 13.6185 483.68 201.47
##
## Step: AIC=197.87
## rating ~ age + gender + race + genre + happiness + bpm + length_sec +
## age:happiness + race:genre + race:bpm + race:length_sec +
## gender:genre
##
## Df Sum of Sq RSS AIC
## - gender:genre 2 6.0005 477.95 196.75
## - age:happiness 1 3.9888 475.93 197.79
## <none> 471.95 197.87
## - race:length_sec 1 5.4211 477.37 198.48
## - race:bpm 1 7.4563 479.40 199.45
## - race:genre 2 13.6851 485.63 200.39
##
## Step: AIC=196.75
## rating ~ age + gender + race + genre + happiness + bpm + length_sec +
## age:happiness + race:genre + race:bpm + race:length_sec
##
## Df Sum of Sq RSS AIC
## - gender 1 3.2575 481.20 196.30
## - age:happiness 1 3.3913 481.34 196.37
## <none> 477.95 196.75
## - race:length_sec 1 5.9140 483.86 197.56
## - race:bpm 1 6.8757 484.82 198.01
## - race:genre 2 12.0192 489.97 198.42
##
## Step: AIC=196.3
## rating ~ age + race + genre + happiness + bpm + length_sec +
## age:happiness + race:genre + race:bpm + race:length_sec
##
## Df Sum of Sq RSS AIC
## - age:happiness 1 3.3045 484.51 195.86
## <none> 481.20 196.30

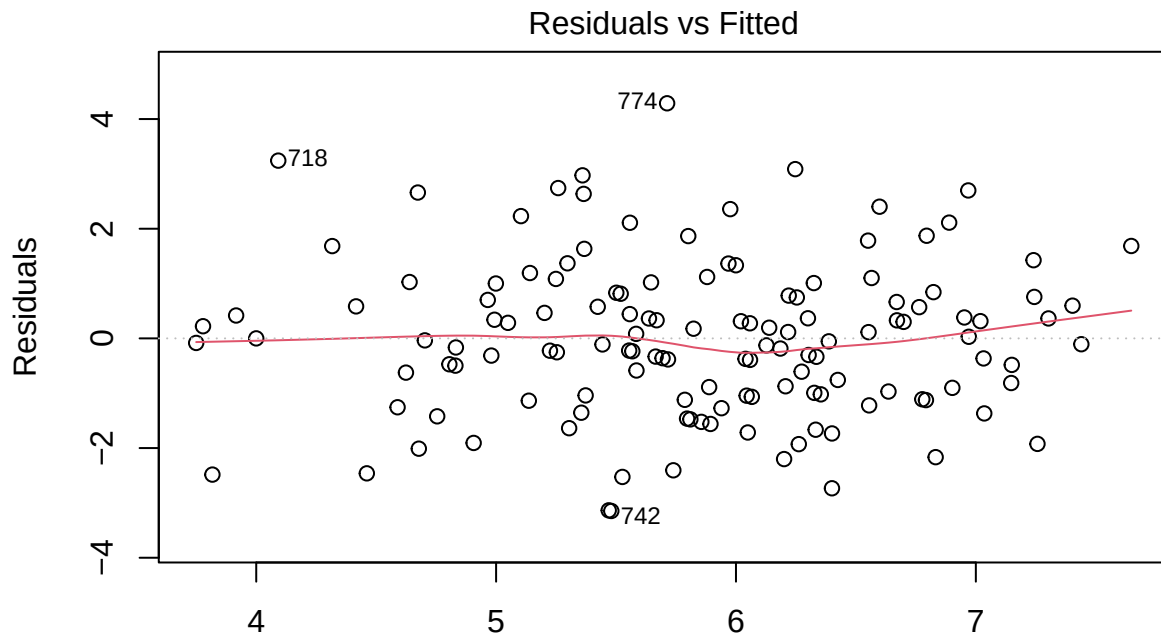
```

```

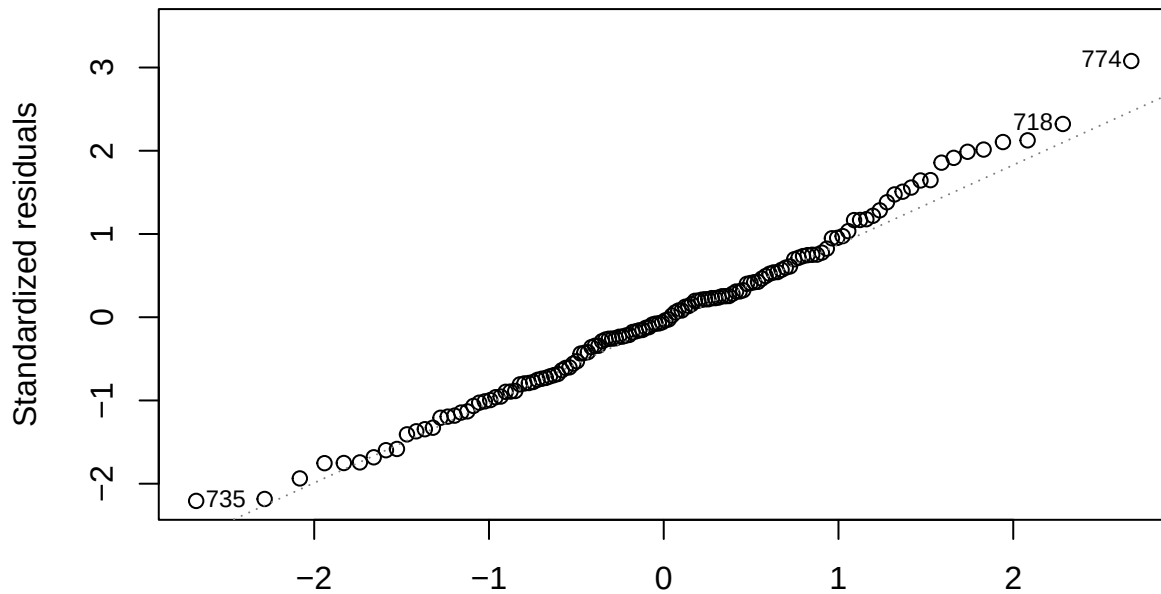
## - race:length_sec 1 5.1720 486.38 196.74
## - race:bpm 1 6.5740 487.78 197.40
## - race:genre 2 13.6543 494.86 198.68
##
## Step: AIC=195.86
## rating ~ age + race + genre + happiness + bpm + length_sec +
## race:genre + race:bpm + race:length_sec
##
## Df Sum of Sq RSS AIC
## - age 1 0.1370 484.65 193.93
## - happiness 1 0.9283 485.44 194.30
## <none> 484.51 195.86
## - race:length_sec 1 5.1063 489.61 196.25
## - race:bpm 1 7.0455 491.55 197.16
## - race:genre 2 13.8550 498.36 198.29
##
## Step: AIC=193.93
## rating ~ race + genre + happiness + bpm + length_sec + race:genre +
## race:bpm + race:length_sec
##
## Df Sum of Sq RSS AIC
## - happiness 1 0.9290 485.57 192.37
## <none> 484.65 193.93
## - race:length_sec 1 5.1852 489.83 194.35
## - race:bpm 1 7.3218 491.97 195.35
## - race:genre 2 13.7380 498.38 196.30
##
## Step: AIC=192.37
## rating ~ race + genre + bpm + length_sec + race:genre + race:bpm +
## race:length_sec
##
## Df Sum of Sq RSS AIC
## <none> 485.57 192.37
## - race:length_sec 1 5.0687 490.64 192.73
## - race:bpm 1 6.8395 492.41 193.55
## - race:genre 2 13.6538 499.23 194.69
## [1] "1980s"

## Warning: not plotting observations with leverage one:
## 14

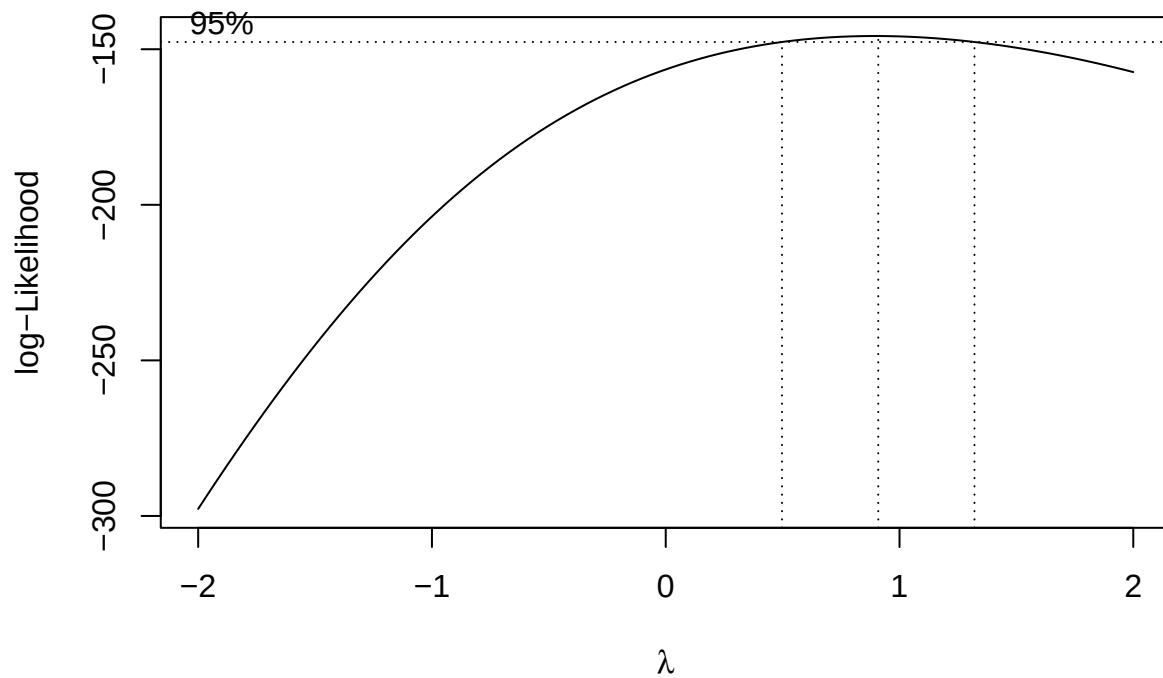
```



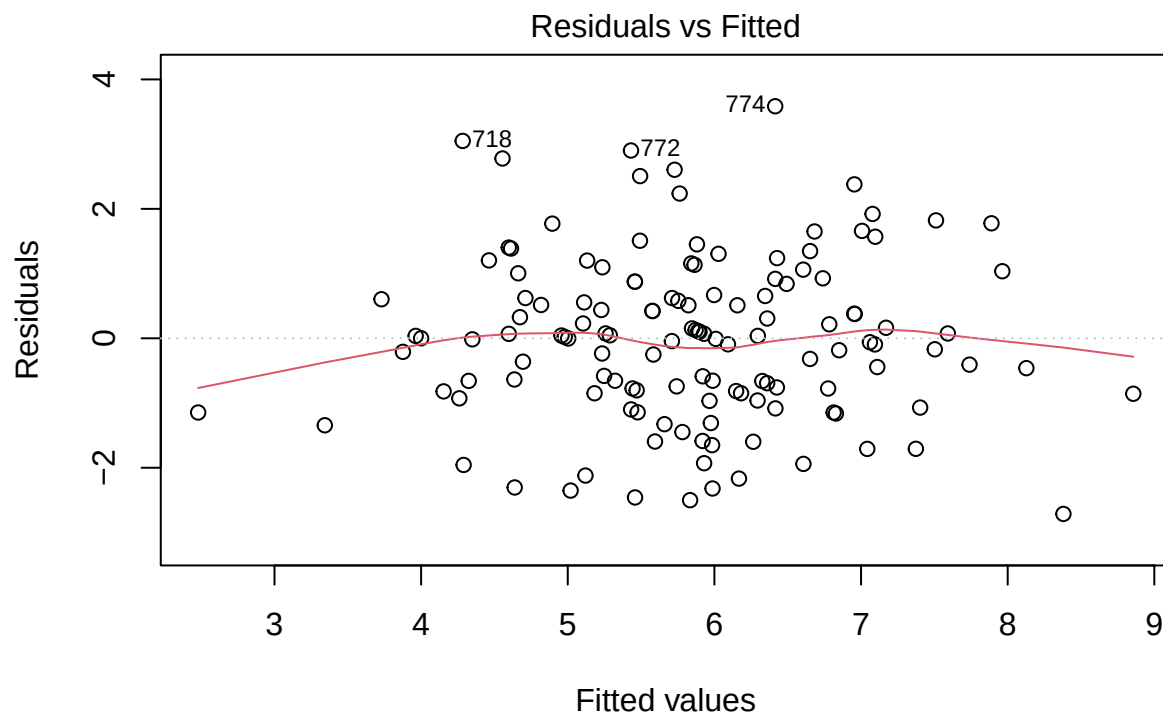
Fitted values
 $\text{lm}(\text{rating} \sim \text{age} + \text{race} + \text{gender} + \text{genre} + \text{energy} + \text{danceability} + \text{happiness} \dots)$
 Q-Q Residuals



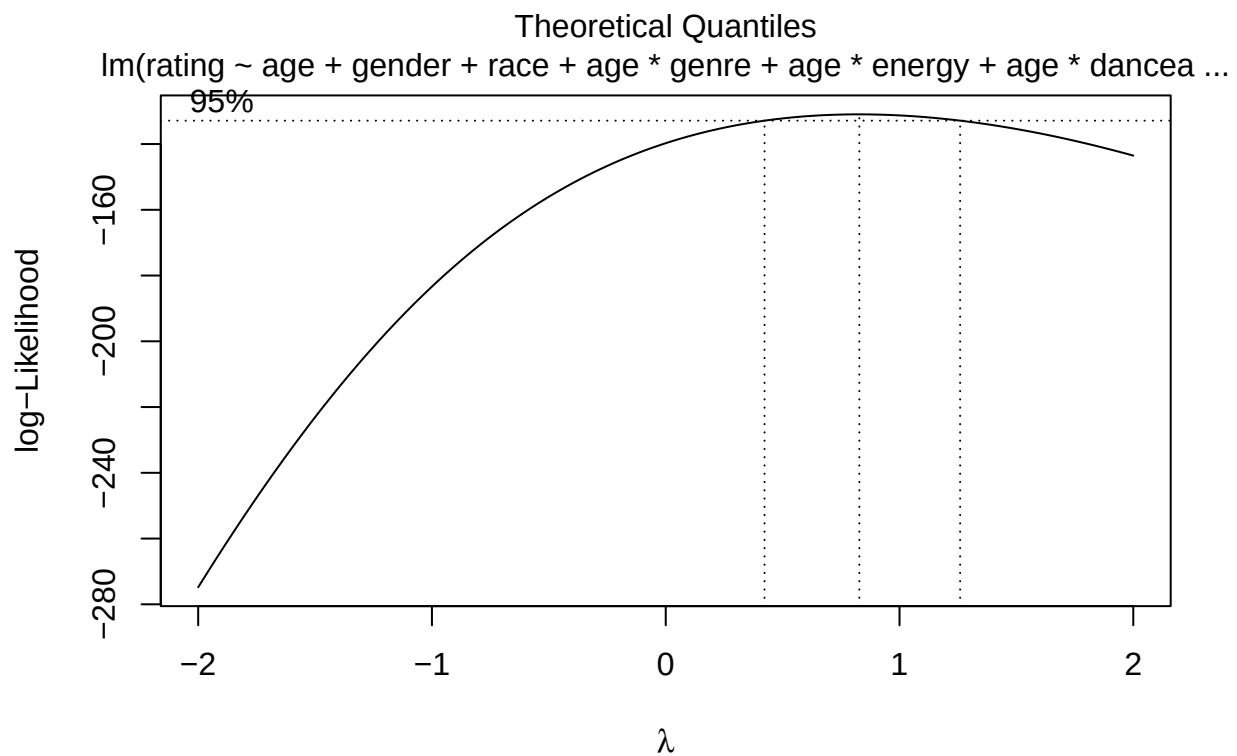
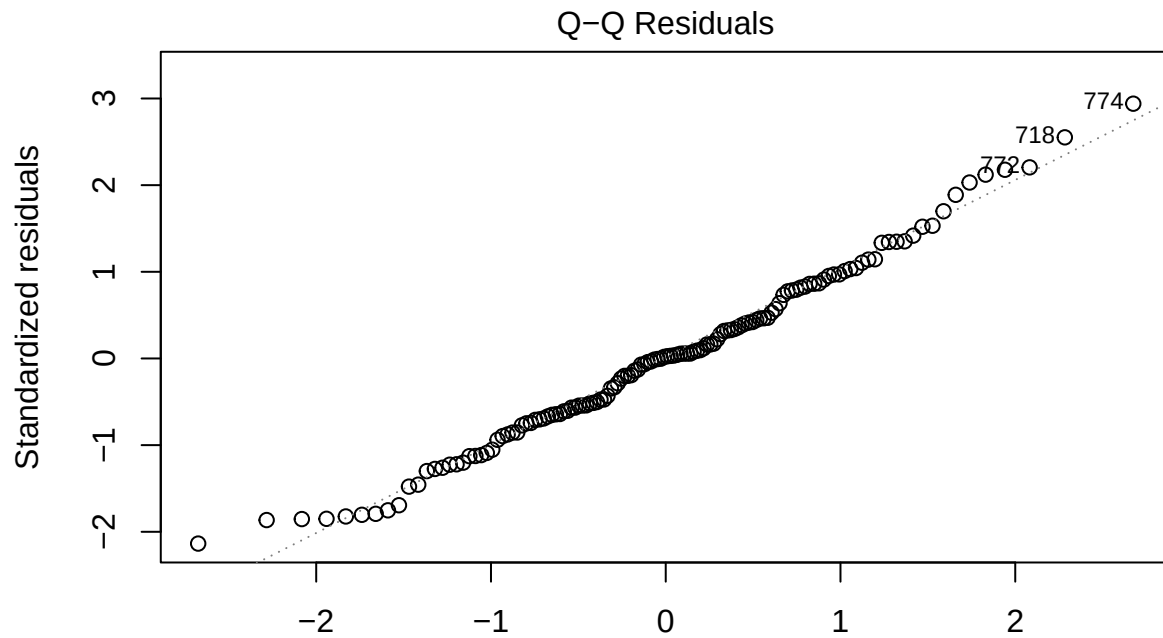
Theoretical Quantiles
 $\text{lm}(\text{rating} \sim \text{age} + \text{race} + \text{gender} + \text{genre} + \text{energy} + \text{danceability} + \text{happiness} \dots)$



```
## Warning: not plotting observations with leverage one:
## 14
```



lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...



```
## Start: AIC=138.39
## rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + +gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##
```



```

##              Df Sum of Sq    RSS    AIC
## - age:genre      2      1.4073 218.91 135.26
## - race:genre      2      1.9738 219.48 135.61
## - age:happiness   1      0.0000 217.50 136.39
## - race:loudness_db 1      0.0001 217.50 136.39
## - race:happiness   1      0.0116 217.52 136.39
## - race:energy      1      0.0527 217.56 136.42
## - gender:energy    1      0.2695 217.77 136.55
## - race:danceability 1      0.3419 217.85 136.60
## - age:bpm          1      0.4545 217.96 136.67
## - gender:bpm       1      0.6115 218.12 136.77
## - gender:genre     2      4.0892 221.59 136.90
## - gender:happiness  1      1.1061 218.61 137.07
## - gender:danceability 1      1.6610 219.16 137.41
## - age:length_sec   1      1.8941 219.40 137.56
## - gender:loudness_db 1      2.6071 220.11 138.00
## - gender:length_sec 1      2.7471 220.25 138.08
## - age:danceability  1      2.9717 220.48 138.22
## <none>                217.50 138.39
## - race:length_sec   1      4.0141 221.52 138.86
## - race:bpm          1      4.1272 221.63 138.93
## - age:energy        1      5.8923 223.40 140.00
## - age:loudness_db   1     13.3515 230.86 144.43
##
## Step:  AIC=135.26
## rating ~ age + gender + race + genre + energy + danceability +
##          happiness + bpm + loudness_db + length_sec + age:energy +
##          age:danceability + age:happiness + age:bpm + age:loudness_db +
##          age:length_sec + race:genre + race:energy + race:danceability +
##          race:happiness + race:bpm + race:loudness_db + race:length_sec +
##          gender:genre + gender:energy + gender:danceability + gender:happiness +
##          gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:genre      2      2.5187 221.43 132.80
## - race:loudness_db  1      0.0090 218.92 133.26
## - race:happiness   1      0.0153 218.93 133.27
## - age:happiness    1      0.0256 218.94 133.27
## - race:energy      1      0.0468 218.96 133.29
## - gender:energy    1      0.1961 219.11 133.38
## - age:bpm          1      0.2465 219.16 133.41
## - race:danceability 1      0.3715 219.28 133.49
## - gender:bpm       1      0.6364 219.55 133.65
## - gender:happiness  1      1.0560 219.97 133.91
## - gender:danceability 1      1.6690 220.58 134.28
## - gender:loudness_db 1      2.3501 221.26 134.70
## - age:length_sec   1      2.7297 221.64 134.93
## <none>                218.91 135.26
## - gender:genre     2      6.6194 225.53 135.28
## - gender:length_sec  1      3.4204 222.33 135.35
## - race:bpm         1      3.8490 222.76 135.61
## - race:length_sec   1      3.8789 222.79 135.63
## - age:energy        1      7.4455 226.36 137.77
## - age:danceability  1      7.4825 226.39 137.79

```

```

## - age:loudness_db      1    15.2561 234.17 142.35
##
## Step: AIC=132.8
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + age:happiness + age:bpm + age:loudness_db +
##      age:length_sec + race:energy + race:danceability + race:happiness +
##      race:bpm + race:loudness_db + race:length_sec + gender:genre +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:bpm + gender:loudness_db + gender:length_sec
##
##
##      Df Sum of Sq    RSS    AIC
## - race:loudness_db      1      0.0193 221.45 130.81
## - race:happiness        1      0.0473 221.48 130.83
## - race:energy           1      0.0674 221.50 130.84
## - age:bpm              1      0.2249 221.66 130.94
## - gender:energy         1      0.4478 221.88 131.07
## - age:happiness         1      0.4952 221.93 131.10
## - gender:bpm           1      0.6623 222.09 131.21
## - gender:happiness      1      0.8000 222.23 131.29
## - race:danceability     1      1.2493 222.68 131.56
## - gender:danceability   1      2.5539 223.98 132.35
## - age:length_sec        1      3.0754 224.50 132.66
## - race:bpm             1      3.2492 224.68 132.77
## <none>                  221.43 132.80
## - gender:genre          2      6.8660 228.30 132.93
## - gender:loudness_db    1      3.9121 225.34 133.17
## - gender:length_sec     1      4.2846 225.72 133.39
## - race:length_sec       1      5.4507 226.88 134.09
## - age:energy            1      6.2558 227.69 134.56
## - age:danceability      1      7.3094 228.74 135.19
## - age:loudness_db       1     14.3725 235.80 139.29
##
## Step: AIC=130.81
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + age:happiness + age:bpm + age:loudness_db +
##      age:length_sec + race:energy + race:danceability + race:happiness +
##      race:bpm + race:length_sec + gender:genre + gender:energy +
##      gender:danceability + gender:happiness + gender:bpm + gender:loudness_db +
##      gender:length_sec
##
##
##      Df Sum of Sq    RSS    AIC
## - race:energy           1      0.0484 221.50 128.84
## - race:happiness        1      0.0485 221.50 128.84
## - age:bpm              1      0.2510 221.70 128.97
## - gender:energy         1      0.4963 221.95 129.12
## - age:happiness         1      0.5329 221.98 129.14
## - gender:bpm           1      0.6513 222.10 129.21
## - gender:happiness      1      0.8008 222.25 129.30
## - race:danceability     1      1.5682 223.02 129.77
## - gender:danceability   1      2.6802 224.13 130.44
## - age:length_sec        1      3.1452 224.59 130.72
## <none>                  221.45 130.81

```

```

## - race:bpm          1    3.3796 224.83 130.86
## - gender:genre      2    6.9326 228.38 130.97
## - gender:loudness_db 1    3.9778 225.43 131.22
## - gender:length_sec 1    4.3779 225.83 131.46
## - race:length_sec   1    5.4732 226.92 132.11
## - age:energy        1    6.3724 227.82 132.64
## - age:danceability  1    7.3109 228.76 133.20
## - age:loudness_db   1   14.8871 236.34 137.60
##
## Step: AIC=128.84
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:happiness + age:bpm + age:loudness_db +
## age:length_sec + race:danceability + race:happiness + race:bpm +
## race:length_sec + gender:genre + gender:energy + gender:danceability +
## gender:happiness + gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:happiness      1    0.1872 221.69 126.96
## - age:bpm             1    0.2447 221.74 126.99
## - gender:energy       1    0.5987 222.10 127.21
## - gender:bpm          1    0.6284 222.13 127.23
## - age:happiness       1    0.7101 222.21 127.28
## - gender:happiness    1    0.8352 222.33 127.35
## - race:danceability   1    1.5605 223.06 127.79
## - gender:danceability 1    2.6462 224.14 128.45
## - age:length_sec      1    3.1174 224.62 128.73
## <none>                221.50 128.84
## - race:bpm           1    3.3338 224.83 128.86
## - gender:genre        2    6.8872 228.38 128.98
## - gender:loudness_db  1    4.1535 225.65 129.35
## - gender:length_sec   1    4.3350 225.83 129.46
## - race:length_sec     1    5.5009 227.00 130.16
## - age:energy          1    7.2624 228.76 131.20
## - age:danceability    1    7.4297 228.93 131.30
## - age:loudness_db     1   14.8550 236.35 135.61
##
## Step: AIC=126.96
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:happiness + age:bpm + age:loudness_db +
## age:length_sec + race:danceability + race:bpm + race:length_sec +
## gender:genre + gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - age:bpm           1    0.3004 221.99 125.14
## - gender:energy      1    0.6378 222.32 125.34
## - gender:bpm         1    0.6701 222.35 125.36
## - age:happiness      1    0.6978 222.38 125.38
## - gender:happiness   1    0.8244 222.51 125.46
## - gender:danceability 1    2.7098 224.40 126.60
## - age:length_sec     1    2.9973 224.68 126.77
## - race:bpm          1    3.1516 224.84 126.86

```

```

## <none>                221.69 126.96
## - race:danceability    1    3.3507 225.04 126.98
## - gender:genre         2    7.1361 228.82 127.23
## - gender:length_sec    1    4.2697 225.95 127.53
## - gender:loudness_db   1    4.4528 226.14 127.64
## - race:length_sec      1    5.4301 227.12 128.22
## - age:danceability     1    7.3944 229.08 129.39
## - age:energy           1    7.5176 229.20 129.46
## - age:loudness_db      1   15.1325 236.82 133.87
##
## Step:  AIC=125.14
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:energy +
##         age:danceability + age:happiness + age:loudness_db + age:length_sec +
##         race:danceability + race:bpm + race:length_sec + gender:genre +
##         gender:energy + gender:danceability + gender:happiness +
##         gender:bpm + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq   RSS   AIC
## - gender:bpm      1    0.4598 222.44 123.42
## - age:happiness    1    0.6830 222.67 123.56
## - gender:energy    1    0.6873 222.67 123.56
## - gender:happiness 1    0.9378 222.92 123.71
## - gender:danceability 1    2.8667 224.85 124.87
## - age:length_sec   1    2.8738 224.86 124.88
## - race:danceability 1    3.2933 225.28 125.13
## <none>                221.99 125.14
## - race:bpm        1    3.8355 225.82 125.45
## - gender:genre     2    7.9091 229.89 125.87
## - gender:loudness_db 1    4.6268 226.61 125.92
## - gender:length_sec 1    4.6550 226.64 125.94
## - race:length_sec   1    5.9976 227.98 126.74
## - age:energy        1    7.7882 229.77 127.80
## - age:danceability  1    8.4810 230.47 128.20
## - age:loudness_db   1   15.6726 237.66 132.35
##
## Step:  AIC=123.42
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:energy +
##         age:danceability + age:happiness + age:loudness_db + age:length_sec +
##         race:danceability + race:bpm + race:length_sec + gender:genre +
##         gender:energy + gender:danceability + gender:happiness +
##         gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq   RSS   AIC
## - age:happiness    1    0.5789 223.02 121.77
## - gender:energy     1    0.8700 223.31 121.95
## - gender:happiness  1    1.0454 223.49 122.05
## - age:length_sec    1    2.8525 225.30 123.14
## - gender:danceability 1    3.1054 225.55 123.29
## - race:danceability 1    3.1871 225.63 123.34
## <none>                222.44 123.42
## - race:bpm         1    3.6837 226.13 123.64
## - gender:length_sec 1    4.8237 227.27 124.32

```

```

## - gender:loudness_db 1 5.1469 227.59 124.51
## - gender:genre 2 8.5771 231.02 124.53
## - race:length_sec 1 6.6767 229.12 125.41
## - age:energy 1 7.9051 230.35 126.13
## - age:danceability 1 8.4988 230.94 126.48
## - age:loudness_db 1 15.6733 238.12 130.61
##
## Step: AIC=121.77
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:loudness_db + age:length_sec + race:danceability +
## race:bpm + race:length_sec + gender:genre + gender:energy +
## gender:danceability + gender:happiness + gender:loudness_db +
## gender:length_sec
##
## Df Sum of Sq RSS AIC
## - gender:energy 1 0.6848 223.71 120.18
## - gender:happiness 1 1.0515 224.07 120.41
## - age:length_sec 1 2.4335 225.46 121.24
## - gender:danceability 1 2.8685 225.89 121.50
## - race:danceability 1 3.1073 226.13 121.64
## <none> 223.02 121.77
## - race:bpm 1 3.4976 226.52 121.87
## - gender:length_sec 1 4.4054 227.43 122.41
## - gender:loudness_db 1 4.8195 227.84 122.66
## - gender:genre 2 8.3224 231.35 122.72
## - race:length_sec 1 6.5107 229.53 123.66
## - age:danceability 1 8.1387 231.16 124.61
## - age:energy 1 12.4540 235.48 127.11
## - age:loudness_db 1 16.2506 239.28 129.26
##
## Step: AIC=120.18
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:loudness_db + age:length_sec + race:danceability +
## race:bpm + race:length_sec + gender:genre + gender:danceability +
## gender:happiness + gender:loudness_db + gender:length_sec
##
## Df Sum of Sq RSS AIC
## - gender:happiness 1 0.4453 224.15 118.45
## - age:length_sec 1 2.1938 225.90 119.50
## - gender:danceability 1 2.9711 226.68 119.97
## - race:danceability 1 3.2804 226.99 120.15
## <none> 223.71 120.18
## - race:bpm 1 3.6180 227.33 120.35
## - gender:loudness_db 1 4.1797 227.89 120.68
## - gender:length_sec 1 4.3795 228.09 120.80
## - gender:genre 2 8.2571 231.97 121.08
## - race:length_sec 1 6.6801 230.39 122.16
## - age:danceability 1 8.1040 231.81 122.99
## - age:energy 1 12.3900 236.10 125.46
## - age:loudness_db 1 15.9089 239.62 127.46
##
## Step: AIC=118.45

```

```

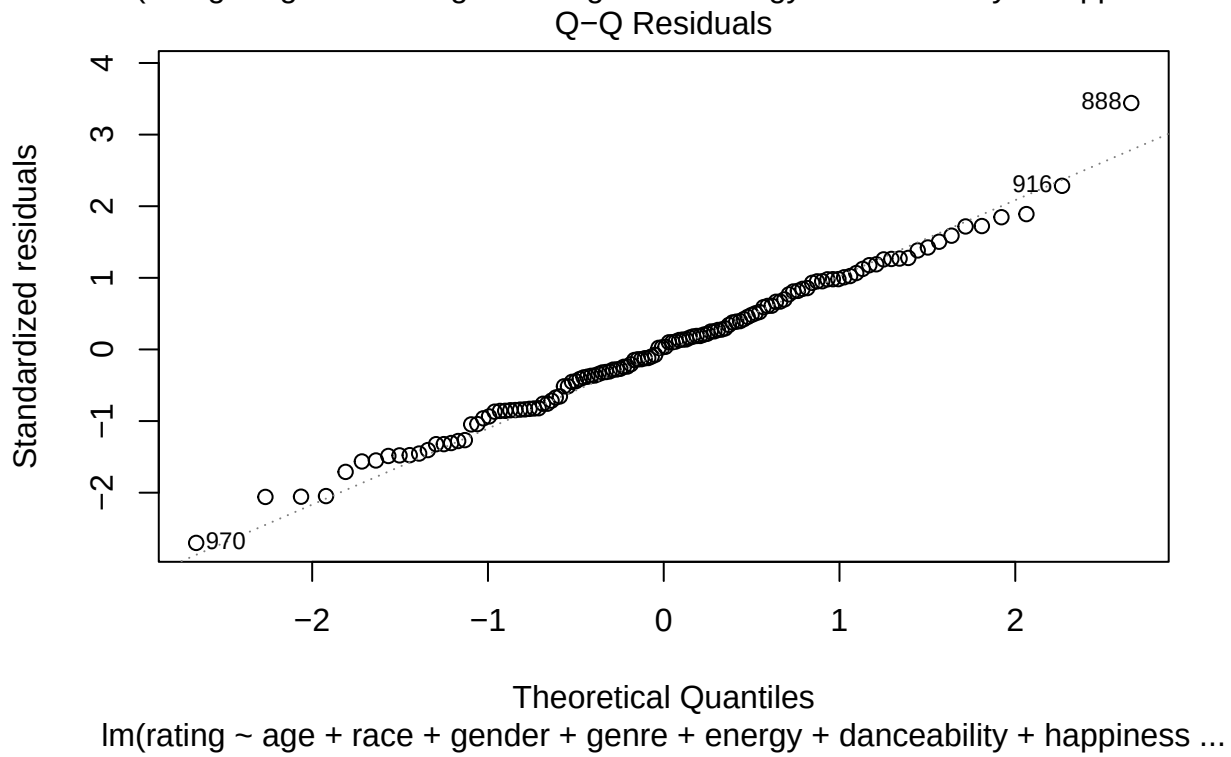
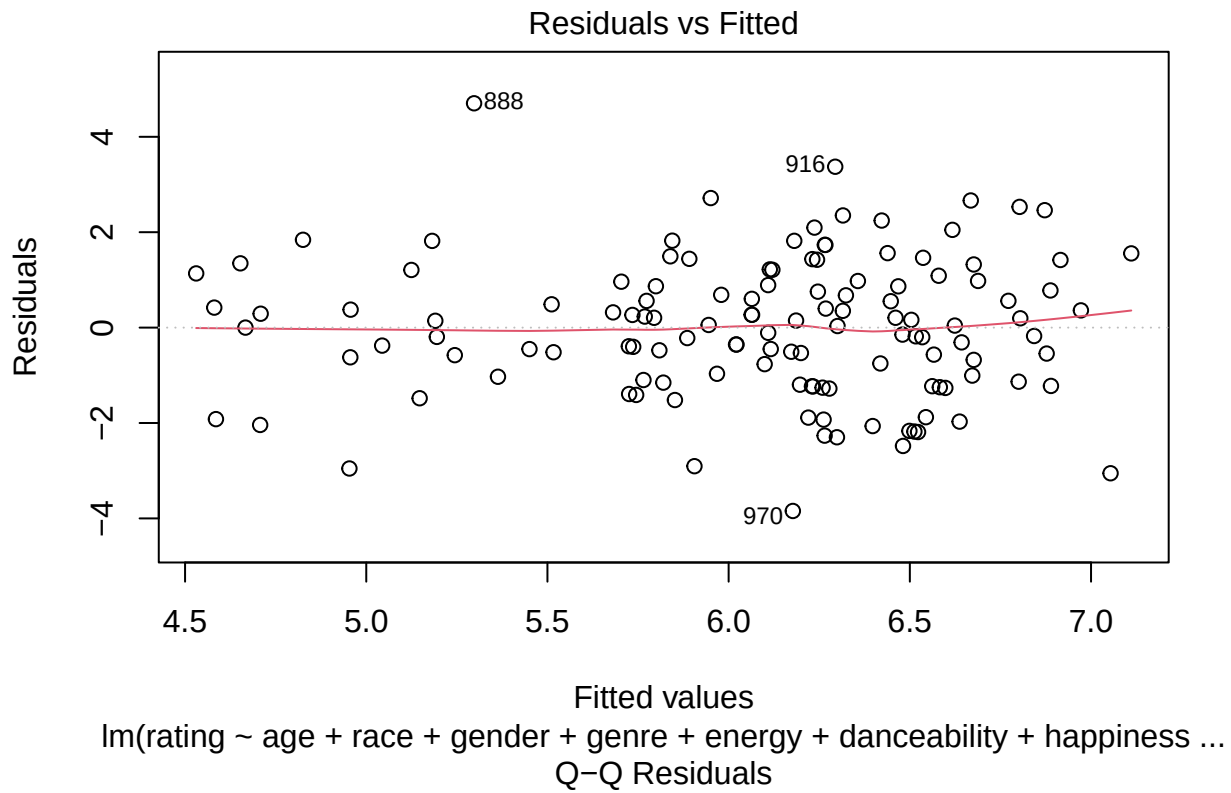
## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:energy +
##   age:danceability + age:loudness_db + age:length_sec + race:danceability +
##   race:bpm + race:length_sec + gender:genre + gender:danceability +
##   gender:loudness_db + gender:length_sec
##
##           Df Sum of Sq   RSS   AIC
## - age:length_sec      1    1.8649 226.02 117.57
## - race:danceability    1    3.2937 227.45 118.42
## <none>                  224.15 118.45
## - race:bpm            1    3.5714 227.72 118.59
## - gender:length_sec    1    3.9799 228.13 118.83
## - gender:genre         2    7.9761 232.13 119.17
## - gender:danceability  1    4.8451 229.00 119.34
## - gender:loudness_db   1    5.0982 229.25 119.49
## - race:length_sec      1    6.3611 230.51 120.23
## - age:danceability     1    8.3086 232.46 121.37
## - happiness            1    9.3752 233.53 121.98
## - age:energy           1   13.2956 237.45 124.23
## - age:loudness_db      1   16.6391 240.79 126.12
##
## Step:  AIC=117.57
## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:energy +
##   age:danceability + age:loudness_db + race:danceability +
##   race:bpm + race:length_sec + gender:genre + gender:danceability +
##   gender:loudness_db + gender:length_sec
##
##           Df Sum of Sq   RSS   AIC
## - race:bpm            1    2.9798 229.00 117.34
## <none>                  226.02 117.57
## - race:danceability    1    3.8262 229.84 117.84
## - gender:genre         2    7.8278 233.85 118.17
## - gender:danceability  1    4.4934 230.51 118.23
## - gender:loudness_db   1    5.0710 231.09 118.57
## - race:length_sec      1    5.1283 231.15 118.60
## - gender:length_sec    1    5.5585 231.58 118.85
## - happiness            1    8.8569 234.88 120.76
## - age:danceability     1    9.5638 235.58 121.17
## - age:energy           1   11.6161 237.63 122.34
## - age:loudness_db      1   15.2086 241.23 124.36
##
## Step:  AIC=117.34
## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:energy +
##   age:danceability + age:loudness_db + race:danceability +
##   race:length_sec + gender:genre + gender:danceability + gender:loudness_db +
##   gender:length_sec
##
##           Df Sum of Sq   RSS   AIC
## - bpm                1    1.7671 230.77 116.38
## <none>                  229.00 117.34
## - gender:danceability  1    4.2421 233.24 117.82
## - gender:genre         2    7.8097 236.81 117.87

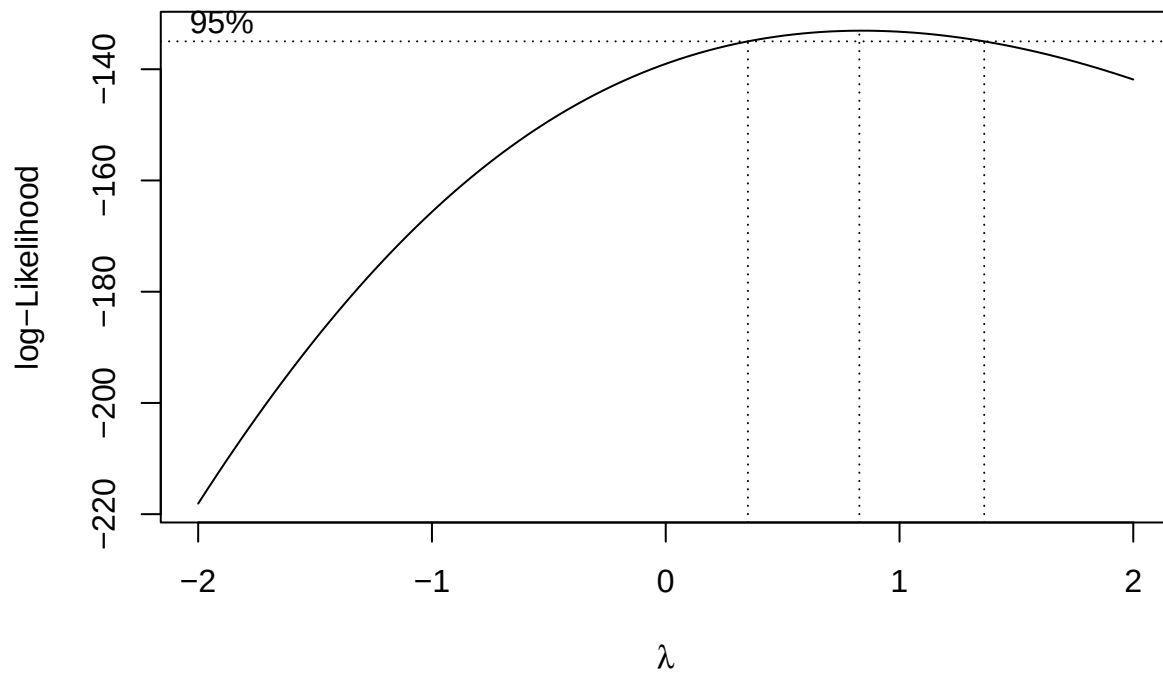
```

```

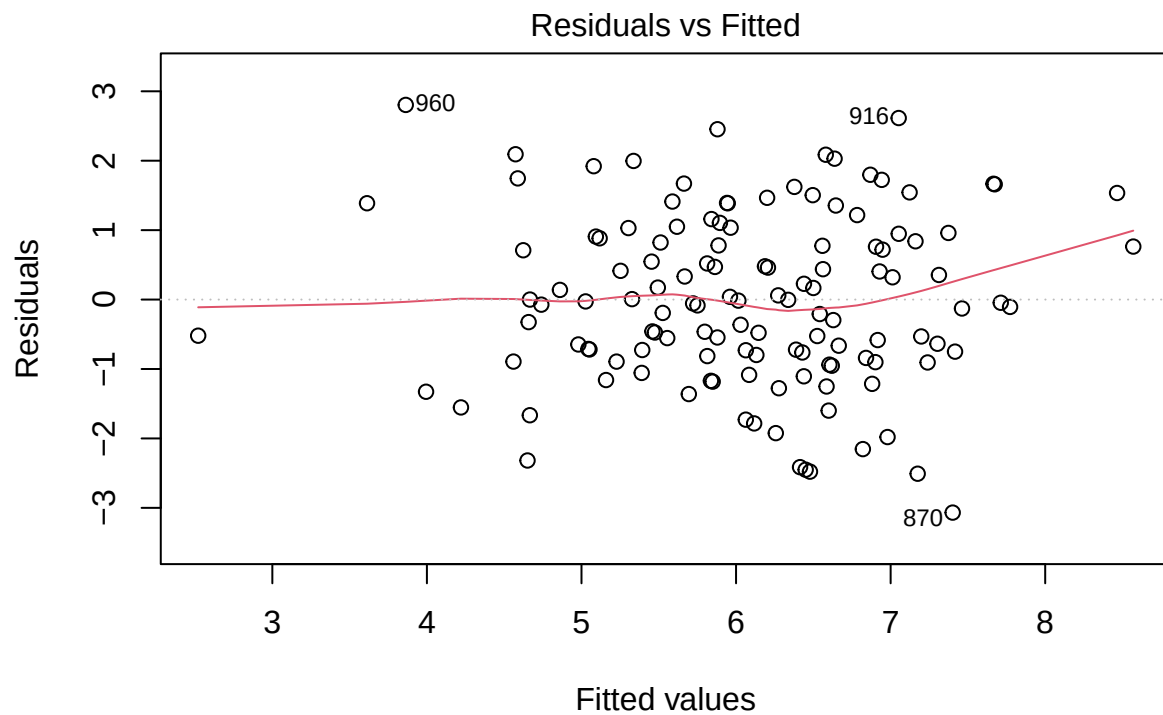
## - race:danceability      1      5.0409 234.04 118.28
## - gender:loudness_db     1      5.3476 234.35 118.46
## - race:length_sec        1      5.7553 234.75 118.69
## - gender:length_sec      1      7.4133 236.41 119.64
## - age:danceability       1      8.5272 237.53 120.28
## - happiness              1     10.1922 239.19 121.22
## - age:energy             1     11.0432 240.04 121.70
## - age:loudness_db        1     14.6957 243.69 123.74
##
## Step:  AIC=116.38
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + loudness_db + length_sec + age:energy + age:danceability +
##         age:loudness_db + race:danceability + race:length_sec + gender:genre +
##         gender:danceability + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## <none>                230.77 116.38
## - gender:genre        2      7.5708 238.34 116.73
## - race:danceability    1      4.3094 235.07 116.88
## - gender:danceability  1      5.4605 236.23 117.53
## - race:length_sec      1      5.6093 236.38 117.62
## - gender:loudness_db   1      5.6884 236.45 117.67
## - gender:length_sec    1      7.4700 238.24 118.68
## - age:danceability     1      8.7287 239.50 119.39
## - age:energy           1     10.4462 241.21 120.35
## - happiness            1     10.7471 241.51 120.52
## - age:loudness_db      1     14.6406 245.41 122.68
## [1] "1990s"
##
## Warning: not plotting observations with leverage one:
##      4

```

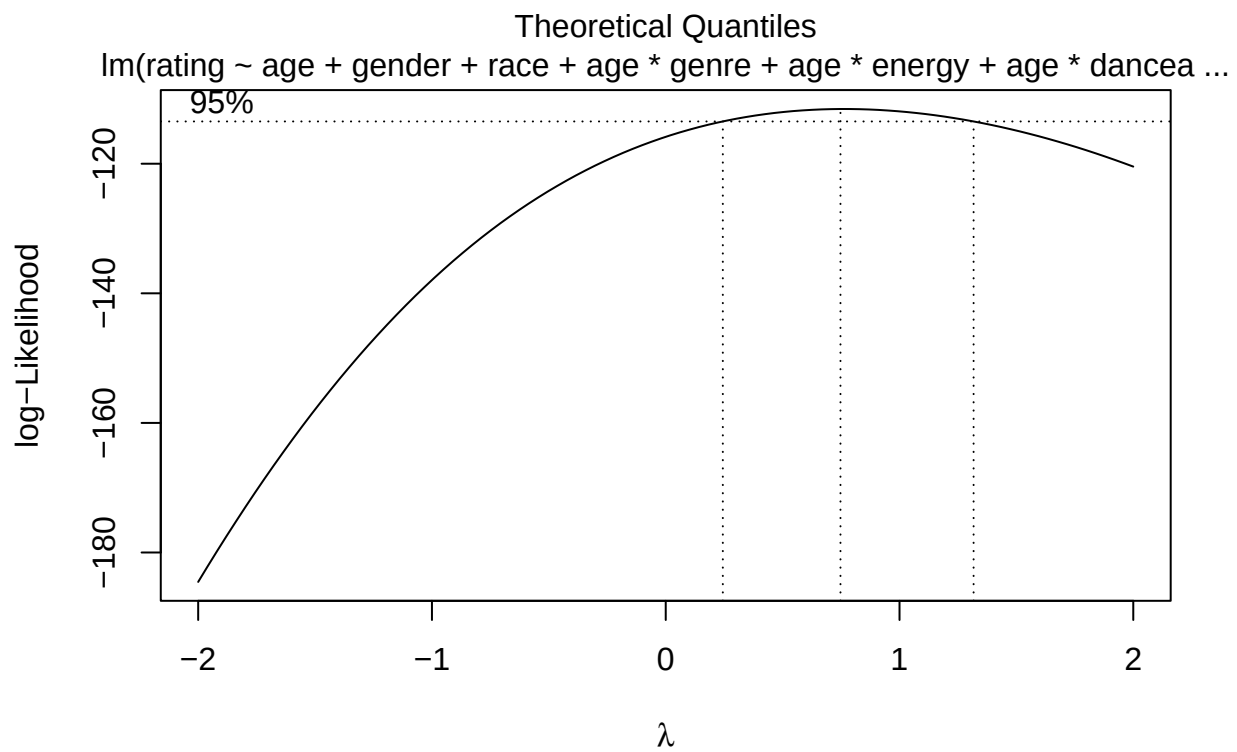
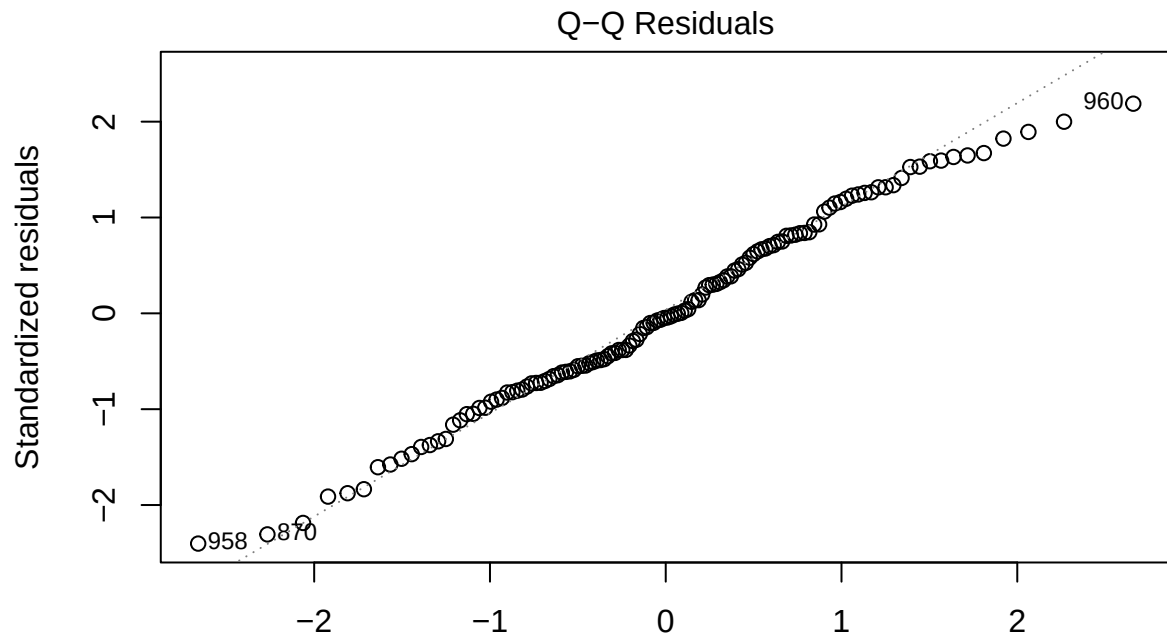




```
## Warning: not plotting observations with leverage one:
## 4
```



lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...



```
## Start: AIC=124.77
## rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + +gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##
```

```

##          Df Sum of Sq    RSS    AIC
## - gender:genre      2    0.2909 194.49 120.96
## - gender:length_sec  1    0.0570 194.26 122.81
## - age:energy         1    0.1216 194.32 122.85
## - age:bpm           1    0.1778 194.38 122.89
## - age:happiness     1    0.4584 194.66 123.07
## - gender:loudness_db 1    0.4794 194.68 123.09
## - age:length_sec    1    0.9614 195.16 123.41
## - race:energy       1    1.2819 195.48 123.62
## - age:genre         2    4.3972 198.60 123.66
## - age:loudness_db   1    1.3852 195.58 123.69
## - gender:energy     1    1.4248 195.62 123.71
## - race:loudness_db  1    1.7680 195.97 123.94
## - age:danceability  1    2.1475 196.35 124.19
## - race:genre        1    2.4169 196.62 124.36
## <none>              194.20 124.77
## - race:length_sec   1    4.0842 198.28 125.45
## - gender:happiness  1    4.4298 198.63 125.68
## - race:happiness    1    5.1869 199.38 126.17
## - gender:bpm        1    5.6035 199.80 126.44
## - gender:danceability 1    8.3414 202.54 128.19
## - race:danceability 1   12.6344 206.83 130.90
## - race:bpm         1   13.9941 208.19 131.75
##
## Step:  AIC=120.96
## rating ~ age + gender + race + genre + energy + danceability +
##          happiness + bpm + loudness_db + length_sec + age:genre +
##          age:energy + age:danceability + age:happiness + age:bpm +
##          age:loudness_db + age:length_sec + race:genre + race:energy +
##          race:danceability + race:happiness + race:bpm + race:loudness_db +
##          race:length_sec + gender:energy + gender:danceability + gender:happiness +
##          gender:bpm + gender:loudness_db + gender:length_sec
##
##          Df Sum of Sq    RSS    AIC
## - gender:length_sec  1    0.1410 194.63 119.06
## - age:energy         1    0.1432 194.63 119.06
## - age:bpm           1    0.2038 194.69 119.10
## - age:happiness     1    0.5093 195.00 119.30
## - gender:loudness_db 1    0.5341 195.02 119.32
## - age:length_sec    1    0.8897 195.38 119.55
## - race:energy       1    1.0952 195.59 119.69
## - age:loudness_db   1    1.3811 195.87 119.88
## - age:genre         2    4.5885 199.08 119.97
## - gender:energy     1    1.6752 196.16 120.07
## - race:loudness_db  1    1.6796 196.17 120.07
## - age:danceability  1    2.1850 196.68 120.40
## - race:genre        1    2.4594 196.95 120.58
## <none>              194.49 120.96
## - race:happiness    1    4.9200 199.41 122.19
## - race:length_sec   1    4.9339 199.42 122.19
## - gender:happiness  1    4.9712 199.46 122.22
## - gender:bpm        1    5.4597 199.95 122.53
## - gender:danceability 1    8.2237 202.71 124.31
## - race:danceability 1   12.9157 207.41 127.26

```

```

## - race:bpm          1    14.2374 208.73 128.08
##
## Step:  AIC=119.06
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:energy +
##      race:danceability + race:happiness + race:bpm + race:loudness_db +
##      race:length_sec + gender:energy + gender:danceability + gender:happiness +
##      gender:bpm + gender:loudness_db
##
##
##      Df Sum of Sq    RSS    AIC
## - age:energy          1      0.1503 194.78 117.16
## - age:bpm             1      0.1755 194.81 117.17
## - gender:loudness_db  1      0.5106 195.14 117.39
## - age:happiness       1      0.5595 195.19 117.43
## - age:length_sec      1      1.0576 195.69 117.76
## - race:energy         1      1.1071 195.74 117.79
## - age:loudness_db     1      1.5460 196.18 118.08
## - gender:energy       1      1.6133 196.24 118.12
## - age:genre           2      4.7363 199.37 118.16
## - race:loudness_db    1      2.0322 196.66 118.40
## - age:danceability    1      2.1949 196.83 118.50
## - race:genre          1      2.3311 196.96 118.59
## <none>                                194.63 119.06
## - race:length_sec     1      4.7938 199.42 120.19
## - race:happiness       1      4.7999 199.43 120.20
## - gender:happiness    1      4.8522 199.48 120.23
## - gender:bpm          1      5.7543 200.38 120.81
## - gender:danceability 1      8.1192 202.75 122.33
## - race:danceability   1     12.7794 207.41 125.26
## - race:bpm            1     14.6052 209.24 126.39
##
##
## Step:  AIC=117.16
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:danceability + age:happiness + age:bpm + age:loudness_db +
##      age:length_sec + race:genre + race:energy + race:danceability +
##      race:happiness + race:bpm + race:loudness_db + race:length_sec +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:bpm + gender:loudness_db
##
##
##      Df Sum of Sq    RSS    AIC
## - age:bpm             1      0.2614 195.04 115.33
## - age:happiness       1      0.4189 195.20 115.43
## - gender:loudness_db  1      0.4382 195.22 115.45
## - race:energy         1      1.0180 195.80 115.83
## - age:length_sec      1      1.0802 195.86 115.87
## - gender:energy       1      1.5427 196.32 116.17
## - age:genre           2      4.6494 199.43 116.20
## - race:loudness_db    1      2.0250 196.81 116.49
## - age:danceability    1      2.2189 197.00 116.62
## - race:genre          1      2.2472 197.03 116.64
## - age:loudness_db     1      2.7256 197.51 116.95

```

```

## <none>                                194.78 117.16
## - race:length_sec      1      4.6696 199.45 118.21
## - race:happiness       1      4.7170 199.50 118.24
## - gender:happiness     1      5.1363 199.92 118.51
## - gender:bpm           1      5.6174 200.40 118.82
## - gender:danceability  1      8.6958 203.48 120.79
## - race:danceability    1     12.7055 207.49 123.31
## - race:bpm             1     14.5050 209.29 124.42
##
## Step:  AIC=115.33
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:genre +
##         age:danceability + age:happiness + age:loudness_db + age:length_sec +
##         race:genre + race:energy + race:danceability + race:happiness +
##         race:bpm + race:loudness_db + race:length_sec + gender:energy +
##         gender:danceability + gender:happiness + gender:bpm + gender:loudness_db
##
##              Df Sum of Sq  RSS    AIC
## - gender:loudness_db  1      0.4518 195.49 113.63
## - age:happiness       1      0.5097 195.55 113.67
## - race:energy         1      1.0076 196.05 113.99
## - age:length_sec      1      1.1602 196.20 114.09
## - age:genre           2      4.5286 199.57 114.29
## - gender:energy       1      1.6739 196.72 114.43
## - age:danceability    1      1.9702 197.01 114.63
## - race:loudness_db    1      2.0320 197.07 114.67
## - race:genre          1      2.2829 197.32 114.83
## <none>                                195.04 115.33
## - age:loudness_db     1      3.3223 198.37 115.51
## - race:length_sec     1      4.7071 199.75 116.41
## - race:happiness      1      4.7576 199.80 116.44
## - gender:bpm          1      5.4326 200.47 116.87
## - gender:happiness    1      5.7397 200.78 117.07
## - gender:danceability  1      9.1047 204.15 119.22
## - race:danceability    1     12.6221 207.66 121.42
## - race:bpm            1     14.2503 209.29 122.43
##
## Step:  AIC=113.63
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:genre +
##         age:danceability + age:happiness + age:loudness_db + age:length_sec +
##         race:genre + race:energy + race:danceability + race:happiness +
##         race:bpm + race:loudness_db + race:length_sec + gender:energy +
##         gender:danceability + gender:happiness + gender:bpm
##
##              Df Sum of Sq  RSS    AIC
## - age:happiness       1      0.6697 196.16 112.07
## - race:energy         1      0.8472 196.34 112.19
## - age:length_sec      1      1.0331 196.53 112.31
## - gender:energy       1      1.2479 196.74 112.45
## - age:genre           2      4.5650 200.06 112.61
## - race:loudness_db    1      1.9267 197.42 112.89
## - race:genre          1      1.9962 197.49 112.94
## - age:danceability    1      2.2628 197.76 113.11

```

```

## <none>                                195.49 113.63
## - age:loudness_db      1      3.6882 199.18 114.04
## - race:length_sec      1      4.3490 199.84 114.47
## - race:happiness       1      4.8955 200.39 114.82
## - gender:bpm           1      5.7848 201.28 115.39
## - gender:happiness     1      5.8293 201.32 115.42
## - gender:danceability  1      8.6722 204.17 117.23
## - race:danceability    1     12.5298 208.02 119.64
## - race:bpm             1     14.1900 209.68 120.67
##
## Step:  AIC=112.07
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:genre +
##         age:danceability + age:loudness_db + age:length_sec + race:genre +
##         race:energy + race:danceability + race:happiness + race:bpm +
##         race:loudness_db + race:length_sec + gender:energy + gender:danceability +
##         gender:happiness + gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - age:length_sec      1      0.6126 196.78 110.47
## - race:energy          1      0.9178 197.08 110.67
## - age:genre            2      4.3793 200.54 110.92
## - gender:energy        1      1.3660 197.53 110.96
## - age:danceability     1      1.9279 198.09 111.33
## - race:loudness_db     1      2.0138 198.18 111.39
## - race:genre           1      2.2968 198.46 111.57
## <none>                                196.16 112.07
## - age:loudness_db      1      3.2745 199.44 112.20
## - race:length_sec      1      4.5901 200.75 113.05
## - race:happiness       1      4.8985 201.06 113.25
## - gender:bpm           1      5.6194 201.78 113.71
## - gender:happiness     1      5.8449 202.01 113.86
## - gender:danceability  1      8.2718 204.44 115.40
## - race:danceability    1     13.9120 210.08 118.91
## - race:bpm             1     14.6554 210.82 119.36
##
## Step:  AIC=110.47
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:genre +
##         age:danceability + age:loudness_db + race:genre + race:energy +
##         race:danceability + race:happiness + race:bpm + race:loudness_db +
##         race:length_sec + gender:energy + gender:danceability + gender:happiness +
##         gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - age:genre            2      3.8284 200.60 108.96
## - race:energy          1      1.0157 197.79 109.14
## - gender:energy        1      1.4213 198.20 109.40
## - race:genre           1      1.9124 198.69 109.72
## - race:loudness_db     1      1.9883 198.76 109.77
## - age:danceability     1      2.4006 199.18 110.03
## <none>                                196.78 110.47
## - age:loudness_db      1      3.1558 199.93 110.52
## - race:length_sec      1      4.6725 201.45 111.50

```

```

## - race:happiness      1      4.8863 201.66 111.64
## - gender:happiness    1      5.7000 202.48 112.16
## - gender:bpm          1      6.4126 203.19 112.61
## - gender:danceability 1      7.9788 204.75 113.60
## - race:danceability   1     13.8881 210.66 117.27
## - race:bpm            1     14.4573 211.23 117.62
##
## Step: AIC=108.96
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:danceability +
## age:loudness_db + race:genre + race:energy + race:danceability +
## race:happiness + race:bpm + race:loudness_db + race:length_sec +
## gender:energy + gender:danceability + gender:happiness +
## gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - race:energy      1      1.1880 201.79 107.72
## - gender:energy     1      1.6288 202.23 108.00
## - race:loudness_db  1      2.8090 203.41 108.75
## - race:genre        1      3.0688 203.67 108.92
## <none>              200.60 108.96
## - age:loudness_db   1      3.7711 204.38 109.36
## - race:length_sec   1      3.8413 204.45 109.40
## - race:happiness     1      4.6947 205.30 109.94
## - gender:happiness   1      5.1011 205.71 110.20
## - age:danceability   1      5.1258 205.73 110.21
## - gender:bpm         1      7.4709 208.08 111.67
## - gender:danceability 1      8.4997 209.10 112.31
## - race:danceability  1     15.2674 215.87 116.42
## - race:bpm          1     16.8572 217.46 117.36
##
## Step: AIC=107.72
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:danceability +
## age:loudness_db + race:genre + race:danceability + race:happiness +
## race:bpm + race:loudness_db + race:length_sec + gender:energy +
## gender:danceability + gender:happiness + gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - race:loudness_db  1      1.6555 203.45 106.77
## - gender:energy     1      2.0790 203.87 107.04
## - race:genre        1      2.9970 204.79 107.62
## <none>              201.79 107.72
## - age:loudness_db   1      3.3881 205.18 107.87
## - race:happiness     1      3.6076 205.40 108.00
## - race:length_sec   1      3.7446 205.54 108.09
## - age:danceability   1      5.2854 207.08 109.05
## - gender:happiness   1      5.9235 207.72 109.45
## - gender:bpm         1      7.7831 209.58 110.60
## - gender:danceability 1      8.3421 210.13 110.94
## - race:danceability  1     14.8282 216.62 114.86
## - race:bpm          1     18.2821 220.07 116.91
##
## Step: AIC=106.77

```

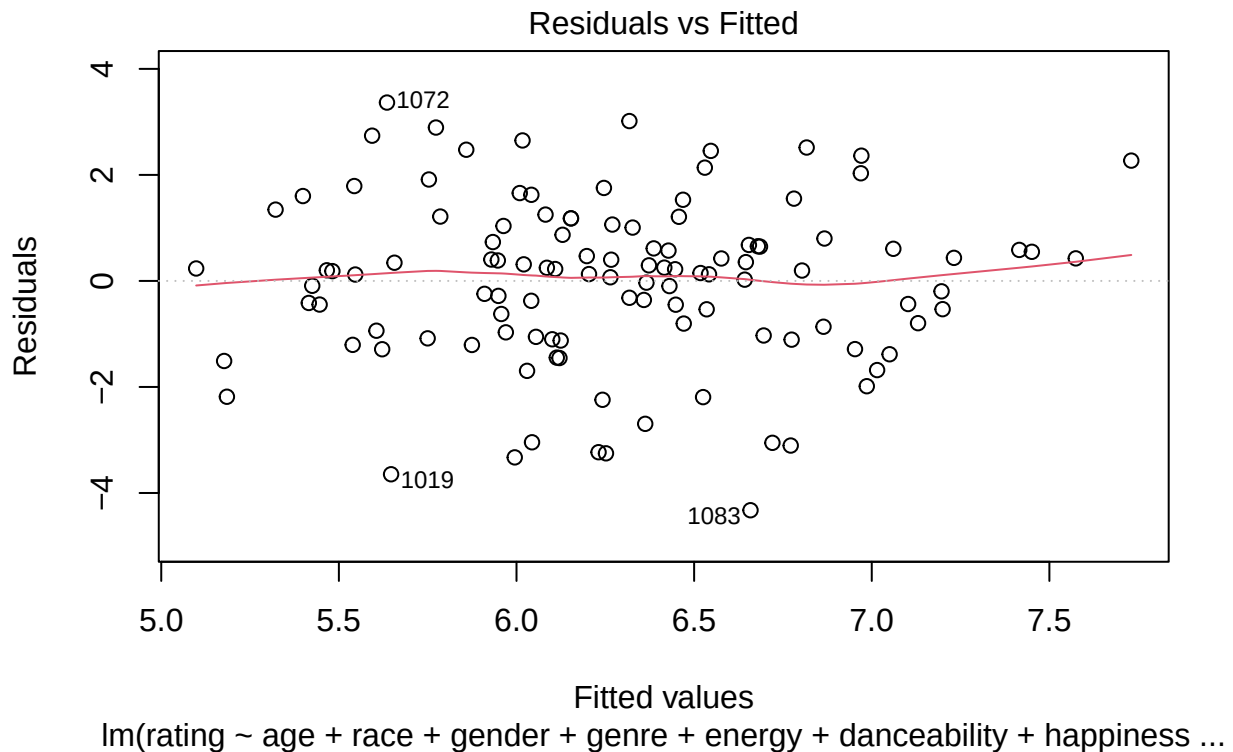
```

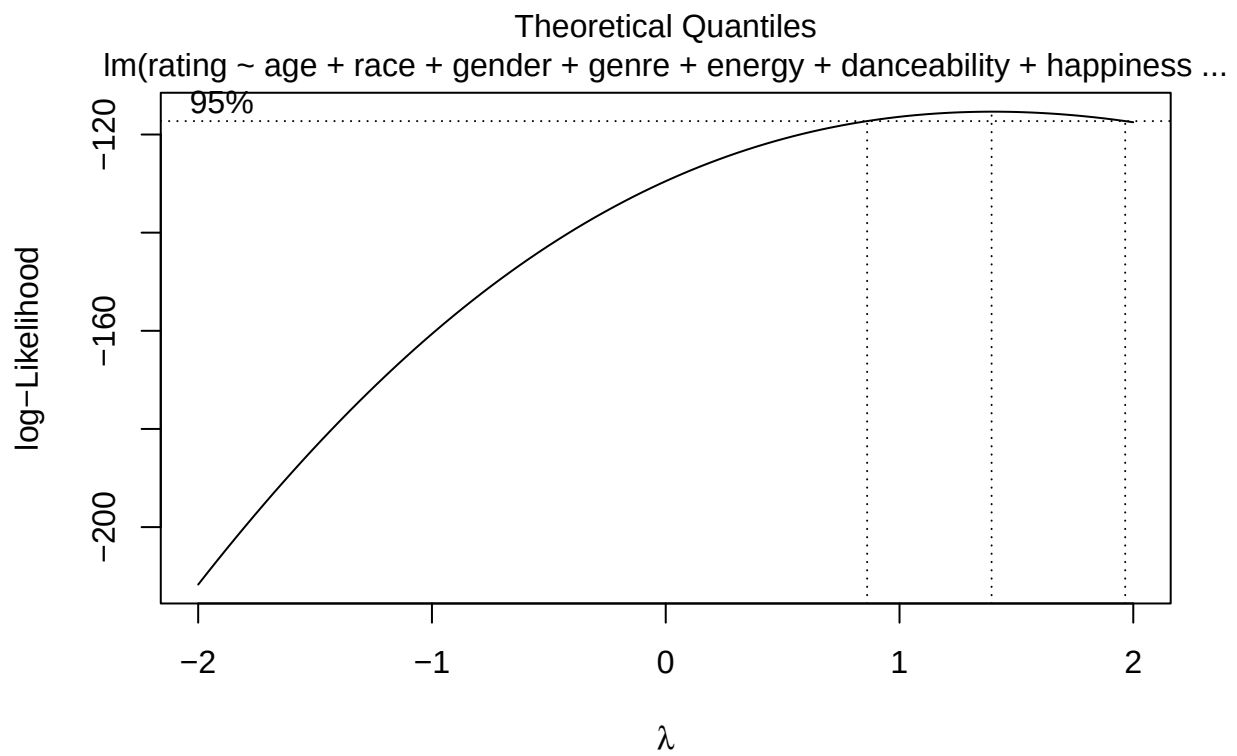
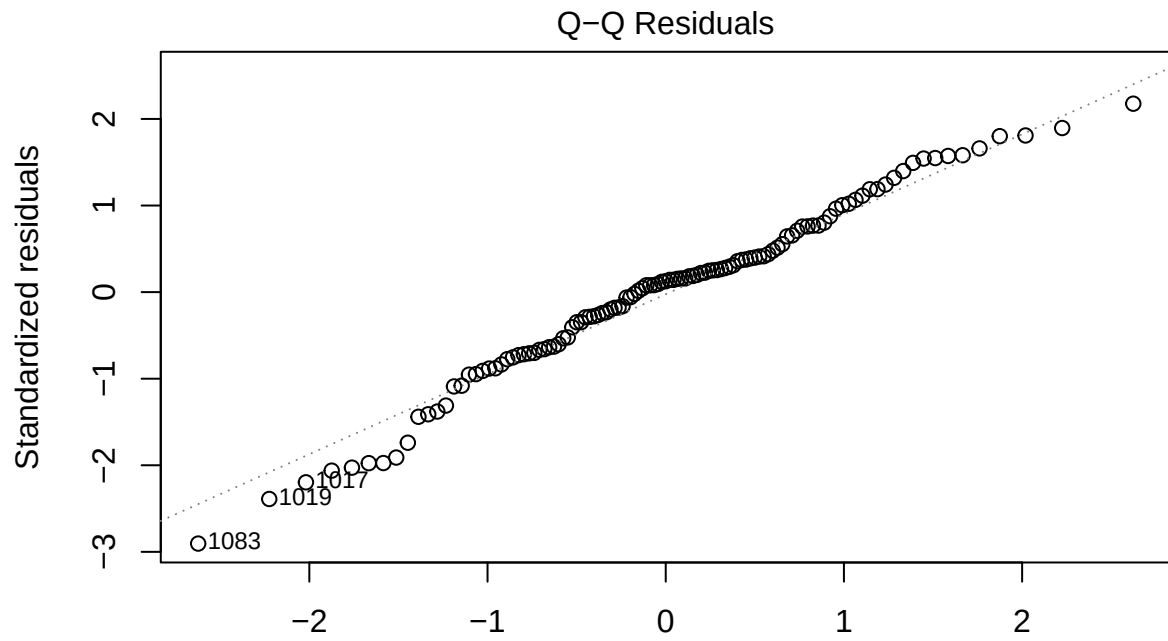
## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:danceability +
##   age:loudness_db + race:genre + race:danceability + race:happiness +
##   race:bpm + race:length_sec + gender:energy + gender:danceability +
##   gender:happiness + gender:bpm
##
##
##      Df Sum of Sq  RSS   AIC
## - gender:energy      1    1.7232 205.17 105.86
## - age:loudness_db     1    2.6363 206.09 106.43
## <none>                  203.45 106.77
## - race:genre          1    3.5983 207.05 107.03
## - race:happiness       1    3.9824 207.43 107.27
## - race:length_sec     1    4.7272 208.18 107.73
## - age:danceability    1    4.8824 208.33 107.83
## - gender:happiness    1    5.6743 209.12 108.32
## - gender:bpm          1    6.7822 210.23 109.00
## - gender:danceability 1    8.2410 211.69 109.89
## - race:danceability   1   13.9885 217.44 113.35
## - race:bpm            1   16.7094 220.16 114.95
##
## Step: AIC=105.86
## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:danceability +
##   age:loudness_db + race:genre + race:danceability + race:happiness +
##   race:bpm + race:length_sec + gender:danceability + gender:happiness +
##   gender:bpm
##
##      Df Sum of Sq  RSS   AIC
## - age:loudness_db     1    2.9405 208.11 105.70
## <none>                  205.17 105.86
## - race:genre          1    3.2193 208.39 105.87
## - race:happiness       1    3.7242 208.90 106.18
## - gender:happiness    1    4.0686 209.24 106.39
## - energy              1    4.9055 210.08 106.91
## - age:danceability    1    5.3977 210.57 107.21
## - gender:bpm          1    5.4537 210.62 107.25
## - race:length_sec     1    5.6004 210.77 107.33
## - gender:danceability 1    7.3115 212.48 108.38
## - race:danceability   1   13.1129 218.28 111.85
## - race:bpm            1   16.2288 221.40 113.68
##
## Step: AIC=105.7
## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:danceability +
##   race:genre + race:danceability + race:happiness + race:bpm +
##   race:length_sec + gender:danceability + gender:happiness +
##   gender:bpm
##
##      Df Sum of Sq  RSS   AIC
## - loudness_db         1    0.5011 208.61 104.01
## <none>                  208.11 105.70
## - race:happiness      1    3.6051 211.72 105.91
## - race:genre           1    3.6755 211.79 105.95
## - gender:happiness     1    3.8465 211.96 106.06

```

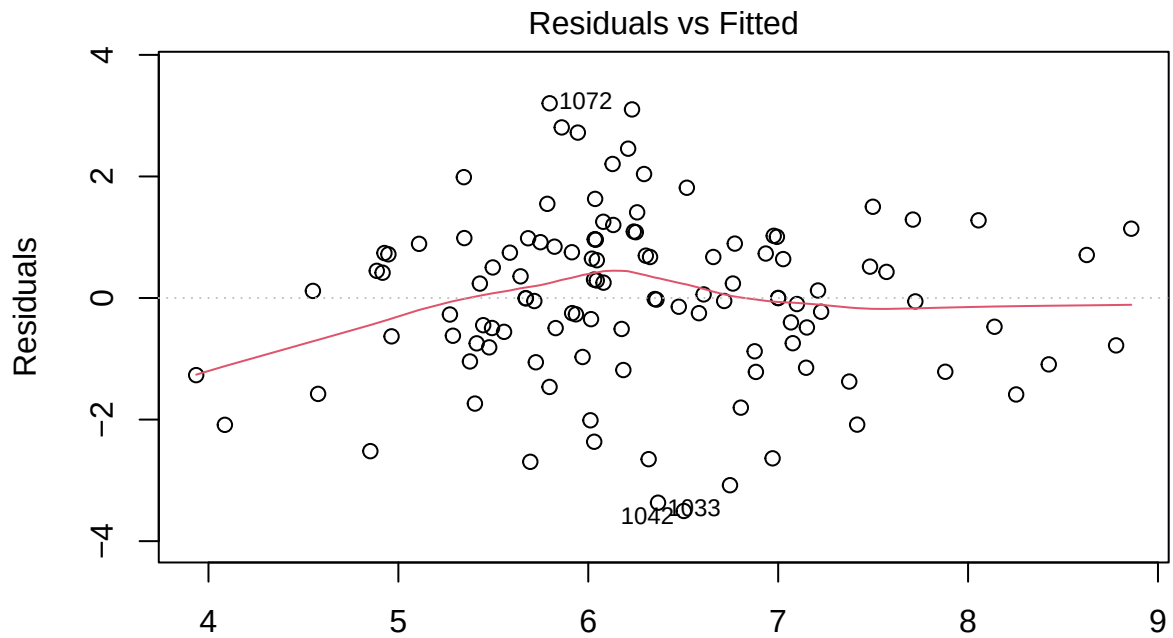


```
## - race:length_sec      1      5.2207 213.33 106.89
## - energy               1      5.2381 213.35 106.90
## - gender:bpm           1      5.4444 213.56 107.03
## - age:danceability     1      5.7563 213.87 107.22
## - gender:danceability  1      6.3411 214.45 107.57
## - race:danceability    1     13.7771 221.89 111.97
## - race:bpm             1     15.7977 223.91 113.14
##
## Step: AIC=104.01
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + length_sec + age:danceability + race:genre +
## race:danceability + race:happiness + race:bpm + race:length_sec +
## gender:danceability + gender:happiness + gender:bpm
##
##              Df Sum of Sq  RSS   AIC
## <none>                 208.61 104.01
## - race:happiness       1    3.3876 212.00 104.08
## - race:genre           1    3.6228 212.24 104.23
## - gender:happiness     1    4.1468 212.76 104.55
## - energy               1    5.3544 213.97 105.28
## - age:danceability     1    5.6004 214.21 105.42
## - race:length_sec      1    5.6508 214.26 105.45
## - gender:danceability  1    5.9037 214.52 105.61
## - gender:bpm           1    6.1967 214.81 105.78
## - race:danceability    1   13.7692 222.38 110.25
## - race:bpm             1   15.5847 224.20 111.30
## [1] "2000s"
```

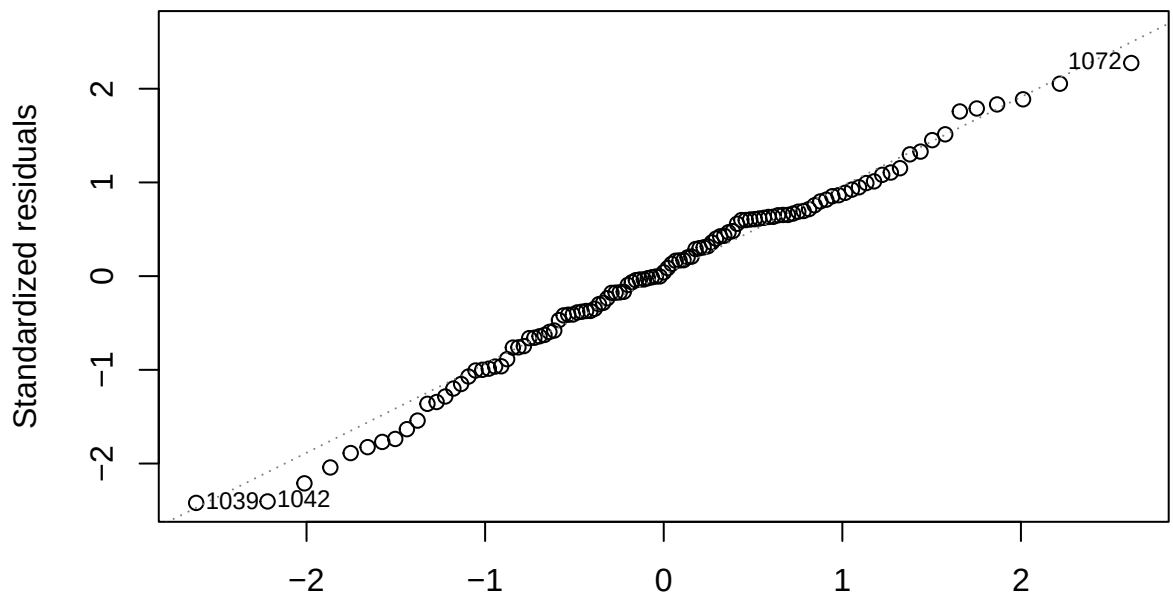




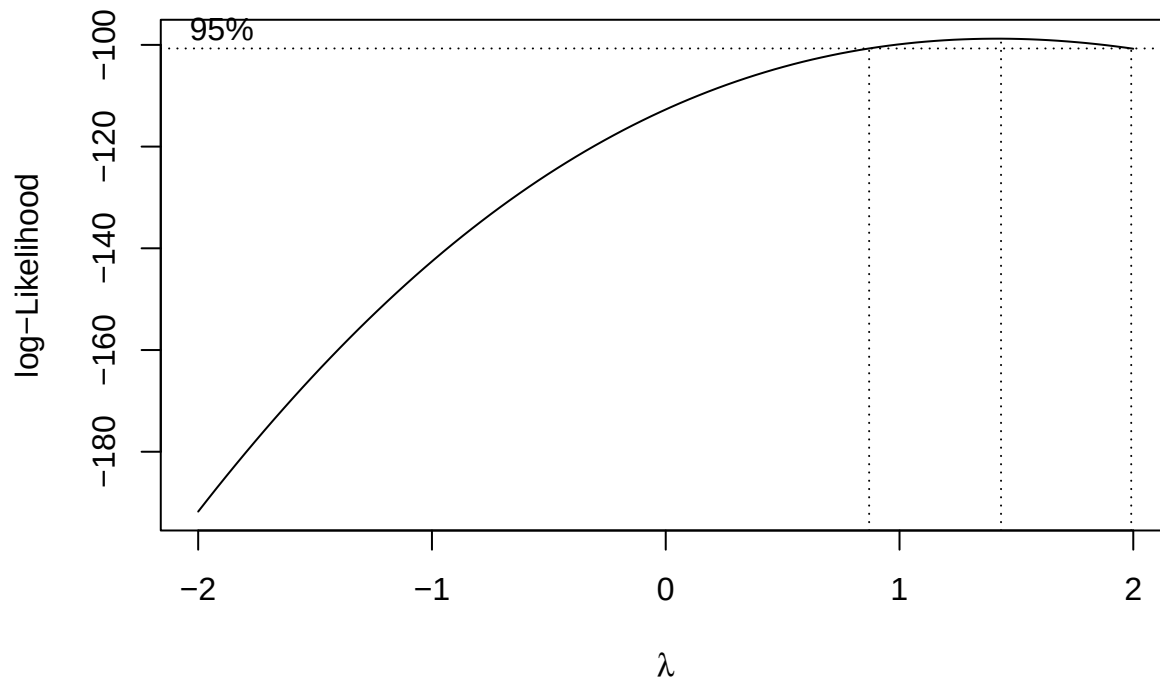
```
## Warning: not plotting observations with leverage one:
## 41, 104
```



Fitted values
 $\text{lm}(\text{rating} \sim \text{age} + \text{gender} + \text{race} + \text{age} * \text{genre} + \text{age} * \text{energy} + \text{age} * \text{dancea} \dots)$
 Q-Q Residuals



Theoretical Quantiles
 $\text{lm}(\text{rating} \sim \text{age} + \text{gender} + \text{race} + \text{age} * \text{genre} + \text{age} * \text{energy} + \text{age} * \text{dancea} \dots)$



```
## Start:  AIC=135.23
## rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + +gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##
##           Df Sum of Sq  RSS   AIC
## - race:length_sec      1    0.0490 206.39 133.26
## - age:loudness_db      1    0.1161 206.46 133.29
## - gender:danceability  1    0.3076 206.65 133.40
## - race:energy          1    0.3309 206.67 133.41
## - age:danceability     1    0.3807 206.72 133.44
## - gender:loudness_db   1    0.5786 206.92 133.55
## - race:bpm             1    0.6584 207.00 133.60
## - gender:energy        1    0.6699 207.01 133.60
## - race:genre           1    0.9574 207.30 133.76
## - race:happiness       1    1.0044 207.35 133.79
## - race:loudness_db     1    1.3673 207.71 133.99
## - gender:happiness     1    1.3906 207.73 134.00
## - age:happiness        1    1.4815 207.82 134.05
## - gender:length_sec    1    1.6502 207.99 134.15
## - race:danceability    1    1.7143 208.06 134.18
## - gender:bpm           1    1.9349 208.28 134.30
## - age:bpm              1    3.3788 209.72 135.10
## <none>                  0    206.34 135.23
## - age:energy           1    3.6401 209.98 135.24
## - gender:genre         1    4.1682 210.51 135.53
## - age:length_sec       1    4.6850 211.03 135.81
## - age:genre            1   14.3419 220.68 140.96
```

```

##
## Step: AIC=133.26
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:danceability + age:happiness + age:bpm +
## age:loudness_db + age:length_sec + race:genre + race:energy +
## race:danceability + race:happiness + race:bpm + race:loudness_db +
## gender:genre + gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
##
##      Df Sum of Sq  RSS   AIC
## - age:loudness_db      1    0.1139 206.50 131.32
## - race:energy           1    0.2968 206.69 131.42
## - gender:danceability   1    0.3235 206.71 131.44
## - age:danceability      1    0.3640 206.75 131.46
## - gender:loudness_db    1    0.5707 206.96 131.57
## - race:bpm             1    0.6213 207.01 131.60
## - gender:energy         1    0.7372 207.13 131.67
## - race:genre           1    0.9750 207.37 131.80
## - race:happiness        1    0.9902 207.38 131.81
## - gender:happiness      1    1.3678 207.76 132.02
## - race:loudness_db      1    1.4274 207.82 132.05
## - age:happiness         1    1.5048 207.90 132.09
## - race:danceability     1    1.6843 208.07 132.19
## - gender:bpm           1    1.9496 208.34 132.34
## - gender:length_sec     1    2.0186 208.41 132.38
## - age:bpm              1    3.4745 209.87 133.18
## <none>                  1    206.39 133.26
## - age:energy           1    3.8029 210.19 133.36
## - gender:genre          1    4.1221 210.51 133.53
## - age:length_sec        1    5.2419 211.63 134.14
## - age:genre            1   14.3108 220.70 138.97
##
## Step: AIC=131.32
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:danceability + age:happiness + age:bpm +
## age:length_sec + race:genre + race:energy + race:danceability +
## race:happiness + race:bpm + race:loudness_db + gender:genre +
## gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
##
##      Df Sum of Sq  RSS   AIC
## - gender:danceability   1    0.2932 206.80 129.48
## - gender:loudness_db    1    0.4647 206.97 129.58
## - race:energy           1    0.4876 206.99 129.59
## - age:danceability      1    0.5684 207.07 129.64
## - race:bpm             1    0.6983 207.20 129.71
## - gender:energy         1    0.9055 207.41 129.82
## - race:happiness        1    0.9794 207.48 129.86
## - race:genre           1    0.9916 207.50 129.87
## - gender:happiness      1    1.4148 207.92 130.10
## - age:happiness         1    1.4310 207.94 130.11
## - race:danceability     1    1.9473 208.45 130.40

```

```

## - gender:length_sec      1      1.9658 208.47 130.41
## - gender:bpm             1      1.9889 208.49 130.42
## - race:loudness_db       1      2.1307 208.64 130.50
## - age:bpm                1      3.3972 209.90 131.20
## <none>                    206.50 131.32
## - gender:genre           1      4.0628 210.57 131.56
## - age:energy              1      5.2417 211.75 132.20
## - age:length_sec         1      5.2885 211.79 132.23
## - age:genre              1     14.9934 221.50 137.38
##
## Step:  AIC=129.48
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:length_sec + race:genre + race:energy + race:danceability +
##      race:happiness + race:bpm + race:loudness_db + gender:genre +
##      gender:energy + gender:happiness + gender:bpm + gender:loudness_db +
##      gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:loudness_db  1      0.3875 207.19 127.70
## - age:danceability    1      0.4720 207.27 127.75
## - race:energy          1      0.5310 207.33 127.78
## - race:bpm            1      0.6802 207.48 127.86
## - race:happiness       1      0.8607 207.66 127.96
## - race:genre           1      1.0030 207.80 128.04
## - gender:energy        1      1.0516 207.85 128.07
## - age:happiness        1      1.3840 208.18 128.25
## - race:danceability    1      1.7644 208.56 128.46
## - gender:happiness     1      1.9037 208.70 128.54
## - gender:bpm          1      1.9718 208.77 128.57
## - race:loudness_db     1      2.0605 208.86 128.62
## - gender:length_sec    1      2.1253 208.92 128.66
## - age:bpm              1      3.4085 210.21 129.36
## <none>                  206.80 129.48
## - gender:genre         1      4.5263 211.32 129.97
## - age:energy            1      5.0691 211.87 130.27
## - age:length_sec       1      5.9118 212.71 130.72
## - age:genre            1     16.0110 222.81 136.06
##
## Step:  AIC=127.7
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:length_sec + race:genre + race:energy + race:danceability +
##      race:happiness + race:bpm + race:loudness_db + gender:genre +
##      gender:energy + gender:happiness + gender:bpm + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - age:danceability    1      0.5231 207.71 125.99
## - race:energy          1      0.5411 207.73 126.00
## - race:bpm            1      0.6923 207.88 126.08
## - race:happiness       1      1.0097 208.19 126.26
## - race:genre          1      1.2165 208.40 126.37

```

```

## - age:happiness      1      1.3071 208.49 126.42
## - gender:happiness   1      1.7375 208.92 126.66
## - race:danceability  1      1.7634 208.95 126.67
## - race:loudness_db   1      1.8873 209.07 126.74
## - gender:bpm         1      1.9759 209.16 126.79
## - gender:length_sec  1      2.5102 209.70 127.08
## - gender:energy      1      2.8893 210.07 127.29
## <none>                207.19 127.70
## - age:bpm            1      3.6769 210.86 127.72
## - gender:genre       1      4.2695 211.46 128.04
## - age:energy         1      5.1219 212.31 128.51
## - age:length_sec     1      6.9332 214.12 129.48
## - age:genre          1     15.9194 223.10 134.21
##
## Step:  AIC=125.99
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:happiness + age:bpm + age:length_sec + race:genre +
##      race:energy + race:danceability + race:happiness + race:bpm +
##      race:loudness_db + gender:genre + gender:energy + gender:happiness +
##      gender:bpm + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:energy      1      0.5083 208.22 124.27
## - race:bpm         1      0.6760 208.38 124.36
## - race:happiness    1      1.0497 208.76 124.57
## - race:genre        1      1.2306 208.94 124.67
## - race:danceability 1      1.3353 209.04 124.72
## - race:loudness_db  1      1.7469 209.46 124.95
## - age:happiness     1      1.8816 209.59 125.03
## - gender:bpm        1      1.8978 209.61 125.03
## - gender:happiness  1      1.9545 209.66 125.07
## - gender:length_sec 1      2.6662 210.38 125.45
## - gender:energy     1      3.0383 210.75 125.66
## - age:bpm           1      3.1538 210.86 125.72
## <none>              207.71 125.99
## - gender:genre      1      4.5753 212.28 126.49
## - age:energy         1      5.0486 212.76 126.75
## - age:length_sec     1      7.5679 215.28 128.10
## - age:genre          1     15.5192 223.23 132.28
##
## Step:  AIC=124.27
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:happiness + age:bpm + age:length_sec + race:genre +
##      race:danceability + race:happiness + race:bpm + race:loudness_db +
##      gender:genre + gender:energy + gender:happiness + gender:bpm +
##      gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:bpm         1      0.4440 208.66 122.51
## - race:happiness    1      0.6932 208.91 122.65
## - race:genre        1      0.9483 209.16 122.79
## - race:danceability 1      1.0990 209.32 122.88

```

```

## - race:loudness_db      1      1.2422 209.46 122.95
## - gender:bpm           1      1.7001 209.92 123.20
## - gender:happiness     1      2.0808 210.30 123.41
## - age:happiness        1      2.2454 210.46 123.50
## - gender:length_sec    1      2.3083 210.53 123.54
## - age:bpm              1      3.1638 211.38 124.00
## <none>                  208.22 124.27
## - gender:energy        1      3.7204 211.94 124.31
## - gender:genre         1      4.6078 212.82 124.79
## - age:energy           1      6.7450 214.96 125.94
## - age:length_sec       1      7.0736 215.29 126.11
## - age:genre            1     16.7728 224.99 131.18
##
## Step:  AIC=122.51
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:happiness + age:bpm + age:length_sec + race:genre +
##      race:danceability + race:happiness + race:loudness_db + gender:genre +
##      gender:energy + gender:happiness + gender:bpm + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:happiness      1      0.8452 209.51 120.98
## - race:danceability    1      1.0044 209.66 121.07
## - race:genre           1      1.1279 209.79 121.13
## - race:loudness_db     1      1.1556 209.82 121.15
## - gender:bpm           1      1.3033 209.96 121.23
## - gender:length_sec    1      1.9962 210.66 121.61
## - age:happiness        1      2.0236 210.69 121.62
## - gender:happiness     1      2.3165 210.98 121.78
## <none>                  208.66 122.51
## - gender:energy        1      3.9952 212.66 122.70
## - age:bpm              1      4.2849 212.95 122.85
## - gender:genre         1      4.8190 213.48 123.14
## - age:energy           1      6.3976 215.06 123.99
## - age:length_sec       1      7.1158 215.78 124.37
## - age:genre            1     16.3379 225.00 129.18
##
## Step:  AIC=120.98
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:happiness + age:bpm + age:length_sec + race:genre +
##      race:danceability + race:loudness_db + gender:genre + gender:energy +
##      gender:happiness + gender:bpm + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:genre           1      0.6079 210.11 119.31
## - race:danceability    1      0.6560 210.16 119.34
## - gender:bpm           1      1.4355 210.94 119.77
## - gender:length_sec    1      2.6481 212.15 120.42
## - race:loudness_db     1      2.7137 212.22 120.46
## - gender:happiness     1      3.4964 213.00 120.88
## - age:happiness        1      3.6142 213.12 120.95
## <none>                  209.51 120.98
## - gender:energy        1      4.0828 213.59 121.20

```



```

## - age:bpm          1    4.2957 213.80 121.31
## - gender:genre     1    5.9723 215.48 122.21
## - age:length_sec   1    6.8513 216.36 122.68
## - age:energy       1    7.6563 217.16 123.11
## - age:genre        1   18.8752 228.38 128.90
##
## Step:  AIC=119.31
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:happiness + age:bpm + age:length_sec + race:danceability +
##      race:loudness_db + gender:genre + gender:energy + gender:happiness +
##      gender:bpm + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:danceability  1    0.7035 210.82 117.70
## - gender:bpm        1    1.3836 211.50 118.07
## - gender:length_sec  1    2.8436 212.96 118.86
## <none>                210.11 119.31
## - race:loudness_db   1    3.7396 213.85 119.34
## - gender:happiness   1    3.8031 213.92 119.38
## - age:happiness      1    3.9144 214.03 119.44
## - age:bpm           1    4.2441 214.36 119.61
## - gender:energy      1    4.7148 214.83 119.86
## - age:length_sec     1    7.7796 217.89 121.49
## - age:energy         1    8.4517 218.57 121.85
## - gender:genre       1   10.0155 220.13 122.67
## - age:genre          1   23.5322 233.65 129.52
##
## Step:  AIC=117.7
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:happiness + age:bpm + age:length_sec + race:loudness_db +
##      gender:genre + gender:energy + gender:happiness + gender:bpm +
##      gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - danceability       1    0.0373 210.85 115.72
## - gender:bpm         1    1.2836 212.10 116.39
## - gender:length_sec  1    2.6326 213.45 117.12
## - race:loudness_db   1    3.4155 214.23 117.55
## - age:happiness      1    3.4385 214.26 117.56
## <none>                210.82 117.70
## - gender:happiness   1    3.7042 214.52 117.70
## - gender:energy      1    4.3291 215.15 118.03
## - age:bpm           1    4.5001 215.32 118.13
## - age:length_sec     1    7.1707 217.99 119.54
## - age:energy         1    8.2803 219.10 120.13
## - gender:genre       1    9.3317 220.15 120.68
## - age:genre          1   22.8801 233.70 127.55
##
## Step:  AIC=115.72
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##      loudness_db + length_sec + age:genre + age:energy + age:happiness +
##      age:bpm + age:length_sec + race:loudness_db + gender:genre +

```

```

##      gender:energy + gender:happiness + gender:bpm + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:bpm      1      1.2652 212.12 114.41
## - gender:length_sec 1      2.6203 213.47 115.14
## - age:happiness    1      3.4164 214.27 115.57
## - race:loudness_db 1      3.4459 214.30 115.58
## <none>                                210.85 115.72
## - gender:happiness 1      3.7546 214.61 115.75
## - gender:energy     1      4.3489 215.20 116.06
## - age:bpm           1      4.5016 215.36 116.15
## - age:length_sec    1      7.8418 218.70 117.92
## - age:energy        1      8.2812 219.14 118.15
## - gender:genre      1      9.4647 220.32 118.77
## - age:genre         1     22.8864 233.74 125.57
##
## Step:  AIC=114.41
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##      loudness_db + length_sec + age:genre + age:energy + age:happiness +
##      age:bpm + age:length_sec + race:loudness_db + gender:genre +
##      gender:energy + gender:happiness + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:length_sec 1      2.3130 214.43 113.65
## - age:happiness      1      2.5749 214.69 113.79
## - race:loudness_db   1      2.9498 215.07 113.99
## <none>                                212.12 114.41
## - gender:happiness   1      3.8730 215.99 114.49
## - age:bpm            1      4.2575 216.38 114.69
## - gender:energy       1      5.3010 217.42 115.24
## - age:energy          1      7.3038 219.42 116.30
## - age:length_sec     1      7.6141 219.73 116.46
## - gender:genre       1      8.7050 220.82 117.03
## - age:genre          1     21.8868 234.01 123.70
##
## Step:  AIC=113.65
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##      loudness_db + length_sec + age:genre + age:energy + age:happiness +
##      age:bpm + age:length_sec + race:loudness_db + gender:genre +
##      gender:energy + gender:happiness
##
##              Df Sum of Sq    RSS    AIC
## - age:happiness      1      2.2305 216.66 112.84
## - race:loudness_db   1      2.9652 217.40 113.23
## <none>                                214.43 113.65
## - gender:happiness   1      4.1606 218.59 113.86
## - age:bpm            1      4.8673 219.30 114.23
## - gender:energy       1      5.7007 220.13 114.67
## - age:energy          1      6.8720 221.31 115.28
## - age:length_sec     1      8.3442 222.78 116.04
## - gender:genre       1      8.8745 223.31 116.32
## - age:genre          1     20.3945 234.83 122.10
##
## Step:  AIC=112.84

```

```

## rating ~ age + gender + race + genre + energy + happiness + bpm +
##   loudness_db + length_sec + age:genre + age:energy + age:bpm +
##   age:length_sec + race:loudness_db + gender:genre + gender:energy +
##   gender:happiness
##
##           Df Sum of Sq   RSS   AIC
## - race:loudness_db  1     2.8189 219.48 112.33
## <none>                                216.66 112.84
## - age:bpm           1     4.6675 221.33 113.29
## - age:energy         1     4.7498 221.41 113.34
## - gender:happiness  1     5.4999 222.16 113.72
## - gender:energy      1     6.3739 223.04 114.18
## - age:length_sec     1     8.8338 225.50 115.44
## - gender:genre       1    10.8930 227.56 116.48
## - age:genre          1    18.5262 235.19 120.28
##
## Step:   AIC=112.33
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##   loudness_db + length_sec + age:genre + age:energy + age:bpm +
##   age:length_sec + gender:genre + gender:energy + gender:happiness
##
##           Df Sum of Sq   RSS   AIC
## - loudness_db     1     0.1716 219.65 110.42
## - race             1     0.2294 219.71 110.45
## - age:energy        1     2.7840 222.27 111.78
## <none>                                219.48 112.33
## - age:bpm          1     4.9981 224.48 112.92
## - gender:happiness  1     5.4417 224.92 113.15
## - gender:energy     1     5.5351 225.02 113.19
## - age:length_sec    1     9.3725 228.85 115.14
## - gender:genre      1     9.5943 229.08 115.25
## - age:genre         1    17.0670 236.55 118.94
##
## Step:   AIC=110.42
## rating ~ age + gender + race + genre + energy + happiness + bpm +
##   length_sec + age:genre + age:energy + age:bpm + age:length_sec +
##   gender:genre + gender:energy + gender:happiness
##
##           Df Sum of Sq   RSS   AIC
## - race           1     0.2188 219.87 108.53
## - age:energy      1     2.7121 222.37 109.83
## <none>            219.65 110.42
## - age:bpm        1     4.8897 224.54 110.95
## - gender:happiness 1     5.3659 225.02 111.19
## - gender:energy   1     5.4898 225.14 111.26
## - age:length_sec  1     9.2176 228.87 113.15
## - gender:genre    1     9.7357 229.39 113.41
## - age:genre       1    18.2207 237.88 117.58
##
## Step:   AIC=108.53
## rating ~ age + gender + genre + energy + happiness + bpm + length_sec +
##   age:genre + age:energy + age:bpm + age:length_sec + gender:genre +
##   gender:energy + gender:happiness
##

```

```

##           Df Sum of Sq   RSS   AIC
## - age:energy      1    2.9154 222.79 108.05
## <none>                219.87 108.53
## - age:bpm         1    4.7691 224.64 109.00
## - gender:happiness 1    5.2167 225.09 109.23
## - gender:energy    1    5.3407 225.21 109.29
## - age:length_sec   1    9.1286 229.00 111.21
## - gender:genre     1    9.6088 229.48 111.45
## - age:genre        1   18.5139 238.39 115.83
##
## Step:  AIC=108.05
## rating ~ age + gender + genre + energy + happiness + bpm + length_sec +
##       age:genre + age:bpm + age:length_sec + gender:genre + gender:energy +
##       gender:happiness
##
##           Df Sum of Sq   RSS   AIC
## <none>                222.79 108.05
## - gender:happiness    1    4.8910 227.68 108.55
## - age:bpm             1    5.0446 227.83 108.62
## - gender:energy        1    8.0804 230.87 110.14
## - age:length_sec       1    9.8753 232.66 111.04
## - gender:genre         1   10.8644 233.65 111.52
## - age:genre            1   15.6786 238.47 113.87
## [1] "2010s"
## Skipping 2020s due to single-level factor: genre
# Loop through each decade in the model_results list
for (decade in names(model_results)) {

  # Check if the 'optimal' model exists for this decade
  if (!is.null(model_results[[decade]]$optimal)) {

    cat("\n", "=====", "\n")
    cat("  SUMMARY FOR OPTIMAL MODEL:", decade, "\n")
    cat("=====", "\n")

    # Print the summary
    print(summary(model_results[[decade]]$optimal))

  } else {
    cat("\nNo optimal model found for:", decade, "\n")
  }
}

##
## =====
##   SUMMARY FOR OPTIMAL MODEL: 1960s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + length_sec + race:genre + race:energy +
##     race:danceability + gender:danceability + gender:happiness,
##     data = subpop)

```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.3064 -1.2960  0.1313  1.2436  4.2045
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      6.235892   2.001360   3.116  0.00214 **
## age             -0.036813   0.021122  -1.743  0.08309 .
## genderNot Male   -1.820687   1.293523  -1.408  0.16101
## raceWhite        1.604179   1.795663   0.893  0.37287
## genreJazz        1.473039   1.332354   1.106  0.27040
## genreOther       1.978211   0.999147   1.980  0.04926 *
## genrePop         0.988531   0.889417   1.111  0.26788
## genreRock        0.260014   0.629236   0.413  0.67994
## energy           -0.005765   0.015336  -0.376  0.70743
## danceability     -0.013030   0.022465  -0.580  0.56264
## happiness        0.004845   0.009013   0.538  0.59154
## length_sec       0.008965   0.003791   2.365  0.01912 *
## raceWhite:genreJazz -1.353373   2.222521  -0.609  0.54334
## raceWhite:genreOther -3.249204   1.187123  -2.737  0.00683 **
## raceWhite:genrePop  -1.786329   0.784348  -2.277  0.02395 *
## raceWhite:genreRock      NA         NA         NA         NA
## raceWhite:energy      0.024399   0.017347   1.407  0.16132
## raceWhite:danceability -0.046982   0.022464  -2.091  0.03791 *
## genderNot Male:danceability 0.076584   0.023552   3.252  0.00137 **
## genderNot Male:happiness -0.030694   0.016106  -1.906  0.05830 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.797 on 178 degrees of freedom
## Multiple R-squared:  0.314, Adjusted R-squared:  0.2446
## F-statistic: 4.527 on 18 and 178 DF, p-value: 4.551e-08
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 1970s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     happiness + bpm + loudness_db + length_sec + age:genre +
##     age:happiness + age:bpm + age:length_sec + race:energy +
##     gender:length_sec, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.1773 -1.0450  0.0378  1.0913  4.6561
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      4.7599806   6.0727470   0.784  0.43397
## age              0.0742298   0.1986635   0.374  0.70902
## genderNot Male   -3.3233372   1.3575270  -2.448  0.01513 *
```

```

## raceWhite          -1.5914109  0.7655701  -2.079  0.03878 *
## genreJazz          2.9396820  2.5394662   1.158  0.24826
## genreOther        -5.4187148  4.2740093  -1.268  0.20618
## genrePop          -3.9847287  4.0475549  -0.984  0.32594
## genreRock         -6.6280029  4.3994636  -1.507  0.13334
## energy            -0.0217547  0.0131035  -1.660  0.09827 .
## happiness          0.0681171  0.0256047   2.660  0.00837 **
## bpm              -0.0531808  0.0330717  -1.608  0.10923
## loudness_db        0.0702293  0.0514647   1.365  0.17374
## length_sec         0.0505438  0.0166865   3.029  0.00274 **
## age:genreJazz             NA             NA             NA             NA
## age:genreOther          0.2154805  0.1350105   1.596  0.11189
## age:genrePop            0.1145362  0.1272669   0.900  0.36910
## age:genreRock           0.2268970  0.1398953   1.622  0.10623
## age:happiness          -0.0020628  0.0008623  -2.392  0.01758 *
## age:bpm                0.0017767  0.0011320   1.570  0.11794
## age:length_sec         -0.0015432  0.0005939  -2.599  0.00998 **
## raceWhite:energy        0.0207272  0.0131130   1.581  0.11537
## genderNot Male:length_sec 0.0151038  0.0060578   2.493  0.01338 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.772 on 224 degrees of freedom
## Multiple R-squared:  0.2958, Adjusted R-squared:  0.233
## F-statistic: 4.705 on 20 and 224 DF,  p-value: 2.095e-09
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 1980s
## =====
##
## Call:
## lm(formula = rating ~ race + genre + bpm + length_sec + race:genre +
##     race:bpm + race:length_sec, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.4534 -0.8562  0.0485  0.9861  3.9808
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.289636    1.681818   1.361  0.1748
## raceWhite      2.987554    1.951337   1.531  0.1272
## genrePop      -0.726606    0.413305  -1.758  0.0801 .
## genreRock     -0.153197    0.492855  -0.311  0.7562
## bpm            0.011539    0.007813   1.477  0.1412
## length_sec     0.011023    0.005228   2.108  0.0361 *
## raceWhite:genrePop  1.329615    0.537348   2.474  0.0141 *
## raceWhite:genreRock  0.819251    0.591083   1.386  0.1672
## raceWhite:bpm    -0.016139    0.009210  -1.752  0.0811 .
## raceWhite:length_sec -0.009072    0.006014  -1.509  0.1329
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

```

## Residual standard error: 1.492 on 218 degrees of freedom
## Multiple R-squared:  0.06512,    Adjusted R-squared:  0.02652
## F-statistic: 1.687 on 9 and 218 DF,  p-value: 0.09338
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 1990s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + loudness_db + length_sec + age:energy +
##     age:danceability + age:loudness_db + race:danceability +
##     race:length_sec + gender:genre + gender:danceability + gender:loudness_db +
##     gender:length_sec, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.7557 -0.9125  0.0000  0.7949  3.6288
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    10.151452   4.862418   2.088 0.039069 *
## age             -0.188701   0.136443  -1.383 0.169390
## genderNot Male    4.498917   3.029352   1.485 0.140299
## raceWhite        5.519565   3.041628   1.815 0.072227 .
## genreOther      -0.174664   1.634218  -0.107 0.915074
## genrePop         0.268361   1.665154   0.161 0.872253
## genreRock        0.835016   1.632311   0.512 0.609961
## energy          -0.102502   0.032920  -3.114 0.002341 **
## danceability     0.119070   0.050887   2.340 0.021045 *
## happiness        0.017499   0.007628   2.294 0.023637 *
## loudness_db      1.147865   0.316756   3.624 0.000437 ***
## length_sec       0.011877   0.007043   1.686 0.094503 .
## age:energy        0.002489   0.001100   2.262 0.025630 *
## age:danceability -0.003376   0.001633  -2.067 0.040979 *
## age:loudness_db  -0.028004   0.010459  -2.678 0.008522 **
## raceWhite:danceability -0.035957 0.024752  -1.453 0.149090
## raceWhite:length_sec -0.014350 0.008659  -1.657 0.100228
## genderNot Male:genreOther 1.606214 0.980833   1.638 0.104286
## genderNot Male:genrePop   0.660151 1.026215   0.643 0.521341
## genderNot Male:genreRock      NA         NA         NA         NA
## genderNot Male:danceability -0.037786 0.023108  -1.635 0.104790
## genderNot Male:loudness_db -0.199855 0.119747  -1.669 0.097891 .
## genderNot Male:length_sec  -0.015603 0.008158  -1.913 0.058336 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.429 on 113 degrees of freedom
## Multiple R-squared:  0.374,    Adjusted R-squared:  0.2577
## F-statistic: 3.215 on 21 and 113 DF,  p-value: 3.432e-05
##
##
## =====

```

```
## SUMMARY FOR OPTIMAL MODEL: 2000s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + bpm + length_sec + age:danceability +
##     race:genre + race:danceability + race:happiness + race:bpm +
##     race:length_sec + gender:danceability + gender:happiness +
##     gender:bpm, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.80175 -0.88847 -0.00231  1.02953  2.93881
##
## Coefficients: (2 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    14.254799   5.689019   2.506 0.013714 *
## age           -0.262392   0.164223  -1.598 0.113013
## genderNot Male -0.551642   2.208098  -0.250 0.803196
## raceWhite     -13.479191   3.724145  -3.619 0.000451 ***
## genreOther      0.362481   1.668695   0.217 0.828443
## genrePop        0.218854   1.608297   0.136 0.892013
## genreRock       0.090807   1.604975   0.057 0.954986
## energy          0.018882   0.011341   1.665 0.098826 .
## danceability   -0.083852   0.065397  -1.282 0.202519
## happiness       0.001354   0.011393   0.119 0.905583
## bpm            -0.014549   0.008120  -1.792 0.075985 .
## length_sec     -0.009048   0.005130  -1.764 0.080619 .
## age:danceability 0.003737   0.002195   1.703 0.091492 .
## raceWhite:genreOther -1.186109  0.866090  -1.369 0.173685
## raceWhite:genrePop      NA         NA      NA      NA
## raceWhite:genreRock     NA         NA      NA      NA
## raceWhite:danceability  0.084078  0.031491   2.670 0.008759 **
## raceWhite:happiness    -0.025028  0.018899  -1.324 0.188199
## raceWhite:bpm          0.035295  0.012426   2.840 0.005385 **
## raceWhite:length_sec   0.019703  0.011520   1.710 0.090064 .
## genderNot Male:danceability -0.041970  0.024007  -1.748 0.083262 .
## genderNot Male:happiness  0.021226  0.014487   1.465 0.145771
## genderNot Male:bpm      0.019019  0.010619   1.791 0.076076 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.39 on 108 degrees of freedom
## Multiple R-squared:  0.3417, Adjusted R-squared:  0.2198
## F-statistic: 2.803 on 20 and 108 DF,  p-value: 0.0003367
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 2010s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + genre + energy + happiness +
##     bpm + length_sec + age:genre + age:bpm + age:length_sec +
```



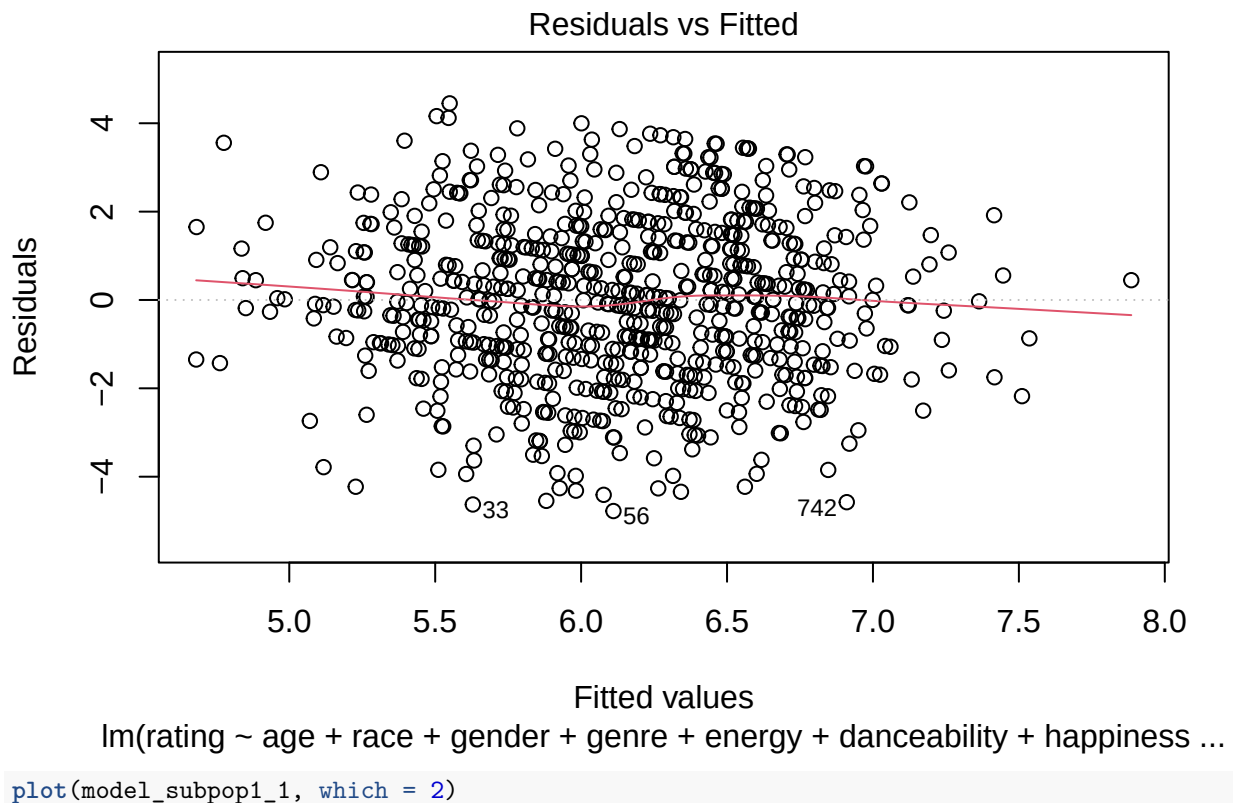
```
##      gender:genre + gender:energy + gender:happiness, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.6153 -0.7497  0.0326  0.8805  3.4548
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.7479129   5.9555613  -0.126  0.90032
## age            0.2078850   0.2138097   0.972  0.33328
## genderNot Male -1.4345612   1.2782808  -1.122  0.26447
## genrePop       4.0067252   1.8431272   2.174  0.03210 *
## genreRock     -1.3841166   6.8130047  -0.203  0.83943
## energy        -0.0346294   0.0140445  -2.466  0.01540 *
## happiness      0.0176167   0.0096788   1.820  0.07176 .
## bpm          -0.0504318   0.0327145  -1.542  0.12637
## length_sec     0.0611063   0.0235217   2.598  0.01081 *
## age:genrePop   -0.1641477   0.0621883  -2.640  0.00965 **
## age:genreRock   0.0665065   0.2087839   0.319  0.75074
## age:bpm        0.0017263   0.0011530   1.497  0.13752
## age:length_sec -0.0017601   0.0008402  -2.095  0.03874 *
## genderNot Male:genrePop  1.5747664   0.7167070   2.197  0.03034 *
## genderNot Male:genreRock      NA         NA         NA         NA
## genderNot Male:energy    0.0408389   0.0215520   1.895  0.06102 .
## genderNot Male:happiness -0.0271834   0.0184388  -1.474  0.14359
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.5 on 99 degrees of freedom
## Multiple R-squared:  0.2797, Adjusted R-squared:  0.1706
## F-statistic: 2.563 on 15 and 99 DF, p-value: 0.002865
```

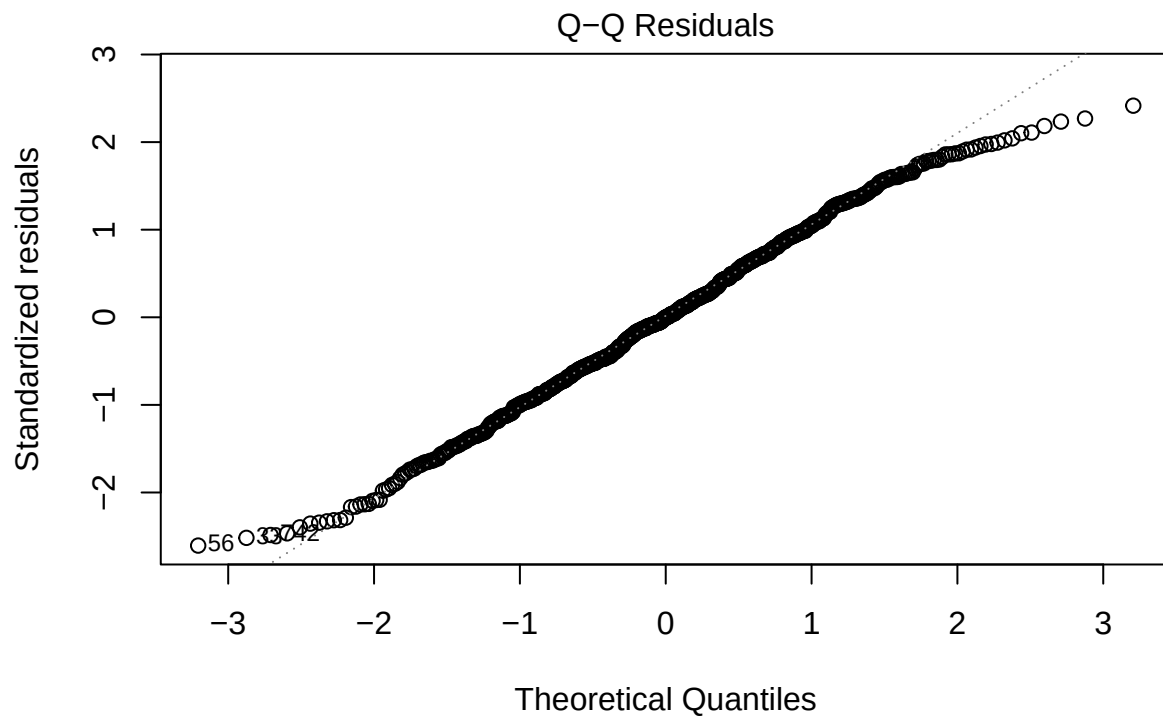
Models - Kian Sharifzadeh

```
model_subpop1_1 <- lm(data = first_subpop, rating ~ age + race + gender + genre + energy + danceability
summary(model_subpop1_1)
```

```
##
## Call:
## lm(formula = rating ~ age + race + gender + genre + energy +
##      danceability + happiness + bpm + loudness_db + length_sec,
##      data = first_subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.7774 -1.2578 -0.0063  1.3111  4.4504
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    8.012405   0.825236   9.709 < 2e-16 ***
## age           -0.030320   0.010595  -2.862  0.00433 **
## raceWhite     -0.648978   0.163236  -3.976 7.72e-05 ***
## genderNot Male  0.001685   0.155477   0.011  0.99135
## genreJazz      1.046241   0.852095   1.228  0.21990
```

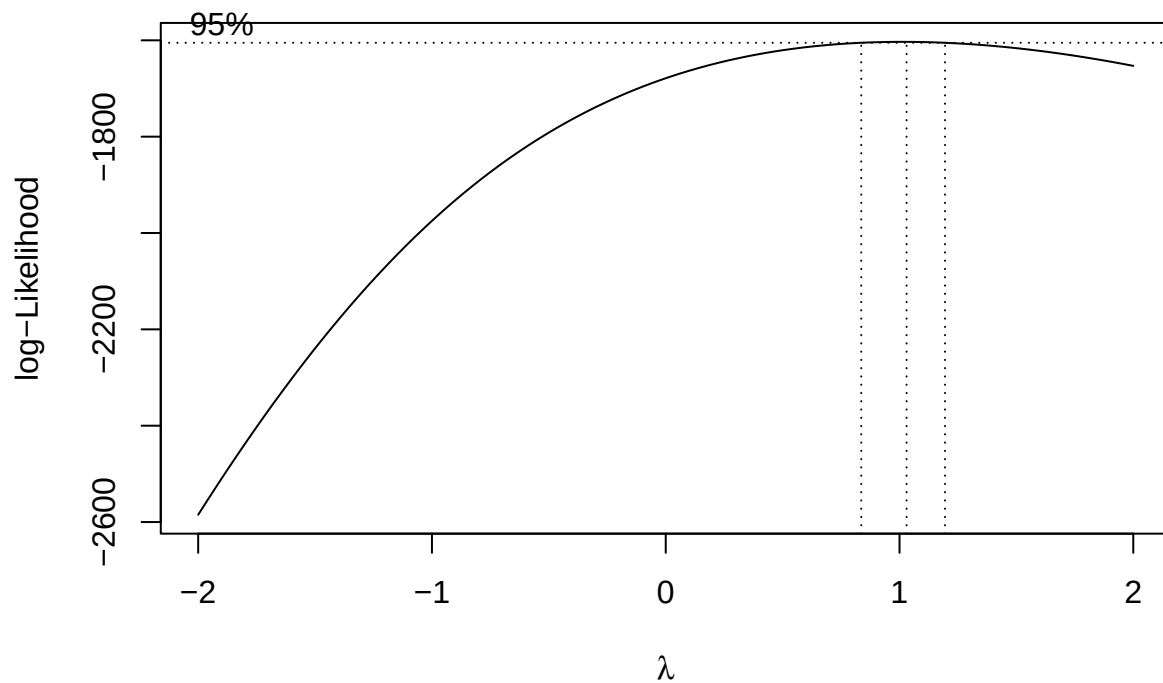
```
## genreOther      0.463522  0.413876  1.120  0.26310
## genrePop       -0.180256  0.386269 -0.467  0.64088
## genreRock       0.526001  0.394097  1.335  0.18239
## energy         -0.008542  0.005863 -1.457  0.14555
## danceability   -0.013090  0.005734 -2.283  0.02271 *
## happiness       0.006972  0.003926  1.776  0.07620 .
## bpm            0.001554  0.002986  0.520  0.60295
## loudness_db     0.063151  0.030448  2.074  0.03842 *
## length_sec      0.002120  0.001475  1.437  0.15111
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.856 on 729 degrees of freedom
## Multiple R-squared:  0.07473,    Adjusted R-squared:  0.05823
## F-statistic: 4.529 on 13 and 729 DF,  p-value: 1.76e-07
#residual anaylsis plots for subpop 1
plot(model_subpop1_1, which = 1)
```





lm(rating ~ age + race + gender + genre + energy + danceability + happiness ...

```
boxcox(model_subpop1_1, plotit=T)
```



```
new_model_1 <- lm(data = cleaned_data, log(rating) ~ age+ gender+ race + log(bpm) + genre*log(energy) +  
summary(new_model_1)
```

```
##  
## Call:  
## lm(formula = log(rating) ~ age + gender + race + log(bpm) + genre *  
##     log(energy) + genre * log(danceability) + happiness + length_sec +
```

```

##      loudness_db + has_brass + has_percussion + has_vocal_or_whistle +
##      year * log(energy) + age * genre, data = cleaned_data)
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -1.68362 -0.16641  0.04248  0.23600  0.66231
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -1.579e+01  1.399e+01  -1.128  0.259480
## age            -1.886e-03  1.057e-02  -0.178  0.858454
## genderNot Male    6.133e-02  2.143e-02   2.862  0.004289 **
## raceWhite       -8.932e-02  2.295e-02  -3.892  0.000105 ***
## log(bpm)         1.922e-02  4.546e-02   0.423  0.672501
## genreJazz        8.513e+00  1.103e+01   0.772  0.440345
## genreOther      -2.536e+00  9.507e-01  -2.668  0.007748 **
## genrePop        -2.312e+00  9.050e-01  -2.555  0.010746 *
## genreRock       -1.424e+00  9.660e-01  -1.474  0.140710
## log(energy)      6.332e+00  3.418e+00   1.852  0.064218 .
## log(danceability) -5.094e-01  2.100e-01  -2.426  0.015432 *
## happiness        1.145e-03  5.617e-04   2.039  0.041661 *
## length_sec       7.151e-04  2.211e-04   3.234  0.001256 **
## loudness_db      1.844e-02  4.770e-03   3.865  0.000117 ***
## has_brassTrue    6.642e-02  8.360e-02   0.795  0.427070
## has_percussionTrue -1.227e-02  2.067e-02  -0.594  0.552859
## has_vocal_or_whistleTrue 1.036e-01  2.377e-02   4.359  1.43e-05 ***
## year            1.022e-02  7.096e-03   1.440  0.150168
## genreJazz:log(energy) -1.507e+00  1.856e+00  -0.812  0.416936
## genreOther:log(energy) 1.163e-01  1.459e-01   0.797  0.425414
## genrePop:log(energy)  9.679e-02  1.342e-01   0.721  0.471028
## genreRock:log(energy) 1.463e-01  1.467e-01   0.997  0.318903
## genreJazz:log(danceability) -3.806e-01  8.979e-01  -0.424  0.671738
## genreOther:log(danceability) 5.201e-01  2.225e-01   2.338  0.019580 *
## genrePop:log(danceability) 5.170e-01  2.205e-01   2.345  0.019216 *
## genreRock:log(danceability) 2.685e-01  2.274e-01   1.180  0.238105
## log(energy):year    -3.305e-03  1.739e-03  -1.900  0.057665 .
## age:genreJazz      -1.975e-02  2.348e-02  -0.841  0.400454
## age:genreOther      4.892e-03  1.084e-02   0.451  0.651753
## age:genrePop       -2.649e-03  1.081e-02  -0.245  0.806507
## age:genreRock      -1.190e-03  1.107e-02  -0.108  0.914381
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3293 on 1123 degrees of freedom
## Multiple R-squared:  0.103, Adjusted R-squared:  0.07899
## F-statistic: 4.296 on 30 and 1123 DF, p-value: 2.619e-13
model_subpop1_2 <- lm(data = first_subpop, rating ~ age + gender + race + age*genre + age*energy + age*
summary(model_subpop1_2)

##
## Call:
## lm(formula = rating ~ age + gender + race + age * genre + age *
##      energy + age * danceability + age * happiness + age * bpm +
##      age * loudness_db + age * length_sec + race * genre + race *

```

```

##      energy + race * danceability + race * happiness + race *
##      bpm + race * loudness_db + race * length_sec + +gender *
##      genre + gender * energy + gender * danceability + gender *
##      happiness + gender * bpm + gender * loudness_db + gender *
##      length_sec, data = first_subpop)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -4.8819 -1.2115 -0.0254  1.2281  4.4727
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.122e+01  4.075e+00   2.753  0.00606 **
## age              -1.025e-01  1.180e-01  -0.868  0.38556
## genderNot Male    9.917e-02  1.661e+00   0.060  0.95241
## raceWhite        -2.046e+00  2.079e+00  -0.984  0.32545
## genreJazz        -2.095e-02  3.139e+00  -0.007  0.99468
## genreOther       -1.395e+00  2.554e+00  -0.546  0.58513
## genrePop         -1.555e+00  2.455e+00  -0.634  0.52657
## genreRock        -9.519e-01  2.531e+00  -0.376  0.70695
## energy           7.785e-03  2.797e-02   0.278  0.78085
## danceability     -6.778e-02  3.036e-02  -2.233  0.02588 *
## happiness        2.095e-02  1.934e-02   1.083  0.27903
## bpm              5.520e-03  1.455e-02   0.379  0.70452
## loudness_db      1.118e-01  1.460e-01   0.766  0.44420
## length_sec      -9.933e-04  7.830e-03  -0.127  0.89909
## age:genreJazz    -1.055e-02  7.959e-02  -0.133  0.89456
## age:genreOther   5.163e-02  6.936e-02   0.744  0.45691
## age:genrePop     8.418e-03  6.482e-02   0.130  0.89672
## age:genreRock   -2.577e-03  6.760e-02  -0.038  0.96960
## age:energy      -9.549e-04  9.949e-04  -0.960  0.33748
## age:danceability 1.781e-03  9.875e-04   1.803  0.07179 .
## age:happiness   -2.467e-04  6.395e-04  -0.386  0.69983
## age:bpm         9.001e-05  4.617e-04   0.195  0.84550
## age:loudness_db -6.034e-04  4.927e-03  -0.122  0.90256
## age:length_sec  5.662e-05  2.685e-04   0.211  0.83308
## raceWhite:genreJazz 1.873e+00  2.108e+00   0.889  0.37447
## raceWhite:genreOther 4.061e-01  1.469e+00   0.277  0.78220
## raceWhite:genrePop 1.540e+00  1.439e+00   1.070  0.28488
## raceWhite:genreRock 1.960e+00  1.479e+00   1.325  0.18546
## raceWhite:energy  1.533e-02  1.287e-02   1.191  0.23407
## raceWhite:danceability -5.342e-03  1.251e-02  -0.427  0.66952
## raceWhite:happiness -4.542e-03  8.705e-03  -0.522  0.60200
## raceWhite:bpm    -1.108e-02  6.886e-03  -1.609  0.10817
## raceWhite:loudness_db -2.007e-02  6.400e-02  -0.314  0.75390
## raceWhite:length_sec 4.570e-03  3.333e-03   1.371  0.17078
## genderNot Male:genreJazz NA         NA         NA         NA
## genderNot Male:genreOther -1.664e+00  9.745e-01  -1.708  0.08810 .
## genderNot Male:genrePop -1.098e+00  9.203e-01  -1.193  0.23330
## genderNot Male:genreRock -1.412e+00  9.677e-01  -1.460  0.14487
## genderNot Male:energy  2.062e-03  1.381e-02   0.149  0.88132
## genderNot Male:danceability 3.385e-02  1.269e-02   2.667  0.00783 **
## genderNot Male:happiness -1.285e-02  8.571e-03  -1.499  0.13424
## genderNot Male:bpm    3.439e-03  6.717e-03   0.512  0.60886

```

```
## genderNot Male:loudness_db -4.889e-02 6.973e-02 -0.701 0.48346
## genderNot Male:length_sec -4.775e-03 3.415e-03 -1.398 0.16246
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.838 on 700 degrees of freedom
## Multiple R-squared:  0.1286, Adjusted R-squared:  0.07634
## F-statistic: 2.46 on 42 and 700 DF, p-value: 1.632e-06
step(model_subpop1_2, direction="backward")
```

```
## Start: AIC=946.08
## rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + +gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##
##           Df Sum of Sq  RSS   AIC
## - age:genre      4      9.737 2374.1 941.14
## - gender:genre    3     12.337 2376.7 943.95
## - age:loudness_db 1       0.051 2364.4 944.10
## - gender:energy   1       0.075 2364.5 944.11
## - age:bpm         1       0.128 2364.5 944.12
## - age:length_sec  1       0.150 2364.5 944.13
## - race:loudness_db 1       0.332 2364.7 944.19
## - age:happiness   1       0.503 2364.9 944.24
## - race:danceability 1       0.616 2365.0 944.28
## - gender:bpm      1       0.885 2365.3 944.36
## - race:happiness   1       0.920 2365.3 944.37
## - gender:loudness_db 1      1.660 2366.1 944.60
## - age:energy       1      3.112 2367.5 945.06
## - race:energy       1      4.791 2369.2 945.59
## - race:length_sec  1      6.350 2370.7 946.08
## <none>                        2364.4 946.08
## - gender:length_sec 1      6.605 2371.0 946.15
## - gender:happiness  1      7.593 2372.0 946.46
## - race:bpm         1      8.740 2373.1 946.82
## - age:danceability  1     10.982 2375.4 947.53
## - gender:danceability 1     24.026 2388.4 951.59
## - race:genre       4     45.833 2410.2 952.35
##
## Step: AIC=941.14
## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:energy +
##   age:danceability + age:happiness + age:bpm + age:loudness_db +
##   age:length_sec + race:genre + race:energy + race:danceability +
##   race:happiness + race:bpm + race:loudness_db + race:length_sec +
##   gender:genre + gender:energy + gender:danceability + gender:happiness +
##   gender:bpm + gender:loudness_db + gender:length_sec
##
##           Df Sum of Sq  RSS   AIC
## - age:loudness_db 1       0.021 2374.2 939.14
```

```

## - age:bpm          1      0.057 2374.2 939.15
## - gender:energy     1      0.128 2374.3 939.18
## - age:happiness     1      0.200 2374.3 939.20
## - race:loudness_db  1      0.249 2374.4 939.21
## - race:danceability 1      0.329 2374.5 939.24
## - gender:bpm        1      0.611 2374.7 939.33
## - race:happiness    1      1.465 2375.6 939.59
## - age:length_sec    1      1.601 2375.7 939.64
## - gender:genre      3     14.913 2389.0 939.79
## - gender:loudness_db 1      2.129 2376.3 939.80
## - age:energy        1      3.523 2377.7 940.24
## - race:energy       1      4.452 2378.6 940.53
## - race:length_sec   1      4.984 2379.1 940.69
## <none>              2374.1 941.14
## - gender:length_sec  1      6.499 2380.6 941.17
## - race:bpm          1      7.803 2381.9 941.57
## - gender:happiness  1      8.175 2382.3 941.69
## - age:danceability  1     12.250 2386.4 942.96
## - race:genre        4     42.193 2416.3 946.22
## - gender:danceability 1     26.426 2400.6 947.36
##
## Step:  AIC=939.14
## rating ~ age + gender + race + genre + energy + danceability +
##         happiness + bpm + loudness_db + length_sec + age:energy +
##         age:danceability + age:happiness + age:bpm + age:length_sec +
##         race:genre + race:energy + race:danceability + race:happiness +
##         race:bpm + race:loudness_db + race:length_sec + gender:genre +
##         gender:energy + gender:danceability + gender:happiness +
##         gender:bpm + gender:loudness_db + gender:length_sec
##
##           Df Sum of Sq  RSS    AIC
## - age:bpm          1      0.053 2374.2 937.16
## - gender:energy     1      0.135 2374.3 937.18
## - age:happiness     1      0.180 2374.3 937.20
## - race:loudness_db  1      0.261 2374.4 937.22
## - race:danceability 1      0.323 2374.5 937.24
## - gender:bpm        1      0.605 2374.8 937.33
## - race:happiness    1      1.498 2375.7 937.61
## - age:length_sec    1      1.729 2375.9 937.68
## - gender:genre      3     14.954 2389.1 937.81
## - gender:loudness_db 1      2.151 2376.3 937.82
## - race:energy       1      4.632 2378.8 938.59
## - race:length_sec   1      4.963 2379.1 938.69
## <none>              2374.2 939.14
## - gender:length_sec  1      6.631 2380.8 939.21
## - race:bpm          1      7.817 2382.0 939.58
## - gender:happiness  1      8.270 2382.4 939.73
## - age:energy        1      8.827 2383.0 939.90
## - age:danceability  1     12.308 2386.5 940.98
## - race:genre        4     42.194 2416.3 944.23
## - gender:danceability 1     26.486 2400.6 945.39
##
## Step:  AIC=937.16
## rating ~ age + gender + race + genre + energy + danceability +

```

```

## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:happiness + age:length_sec + race:genre +
## race:energy + race:danceability + race:happiness + race:bpm +
## race:loudness_db + race:length_sec + gender:genre + gender:energy +
## gender:danceability + gender:happiness + gender:bpm + gender:loudness_db +
## gender:length_sec
##
##          Df Sum of Sq    RSS    AIC
## - gender:energy      1      0.132 2374.3 935.20
## - age:happiness      1      0.155 2374.4 935.21
## - race:loudness_db    1      0.260 2374.5 935.24
## - race:danceability   1      0.315 2374.5 935.26
## - gender:bpm         1      0.627 2374.8 935.36
## - race:happiness     1      1.473 2375.7 935.62
## - age:length_sec     1      1.772 2376.0 935.71
## - gender:genre       3     14.912 2389.1 935.81
## - gender:loudness_db  1      2.146 2376.3 935.83
## - race:energy        1      4.643 2378.8 936.61
## - race:length_sec    1      4.921 2379.1 936.70
## <none>                2374.2 937.16
## - gender:length_sec  1      6.624 2380.8 937.23
## - race:bpm           1      7.867 2382.1 937.62
## - gender:happiness   1      8.235 2382.4 937.73
## - age:energy         1      9.374 2383.6 938.09
## - age:danceability   1     12.274 2386.5 938.99
## - race:genre         4     42.467 2416.7 942.33
## - gender:danceability 1     26.620 2400.8 943.44
##
## Step:  AIC=935.2
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:happiness + age:length_sec + race:genre +
## race:energy + race:danceability + race:happiness + race:bpm +
## race:loudness_db + race:length_sec + gender:genre + gender:danceability +
## gender:happiness + gender:bpm + gender:loudness_db + gender:length_sec
##
##          Df Sum of Sq    RSS    AIC
## - age:happiness      1      0.164 2374.5 933.25
## - race:loudness_db    1      0.242 2374.6 933.28
## - race:danceability   1      0.323 2374.7 933.30
## - gender:bpm         1      0.774 2375.1 933.44
## - race:happiness     1      1.422 2375.8 933.65
## - age:length_sec     1      1.706 2376.0 933.73
## - gender:genre       3     15.433 2389.8 934.01
## - gender:loudness_db  1      3.098 2377.4 934.17
## - race:energy        1      4.541 2378.9 934.62
## - race:length_sec    1      4.903 2379.2 934.73
## <none>                2374.3 935.20
## - gender:length_sec  1      6.578 2380.9 935.26
## - race:bpm           1      7.793 2382.1 935.64
## - gender:happiness   1      9.160 2383.5 936.06
## - age:energy         1      9.367 2383.7 936.13
## - age:danceability   1     12.336 2386.7 937.05
## - race:genre         4     42.590 2416.9 940.41

```



```

## - gender:danceability 1 26.944 2401.3 941.58
##
## Step: AIC=933.25
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:length_sec + race:genre + race:energy +
## race:danceability + race:happiness + race:bpm + race:loudness_db +
## race:length_sec + gender:genre + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
##
##
## Df Sum of Sq RSS AIC
## - race:loudness_db 1 0.253 2374.8 931.33
## - race:danceability 1 0.332 2374.8 931.36
## - gender:bpm 1 0.784 2375.3 931.50
## - race:happiness 1 1.504 2376.0 931.72
## - gender:genre 3 15.570 2390.1 932.11
## - age:length_sec 1 2.998 2377.5 932.19
## - gender:loudness_db 1 3.167 2377.7 932.24
## - race:energy 1 4.651 2379.2 932.71
## - race:length_sec 1 4.857 2379.4 932.77
## <none> 2374.5 933.25
## - gender:length_sec 1 6.503 2381.0 933.28
## - race:bpm 1 7.763 2382.3 933.68
## - gender:happiness 1 9.027 2383.5 934.07
## - age:energy 1 13.781 2388.3 935.55
## - age:danceability 1 16.247 2390.8 936.32
## - race:genre 4 42.449 2416.9 938.42
## - gender:danceability 1 26.849 2401.3 939.61
##
## Step: AIC=931.33
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:length_sec + race:genre + race:energy +
## race:danceability + race:happiness + race:bpm + race:length_sec +
## gender:genre + gender:danceability + gender:happiness + gender:bpm +
## gender:loudness_db + gender:length_sec
##
##
## Df Sum of Sq RSS AIC
## - race:danceability 1 0.367 2375.1 929.45
## - gender:bpm 1 0.852 2375.6 929.60
## - race:happiness 1 1.293 2376.1 929.74
## - gender:genre 3 15.398 2390.2 930.13
## - age:length_sec 1 2.926 2377.7 930.25
## - gender:loudness_db 1 3.043 2377.8 930.28
## - race:length_sec 1 5.183 2379.9 930.95
## - race:energy 1 6.185 2380.9 931.26
## <none> 2374.8 931.33
## - gender:length_sec 1 6.674 2381.4 931.42
## - race:bpm 1 7.646 2382.4 931.72
## - gender:happiness 1 8.929 2383.7 932.12
## - age:energy 1 13.595 2388.3 933.57
## - age:danceability 1 16.233 2391.0 934.39
## - race:genre 4 42.484 2417.2 936.51
## - gender:danceability 1 26.598 2401.3 937.61

```

```

##
## Step: AIC=929.45
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:length_sec + race:genre + race:energy +
## race:happiness + race:bpm + race:length_sec + gender:genre +
## gender:danceability + gender:happiness + gender:bpm + gender:loudness_db +
## gender:length_sec
##
##          Df Sum of Sq    RSS    AIC
## - gender:bpm      1      0.875 2376.0 927.72
## - race:happiness   1      2.605 2377.7 928.26
## - gender:genre     3     15.532 2390.7 928.29
## - age:length_sec   1      2.821 2377.9 928.33
## - gender:loudness_db 1      3.274 2378.4 928.47
## - race:length_sec  1      4.816 2379.9 928.95
## - race:energy      1      6.153 2381.3 929.37
## <none>                2375.1 929.45
## - gender:length_sec 1      7.096 2382.2 929.66
## - race:bpm         1      7.719 2382.8 929.86
## - gender:happiness  1      9.111 2384.2 930.29
## - age:energy        1     13.917 2389.0 931.79
## - age:danceability  1     16.246 2391.4 932.51
## - race:genre        4     44.961 2420.1 935.38
## - gender:danceability 1     28.531 2403.7 936.32
##
## Step: AIC=927.72
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:length_sec + race:genre + race:energy +
## race:happiness + race:bpm + race:length_sec + gender:genre +
## gender:danceability + gender:happiness + gender:loudness_db +
## gender:length_sec
##
##          Df Sum of Sq    RSS    AIC
## - race:happiness   1      2.450 2378.4 926.49
## - age:length_sec   1      2.800 2378.8 926.59
## - gender:genre     3     16.078 2392.1 926.73
## - gender:loudness_db 1      3.308 2379.3 926.75
## - race:length_sec  1      4.822 2380.8 927.23
## - race:energy      1      6.119 2382.1 927.63
## <none>                2376.0 927.72
## - gender:length_sec 1      7.038 2383.0 927.92
## - race:bpm         1      8.182 2384.2 928.27
## - gender:happiness  1      8.633 2384.6 928.41
## - age:energy        1     14.175 2390.2 930.14
## - age:danceability  1     15.991 2392.0 930.70
## - race:genre        4     44.767 2420.8 933.59
## - gender:danceability 1     28.243 2404.2 934.50
##
## Step: AIC=926.49
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:energy +
## age:danceability + age:length_sec + race:genre + race:energy +

```

```

##      race:bpm + race:length_sec + gender:genre + gender:danceability +
##      gender:happiness + gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - age:length_sec      1      2.774 2381.2 925.35
## - gender:genre        3     15.956 2394.4 925.45
## - gender:loudness_db   1      3.658 2382.1 925.63
## - race:energy          1      3.906 2382.3 925.70
## <none>                  2378.4 926.49
## - gender:length_sec    1      7.221 2385.7 926.74
## - race:length_sec      1      7.772 2386.2 926.91
## - gender:happiness     1      8.027 2386.5 926.99
## - race:bpm            1      8.241 2386.7 927.05
## - age:energy          1     13.844 2392.3 928.80
## - age:danceability     1     15.193 2393.6 929.22
## - race:genre          4     46.941 2425.4 933.01
## - gender:danceability  1     28.928 2407.4 933.47
##
## Step:  AIC=925.35
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + race:genre + race:energy + race:bpm +
##      race:length_sec + gender:genre + gender:danceability + gender:happiness +
##      gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:loudness_db   1      3.541 2384.8 924.46
## - gender:genre        3     16.430 2397.7 924.46
## - race:energy          1      3.743 2385.0 924.52
## <none>                  2381.2 925.35
## - gender:length_sec    1      6.705 2387.9 925.44
## - race:bpm            1      7.288 2388.5 925.62
## - gender:happiness     1      7.714 2388.9 925.75
## - race:length_sec      1      8.234 2389.4 925.92
## - age:energy          1     13.100 2394.3 927.43
## - age:danceability     1     20.626 2401.8 929.76
## - race:genre          4     46.560 2427.8 931.74
## - gender:danceability  1     28.985 2410.2 932.34
##
## Step:  AIC=924.46
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + race:genre + race:energy + race:bpm +
##      race:length_sec + gender:genre + gender:danceability + gender:happiness +
##      gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:energy          1      4.446 2389.2 923.84
## - gender:genre        3     17.757 2402.5 923.97
## <none>                  2384.8 924.46
## - race:length_sec      1      7.076 2391.8 924.66
## - race:bpm            1      7.086 2391.8 924.66
## - gender:length_sec    1      7.346 2392.1 924.74
## - gender:happiness     1      9.946 2394.7 925.55

```

```

## - age:energy          1    13.012 2397.8 926.50
## - loudness_db         1    16.375 2401.1 927.54
## - age:danceability    1    20.842 2405.6 928.92
## - race:genre          4    46.867 2431.6 930.92
## - gender:danceability 1    29.449 2414.2 931.57
##
## Step: AIC=923.84
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + race:genre + race:bpm + race:length_sec +
##      gender:genre + gender:danceability + gender:happiness + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - race:bpm      1      5.278 2394.5 923.48
## - gender:genre   3     18.483 2407.7 923.56
## <none>                2389.2 923.84
## - gender:length_sec 1      6.739 2395.9 923.93
## - race:length_sec  1      7.809 2397.0 924.26
## - gender:happiness 1     10.518 2399.7 925.10
## - age:energy      1     12.268 2401.5 925.64
## - loudness_db     1     16.181 2405.4 926.85
## - age:danceability 1     21.781 2411.0 928.58
## - race:genre      4     46.142 2435.3 930.05
## - gender:danceability 1    30.222 2419.4 931.18
##
## Step: AIC=923.48
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:energy +
##      age:danceability + race:genre + race:length_sec + gender:genre +
##      gender:danceability + gender:happiness + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - bpm            1      0.523 2395.0 921.64
## - gender:genre    3     17.826 2412.3 922.99
## <none>                2394.5 923.48
## - gender:length_sec 1      6.753 2401.2 923.57
## - race:length_sec  1      8.277 2402.8 924.04
## - gender:happiness 1     11.019 2405.5 924.89
## - age:energy      1     11.537 2406.0 925.05
## - loudness_db     1     16.289 2410.8 926.52
## - age:danceability 1     21.770 2416.2 928.20
## - race:genre      4     43.852 2438.3 928.96
## - gender:danceability 1    30.661 2425.1 930.93
##
## Step: AIC=921.64
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + age:energy + age:danceability +
##      race:genre + race:length_sec + gender:genre + gender:danceability +
##      gender:happiness + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:genre    3     17.930 2412.9 921.18
## <none>                2395.0 921.64
## - gender:length_sec 1      6.787 2401.8 921.74

```

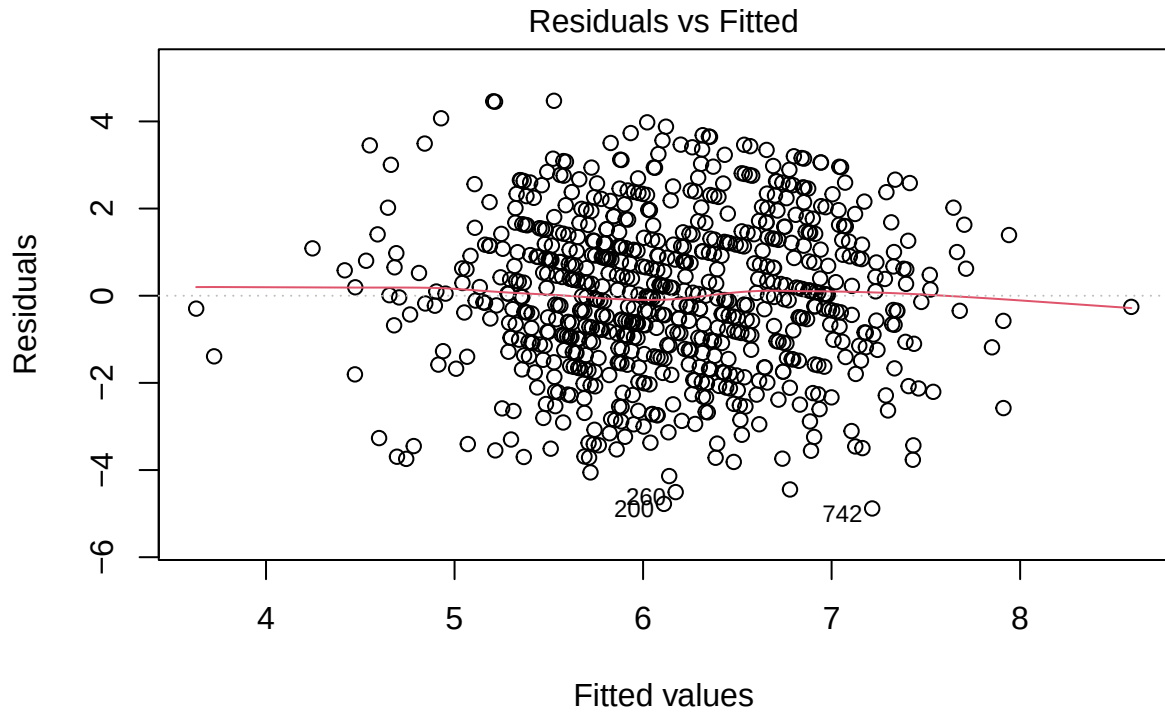
```

## - race:length_sec      1      8.522 2403.5 922.28
## - gender:happiness     1     10.882 2405.9 923.01
## - age:energy           1     11.281 2406.3 923.13
## - loudness_db          1     16.055 2411.1 924.61
## - age:danceability     1     21.604 2416.6 926.31
## - race:genre           4     44.344 2439.3 927.27
## - gender:danceability  1     30.488 2425.5 929.04
##
## Step: AIC=921.18
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + loudness_db + length_sec + age:energy + age:danceability +
## race:genre + race:length_sec + gender:danceability + gender:happiness +
## gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## <none>                        2412.9 921.18
## - race:length_sec      1      7.560 2420.5 921.51
## - age:energy           1     12.072 2425.0 922.89
## - gender:happiness     1     13.906 2426.8 923.45
## - gender:length_sec    1     14.101 2427.0 923.51
## - loudness_db          1     17.532 2430.5 924.56
## - race:genre           4     38.622 2451.6 924.98
## - age:danceability     1     21.255 2434.2 925.70
## - gender:danceability  1     24.641 2437.6 926.73
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
## danceability + happiness + loudness_db + length_sec + age:energy +
## age:danceability + race:genre + race:length_sec + gender:danceability +
## gender:happiness + gender:length_sec, data = first_subpop)
##
## Coefficients:
##              (Intercept)                  age
##              11.3072125                  -0.0803936
##              genderNot Male                raceWhite
##              0.3815542                    -2.7522149
##              genreJazz                    genreOther
##              -0.7124181                   -0.3804099
##              genrePop                     genreRock
##              -1.3932352                   -1.2816427
##              energy                      danceability
##              0.0208016                   -0.0751822
##              happiness                   loudness_db
##              0.0118214                   0.0693646
##              length_sec                  age:energy
##              0.0006032                   -0.0010484
##              age:danceability             raceWhite:genreJazz
##              0.0018960                   1.6225519
##              raceWhite:genreOther         raceWhite:genrePop
##              0.4957498                   1.3548092
##              raceWhite:genreRock         raceWhite:length_sec
##              1.9127645                   0.0044321
## genderNot Male:danceability             genderNot Male:happiness

```

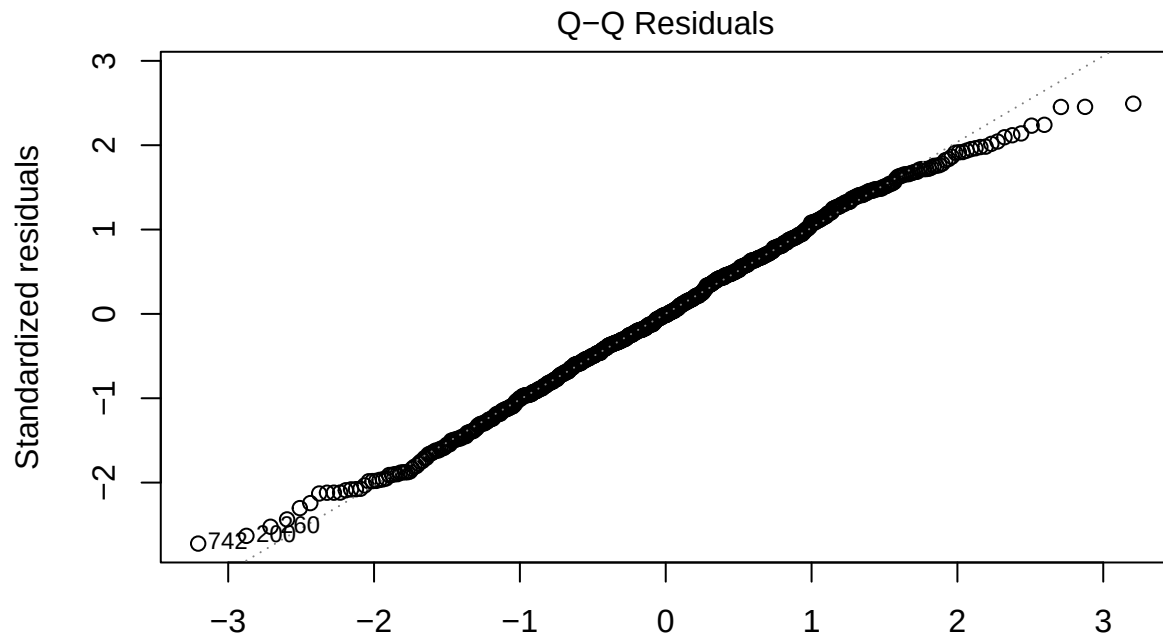
```
##          0.0324124          -0.0148272
## genderNot Male:length_sec
##          -0.0063696
```

```
#residual anaylsis plots for subpop 2
plot(model_subpop1_2, which = 1)
```



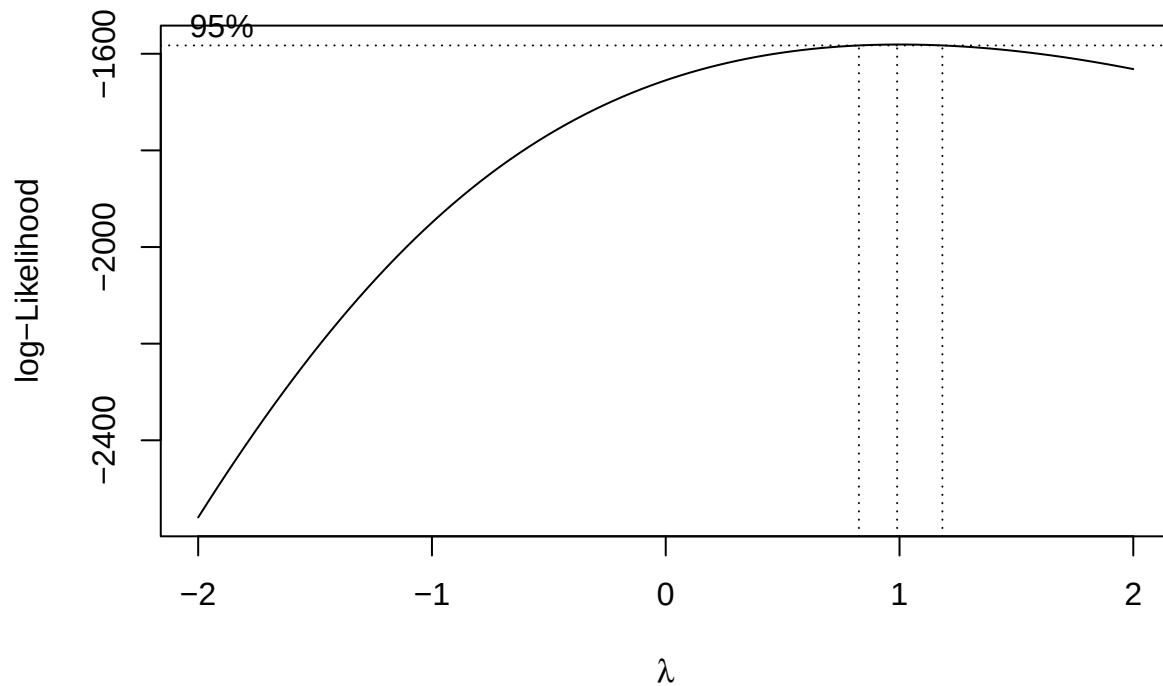
lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...

```
plot(model_subpop1_2, which = 2)
```



lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...

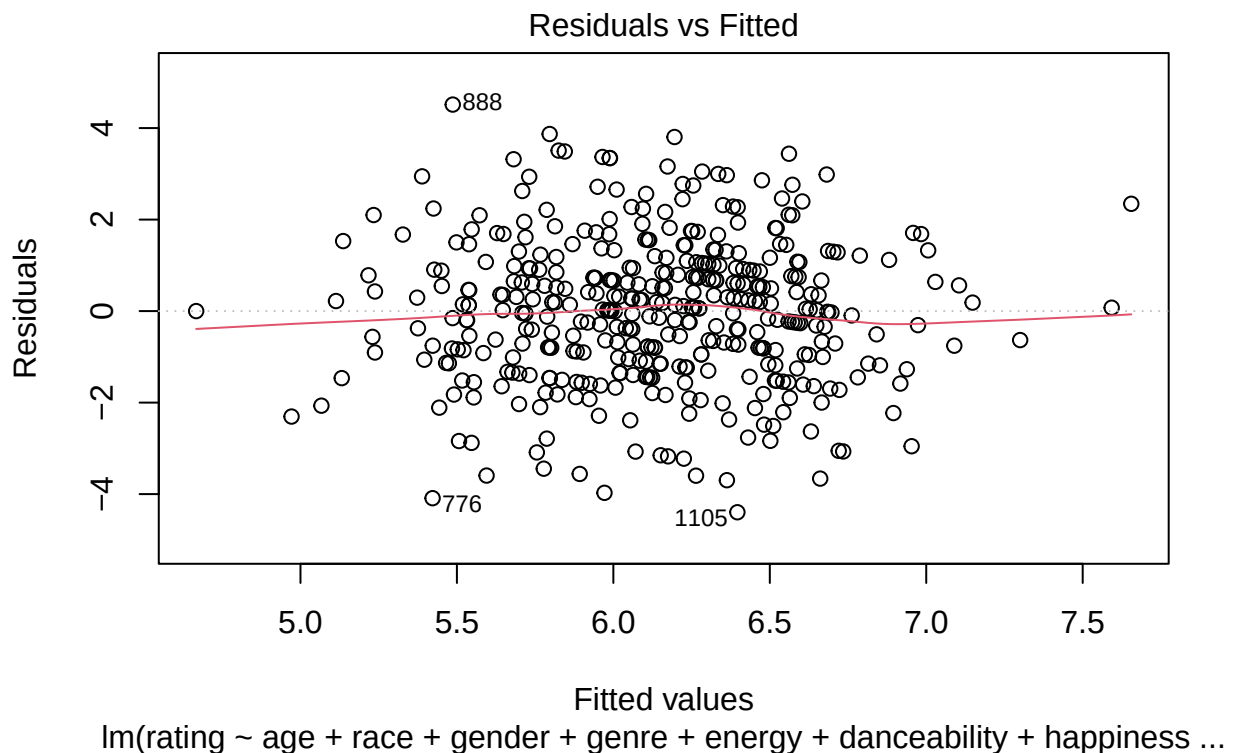
```
boxcox(model_subpop1_2, plotit=T)
```



```
model_subpop2_1 <- lm(data = second_subpop, rating ~ age + race + gender + genre + energy + danceability,
summary(model_subpop2_1)
```

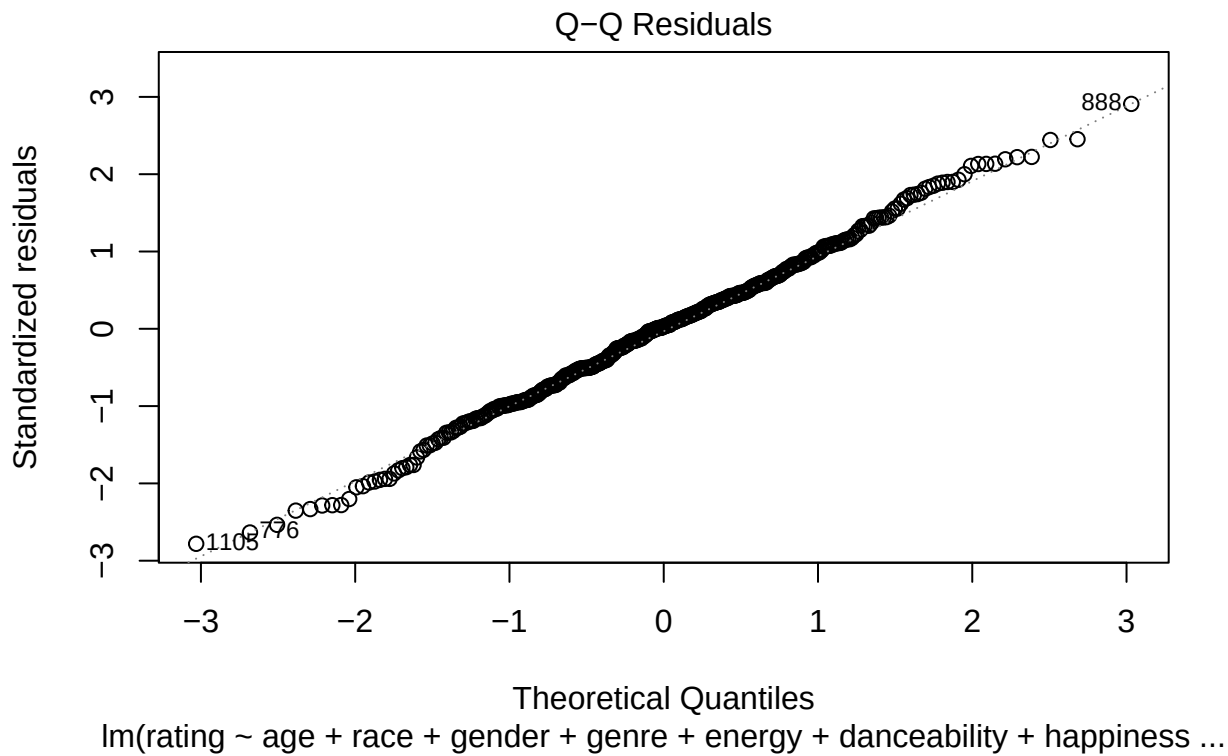
```
##
## Call:
## lm(formula = rating ~ age + race + gender + genre + energy +
##     danceability + happiness + bpm + loudness_db + length_sec,
```

```
##      data = second_subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.3964 -1.0558  0.0392  0.9655  4.5133
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.016752   1.960993   2.558  0.01089 *
## age            -0.001924   0.012044  -0.160  0.87313
## raceWhite      -0.128474   0.187366  -0.686  0.49331
## genderNot Male  0.484635   0.162164   2.989  0.00298 **
## genreOther      0.707159   1.615313   0.438  0.66178
## genrePop        0.748135   1.619543   0.462  0.64438
## genreRock       0.657571   1.637667   0.402  0.68825
## energy         -0.015693   0.007433  -2.111  0.03537 *
## danceability    0.003811   0.007058   0.540  0.58957
## happiness       0.011840   0.004400   2.691  0.00742 **
## bpm            -0.001863   0.003034  -0.614  0.53954
## loudness_db     0.081621   0.050192   1.626  0.10470
## length_sec      0.004682   0.001934   2.420  0.01595 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.595 on 398 degrees of freedom
## Multiple R-squared:  0.07127,    Adjusted R-squared:  0.04327
## F-statistic: 2.545 on 12 and 398 DF,  p-value: 0.002998
plot(model_subpop2_1, which = 1)
```

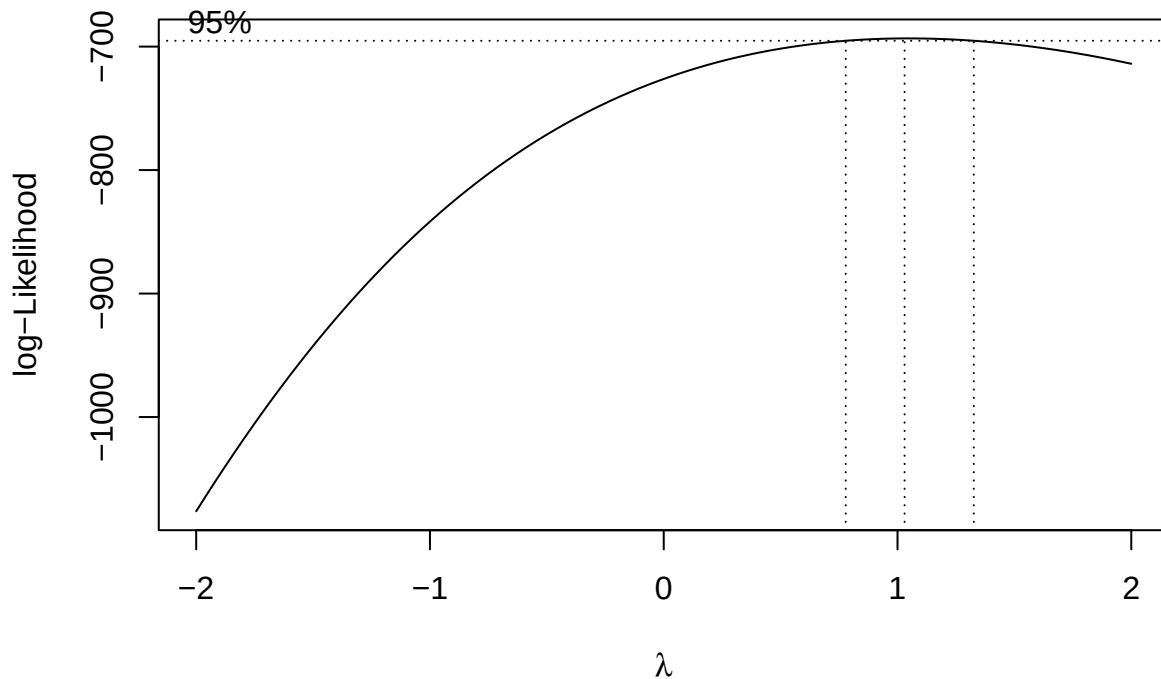



```
plot(model_subpop2_1, which = 2)
```

```
## Warning: not plotting observations with leverage one:
##      89
```



```
boxcox(model_subpop2_1, plotit=T)
```



```
model_subpop2_2 <- lm(data = second_subpop, rating ~ age + gender + race + age*genre + age*energy + age*
summary(model_subpop2_2)
```

```
##
## Call:
## lm(formula = rating ~ age + gender + race + age * genre + age *
##     energy + age * danceability + age * happiness + age * bpm +
##     age * loudness_db + age * length_sec + race * genre + race *
##     energy + race * danceability + race * happiness + race *
##     bpm + race * loudness_db + race * length_sec + +gender *
##     genre + gender * energy + gender * danceability + gender *
##     happiness + gender * bpm + gender * loudness_db + gender *
##     length_sec, data = second_subpop)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-4.2473	-1.0138	0.0664	0.9589	3.6915

```
##
## Coefficients: (3 not defined because of singularities)
##
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.962e+00	4.880e+00	-0.402	0.68790
age	2.331e-01	1.437e-01	1.622	0.10573
genderNot Male	-9.228e-01	2.197e+00	-0.420	0.67474
raceWhite	-1.117e+00	2.418e+00	-0.462	0.64441
genreOther	4.728e+00	2.903e+00	1.628	0.10428
genrePop	6.778e+00	2.969e+00	2.283	0.02300 *
genreRock	1.609e+00	1.681e+00	0.957	0.33913
energy	1.072e-02	2.993e-02	0.358	0.72044
danceability	1.117e-02	3.239e-02	0.345	0.73033
happiness	-6.277e-03	1.940e-02	-0.324	0.74642
bpm	-1.537e-02	1.478e-02	-1.040	0.29899
loudness_db	-8.527e-03	2.325e-01	-0.037	0.97076
length_sec	1.609e-02	9.603e-03	1.676	0.09460 .
age:genreOther	-9.014e-02	5.513e-02	-1.635	0.10289
age:genrePop	-1.817e-01	5.882e-02	-3.089	0.00216 **
age:genreRock	NA	NA	NA	NA
age:energy	-1.092e-03	9.919e-04	-1.100	0.27183
age:danceability	4.116e-05	1.026e-03	0.040	0.96802
age:happiness	6.305e-04	6.349e-04	0.993	0.32131
age:bpm	3.916e-04	4.937e-04	0.793	0.42812
age:loudness_db	4.001e-03	7.673e-03	0.521	0.60238
age:length_sec	-5.426e-04	3.175e-04	-1.709	0.08824 .
raceWhite:genreOther	-6.881e-01	1.096e+00	-0.628	0.53061
raceWhite:genrePop	6.208e-02	1.107e+00	0.056	0.95533
raceWhite:genreRock	NA	NA	NA	NA
raceWhite:energy	-2.158e-02	1.578e-02	-1.368	0.17215
raceWhite:danceability	1.540e-02	1.591e-02	0.968	0.33351
raceWhite:happiness	-6.471e-03	1.026e-02	-0.631	0.52846
raceWhite:bpm	1.194e-02	6.783e-03	1.760	0.07915 .
raceWhite:loudness_db	1.438e-01	1.048e-01	1.372	0.17094
raceWhite:length_sec	7.067e-03	4.177e-03	1.692	0.09149 .
genderNot Male:genreOther	-1.591e-01	8.633e-01	-0.184	0.85384
genderNot Male:genrePop	1.766e-01	9.150e-01	0.193	0.84703
genderNot Male:genreRock	NA	NA	NA	NA
genderNot Male:energy	2.686e-02	1.522e-02	1.765	0.07845 .

```
## genderNot Male:danceability -2.706e-02 1.442e-02 -1.877 0.06128 .
## genderNot Male:happiness 2.947e-03 9.101e-03 0.324 0.74625
## genderNot Male:bpm -1.735e-04 6.152e-03 -0.028 0.97752
## genderNot Male:loudness_db -1.377e-01 1.032e-01 -1.334 0.18311
## genderNot Male:length_sec 2.612e-03 4.072e-03 0.641 0.52162
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.568 on 374 degrees of freedom
## Multiple R-squared: 0.1563, Adjusted R-squared: 0.0751
## F-statistic: 1.925 on 36 and 374 DF, p-value: 0.001511
```

```
step(model_subpop2_2, direction="backward")
```

```
## Start: AIC=405.11
## rating ~ age + gender + race + age * genre + age * energy + age *
## danceability + age * happiness + age * bpm + age * loudness_db +
## age * length_sec + race * genre + race * energy + race *
## danceability + race * happiness + race * bpm + race * loudness_db +
## race * length_sec + +gender * genre + gender * energy + gender *
## danceability + gender * happiness + gender * bpm + gender *
## loudness_db + gender * length_sec
##
##           Df Sum of Sq  RSS   AIC
## - gender:genre      2    1.4843 921.34 401.78
## - gender:bpm        1     0.0020 919.86 403.11
## - age:danceability   1     0.0040 919.86 403.11
## - gender:happiness   1     0.2579 920.12 403.23
## - age:loudness_db    1     0.6687 920.53 403.41
## - race:happiness     1     0.9791 920.84 403.55
## - gender:length_sec  1     1.0120 920.87 403.57
## - age:bpm           1     1.5477 921.41 403.80
## - race:danceability  1     2.3061 922.17 404.14
## - age:happiness      1     2.4257 922.29 404.20
## - race:genre         2     7.1470 927.01 404.29
## - age:energy         1     2.9786 922.84 404.44
## - gender:loudness_db 1     4.3750 924.23 405.06
## <none>                919.86 405.11
## - race:energy        1     4.6023 924.46 405.16
## - race:loudness_db   1     4.6285 924.49 405.18
## - race:length_sec    1     7.0406 926.90 406.25
## - age:length_sec     1     7.1854 927.05 406.31
## - race:bpm           1     7.6224 927.48 406.50
## - gender:energy      1     7.6581 927.52 406.52
## - gender:danceability 1     8.6664 928.53 406.97
## - age:genre         2    28.9260 948.79 413.84
##
## Step: AIC=401.78
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:danceability + age:happiness + age:bpm +
## age:loudness_db + age:length_sec + race:genre + race:energy +
## race:danceability + race:happiness + race:bpm + race:loudness_db +
## race:length_sec + gender:energy + gender:danceability + gender:happiness +
## gender:bpm + gender:loudness_db + gender:length_sec
```

```

##
##           Df Sum of Sq    RSS    AIC
## - gender:bpm      1      0.0096 921.35 399.78
## - age:danceability 1      0.0135 921.36 399.78
## - gender:happiness 1      0.3309 921.68 399.92
## - age:loudness_db  1      0.5635 921.91 400.03
## - race:happiness   1      0.9249 922.27 400.19
## - gender:length_sec 1      1.0352 922.38 400.24
## - age:bpm          1      1.2750 922.62 400.34
## - race:danceability 1      2.4083 923.75 400.85
## - age:happiness    1      2.5064 923.85 400.89
## - age:energy        1      2.8033 924.15 401.02
## - race:genre        2      8.2973 929.64 401.46
## <none>              921.34 401.78
## - race:energy       1      4.6793 926.02 401.86
## - gender:loudness_db 1      4.8048 926.15 401.91
## - race:loudness_db  1      5.1168 926.46 402.05
## - race:length_sec   1      6.7872 928.13 402.79
## - age:length_sec    1      6.9356 928.28 402.86
## - gender:energy     1      7.3142 928.66 403.03
## - race:bpm          1      7.3955 928.74 403.06
## - gender:danceability 1     10.9444 932.29 404.63
## - age:genre         2     30.9634 952.31 411.36
##
## Step:  AIC=399.78
## rating ~ age + gender + race + genre + energy + danceability +
##           happiness + bpm + loudness_db + length_sec + age:genre +
##           age:energy + age:danceability + age:happiness + age:bpm +
##           age:loudness_db + age:length_sec + race:genre + race:energy +
##           race:danceability + race:happiness + race:bpm + race:loudness_db +
##           race:length_sec + gender:energy + gender:danceability + gender:happiness +
##           gender:loudness_db + gender:length_sec
##
##           Df Sum of Sq    RSS    AIC
## - age:danceability  1      0.0132 921.37 397.79
## - gender:happiness   1      0.3392 921.69 397.93
## - age:loudness_db    1      0.5612 921.91 398.03
## - race:happiness     1      0.9188 922.27 398.19
## - gender:length_sec  1      1.0482 922.40 398.25
## - age:bpm            1      1.2726 922.63 398.35
## - race:danceability  1      2.4025 923.76 398.85
## - age:happiness      1      2.4997 923.85 398.89
## - age:energy         1      2.7974 924.15 399.03
## - race:genre         2      8.3276 929.68 399.48
## <none>               921.35 399.78
## - race:energy        1      4.7019 926.06 399.87
## - gender:loudness_db  1      4.8105 926.16 399.92
## - race:loudness_db   1      5.1366 926.49 400.07
## - race:length_sec    1      6.7776 928.13 400.79
## - age:length_sec     1      6.9392 928.29 400.86
## - gender:energy      1      7.3801 928.73 401.06
## - race:bpm           1      7.4006 928.75 401.07
## - gender:danceability 1     11.0170 932.37 402.67
## - age:genre          2     31.4020 952.76 409.55

```

```

##
## Step: AIC=397.79
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:happiness + age:bpm + age:loudness_db +
## age:length_sec + race:genre + race:energy + race:danceability +
## race:happiness + race:bpm + race:loudness_db + race:length_sec +
## gender:energy + gender:danceability + gender:happiness +
## gender:loudness_db + gender:length_sec
##
##
## Df Sum of Sq RSS AIC
## - gender:happiness 1 0.342 921.71 395.94
## - age:loudness_db 1 0.582 921.95 396.05
## - race:happiness 1 0.918 922.29 396.20
## - gender:length_sec 1 1.042 922.41 396.25
## - age:bpm 1 1.264 922.63 396.35
## - race:danceability 1 2.390 923.76 396.85
## - age:energy 1 2.786 924.15 397.03
## - age:happiness 1 2.866 924.23 397.06
## - race:genre 2 8.324 929.69 397.48
## <none> 921.37 397.79
## - race:energy 1 4.694 926.06 397.87
## - gender:loudness_db 1 4.809 926.18 397.93
## - race:loudness_db 1 5.137 926.50 398.07
## - race:length_sec 1 6.857 928.22 398.83
## - age:length_sec 1 7.080 928.45 398.93
## - gender:energy 1 7.379 928.75 399.06
## - race:bpm 1 7.474 928.84 399.11
## - gender:danceability 1 11.127 932.49 400.72
## - age:genre 2 32.202 953.57 407.91
##
## Step: AIC=395.94
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:happiness + age:bpm + age:loudness_db +
## age:length_sec + race:genre + race:energy + race:danceability +
## race:happiness + race:bpm + race:loudness_db + race:length_sec +
## gender:energy + gender:danceability + gender:loudness_db +
## gender:length_sec
##
##
## Df Sum of Sq RSS AIC
## - age:loudness_db 1 0.631 922.34 394.22
## - race:happiness 1 0.834 922.54 394.31
## - gender:length_sec 1 0.942 922.65 394.36
## - age:bpm 1 1.397 923.11 394.56
## - race:danceability 1 2.286 924.00 394.96
## - age:happiness 1 2.736 924.44 395.16
## - age:energy 1 2.898 924.61 395.23
## - race:genre 2 8.603 930.31 395.76
## <none> 921.71 395.94
## - race:energy 1 4.766 926.47 396.06
## - gender:loudness_db 1 5.107 926.82 396.21
## - race:loudness_db 1 5.267 926.98 396.28
## - race:length_sec 1 6.873 928.58 396.99

```

```

## - age:length_sec      1      7.196 928.90 397.13
## - race:bpm            1      7.406 929.11 397.23
## - gender:energy       1     10.861 932.57 398.75
## - gender:danceability 1     11.549 933.26 399.06
## - age:genre           2     32.582 954.29 406.22
##
## Step: AIC=394.22
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:happiness + age:bpm + age:length_sec + race:genre +
## race:energy + race:danceability + race:happiness + race:bpm +
## race:loudness_db + race:length_sec + gender:energy + gender:danceability +
## gender:loudness_db + gender:length_sec
##
##              Df Sum of Sq    RSS    AIC
## - gender:length_sec      1      0.774 923.11 392.56
## - race:happiness          1      0.802 923.14 392.58
## - age:bpm                 1      1.482 923.82 392.88
## - race:danceability       1      2.292 924.63 393.24
## - age:energy              1      2.306 924.65 393.25
## - age:happiness           1      2.365 924.70 393.27
## - race:genre              2      8.983 931.32 394.20
## <none>                    922.34 394.22
## - race:energy             1      4.973 927.31 394.43
## - gender:loudness_db      1      5.447 927.79 394.64
## - race:loudness_db        1      5.577 927.92 394.70
## - race:length_sec         1      6.939 929.28 395.30
## - age:length_sec          1      7.032 929.37 395.34
## - race:bpm                1      7.616 929.96 395.60
## - gender:energy           1     10.863 933.20 397.03
## - gender:danceability     1     11.212 933.55 397.19
## - age:genre               2     32.096 954.44 404.28
##
## Step: AIC=392.56
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:happiness + age:bpm + age:length_sec + race:genre +
## race:energy + race:danceability + race:happiness + race:bpm +
## race:loudness_db + race:length_sec + gender:energy + gender:danceability +
## gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - race:happiness          1      0.7424 923.86 390.89
## - age:bpm                 1      1.3493 924.46 391.16
## - race:danceability       1      2.2669 925.38 391.57
## - age:energy              1      2.3645 925.48 391.62
## - age:happiness           1      2.5312 925.64 391.69
## - race:genre              2      8.5359 931.65 392.35
## <none>                    923.11 392.56
## - race:energy             1      5.1581 928.27 392.85
## - race:loudness_db        1      5.5774 928.69 393.04
## - gender:loudness_db      1      5.7451 928.86 393.11
## - age:length_sec          1      7.1242 930.24 393.72
## - race:bpm                1      7.5345 930.65 393.91

```

```

## - race:length_sec      1      8.6645 931.78 394.40
## - gender:energy        1     10.7990 933.91 395.34
## - gender:danceability  1     11.5194 934.63 395.66
## - age:genre            2     31.5306 954.64 402.37
##
## Step: AIC=390.89
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:happiness + age:bpm + age:length_sec + race:genre +
##      race:energy + race:danceability + race:bpm + race:loudness_db +
##      race:length_sec + gender:energy + gender:danceability + gender:loudness_db
##
##              Df Sum of Sq   RSS   AIC
## - age:bpm      1      1.5613 925.42 389.59
## - race:danceability 1      1.5730 925.43 389.59
## - age:energy    1      2.2163 926.07 389.88
## - age:happiness  1      2.6159 926.47 390.06
## - race:genre    2      8.0500 931.91 390.46
## <none>                      923.86 390.89
## - gender:loudness_db 1      5.8441 929.70 391.49
## - race:loudness_db  1      6.0102 929.87 391.56
## - age:length_sec   1      6.7965 930.65 391.91
## - race:bpm        1      7.7711 931.63 392.34
## - race:energy      1      8.6649 932.52 392.73
## - race:length_sec  1      9.7308 933.59 393.20
## - gender:energy    1     10.6734 934.53 393.62
## - gender:danceability 1     11.4534 935.31 393.96
## - age:genre        2     31.4595 955.32 400.66
##
## Step: AIC=389.59
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:happiness + age:length_sec + race:genre +
##      race:energy + race:danceability + race:bpm + race:loudness_db +
##      race:length_sec + gender:energy + gender:danceability + gender:loudness_db
##
##              Df Sum of Sq   RSS   AIC
## - age:energy      1      1.3646 926.78 388.19
## - race:danceability 1      1.6576 927.07 388.32
## - age:happiness    1      1.8589 927.28 388.41
## - race:genre       2      7.9712 933.39 389.11
## <none>                      925.42 389.59
## - gender:loudness_db 1      5.3774 930.79 389.97
## - age:length_sec    1      5.5786 931.00 390.06
## - race:loudness_db  1      6.3500 931.77 390.40
## - race:bpm         1      7.5795 933.00 390.94
## - race:energy       1      8.7319 934.15 391.45
## - race:length_sec   1      9.5836 935.00 391.82
## - gender:energy     1      9.9288 935.35 391.97
## - gender:danceability 1     11.0635 936.48 392.47
## - age:genre         2     30.0016 955.42 398.70
##
## Step: AIC=388.19
## rating ~ age + gender + race + genre + energy + danceability +

```

```

##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:happiness + age:length_sec + race:genre + race:energy +
##      race:danceability + race:bpm + race:loudness_db + race:length_sec +
##      gender:energy + gender:danceability + gender:loudness_db
##
##      Df Sum of Sq    RSS    AIC
## - age:happiness      1      1.0907 927.87 386.68
## - race:danceability   1      1.6670 928.45 386.93
## - race:genre          2      8.6525 935.43 388.01
## <none>                                926.78 388.19
## - gender:loudness_db  1      5.1876 931.97 388.49
## - race:loudness_db    1      6.2395 933.02 388.95
## - race:bpm            1      7.2407 934.02 389.39
## - age:length_sec      1      8.5719 935.35 389.98
## - race:energy         1      8.7457 935.53 390.05
## - race:length_sec     1      9.6912 936.47 390.47
## - gender:energy       1      9.9734 936.76 390.59
## - gender:danceability 1     11.2363 938.02 391.15
## - age:genre           2     28.7474 955.53 396.75
##
## Step: AIC=386.68
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:length_sec + race:genre + race:energy + race:danceability +
##      race:bpm + race:loudness_db + race:length_sec + gender:energy +
##      gender:danceability + gender:loudness_db
##
##      Df Sum of Sq    RSS    AIC
## - race:danceability   1      1.6792 929.55 385.42
## - race:genre          2      8.2264 936.10 386.31
## <none>                                927.87 386.68
## - gender:loudness_db  1      4.7561 932.63 386.78
## - race:loudness_db    1      6.7349 934.61 387.65
## - race:bpm            1      6.9644 934.84 387.75
## - race:length_sec     1      9.3496 937.22 388.80
## - gender:energy       1      9.4701 937.34 388.85
## - race:energy         1      9.5804 937.45 388.90
## - gender:danceability 1     11.4918 939.36 389.74
## - age:length_sec      1     13.4189 941.29 390.58
## - happiness           1     14.8714 942.74 391.21
## - age:genre           2     29.1701 957.04 395.40
##
## Step: AIC=385.42
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:length_sec + race:genre + race:energy + race:bpm + race:loudness_db +
##      race:length_sec + gender:energy + gender:danceability + gender:loudness_db
##
##      Df Sum of Sq    RSS    AIC
## - race:genre          2      6.9875 936.54 384.50
## <none>                                929.55 385.42
## - gender:loudness_db  1      4.7120 934.26 385.50
## - race:bpm            1      5.8556 935.41 386.00
## - race:loudness_db    1      7.1603 936.71 386.57

```



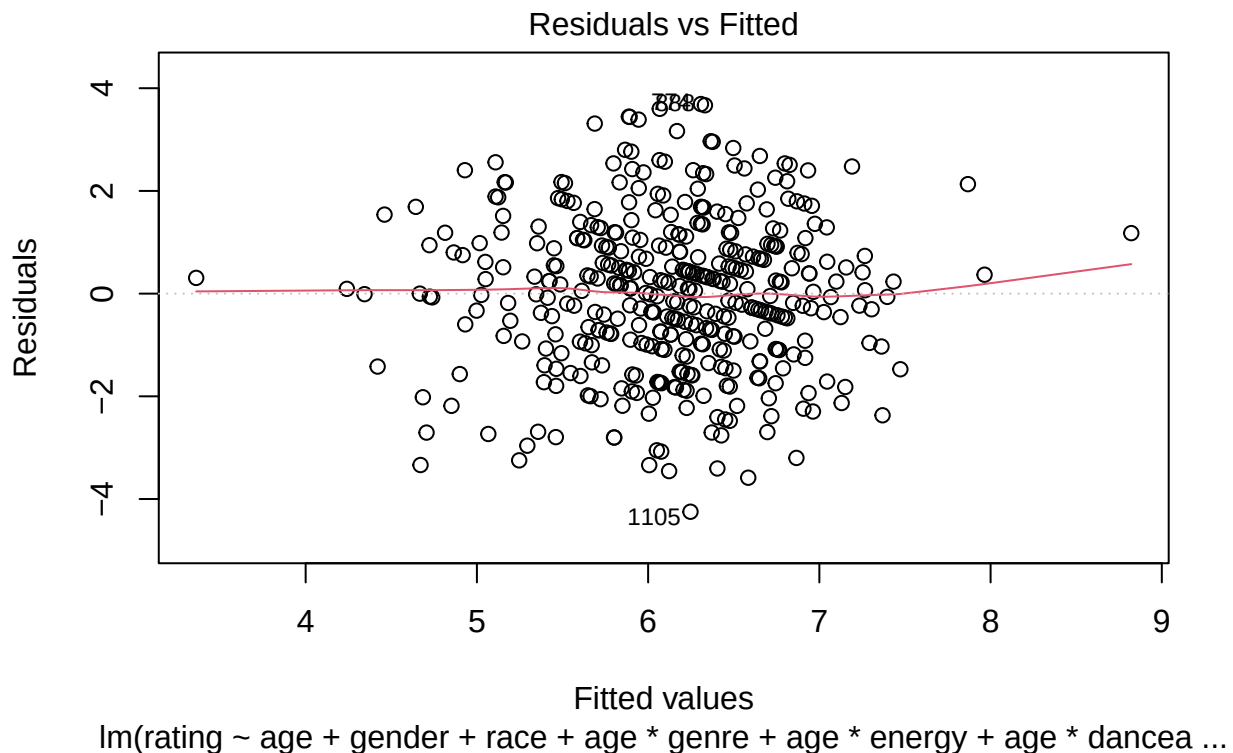
```

## - race:length_sec      1      8.5084 938.06 387.17
## - race:energy          1      8.9631 938.51 387.36
## - gender:energy        1      9.5329 939.08 387.61
## - gender:danceability  1     11.7423 941.29 388.58
## - age:length_sec       1     12.6574 942.21 388.98
## - happiness            1     15.0793 944.63 390.03
## - age:genre            2     28.5969 958.15 393.87
##
## Step: AIC=384.5
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:length_sec + race:energy + race:bpm + race:loudness_db +
##      race:length_sec + gender:energy + gender:danceability + gender:loudness_db
##
##              Df Sum of Sq    RSS    AIC
## - gender:loudness_db  1      4.463 941.00 384.45
## <none>                                936.54 384.50
## - race:bpm            1      5.179 941.72 384.77
## - race:loudness_db    1      6.775 943.31 385.46
## - race:length_sec     1      9.438 945.98 386.62
## - gender:energy       1     10.095 946.63 386.91
## - race:energy         1     10.845 947.38 387.23
## - age:length_sec      1     11.039 947.58 387.32
## - gender:danceability 1     11.255 947.79 387.41
## - happiness           1     16.649 953.19 389.74
## - age:genre           2     32.429 968.97 394.49
##
## Step: AIC=384.45
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:length_sec + race:energy + race:bpm + race:loudness_db +
##      race:length_sec + gender:energy + gender:danceability
##
##              Df Sum of Sq    RSS    AIC
## <none>                                941.00 384.45
## - race:bpm            1      5.4729 946.47 384.84
## - gender:energy       1      5.6331 946.64 384.91
## - race:loudness_db    1      7.2127 948.21 385.59
## - age:length_sec      1      9.3183 950.32 386.50
## - race:length_sec     1      9.7073 950.71 386.67
## - race:energy         1     10.7980 951.80 387.14
## - gender:danceability 1     12.9272 953.93 388.06
## - happiness           1     17.4717 958.47 390.01
## - age:genre           2     30.8859 971.89 393.73
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##      danceability + happiness + bpm + loudness_db + length_sec +
##      age:genre + age:length_sec + race:energy + race:bpm + race:loudness_db +
##      race:length_sec + gender:energy + gender:danceability, data = second_subpop)
##
## Coefficients:
##              (Intercept)                      age

```

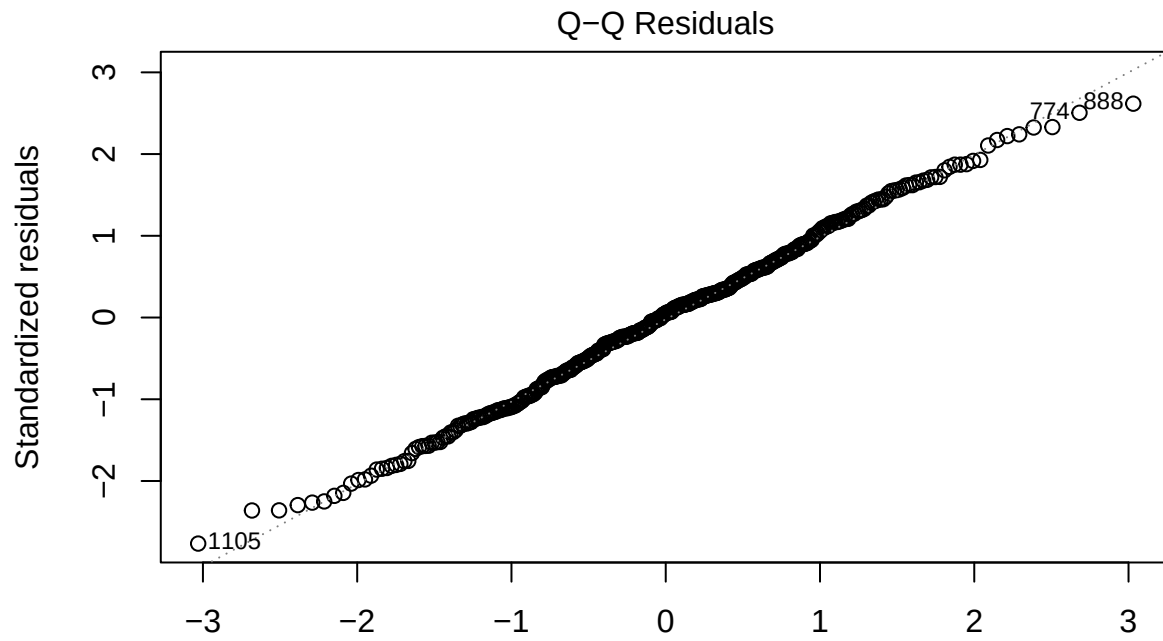
```
##          -1.3452497          0.1796533
##      genderNot Male      raceWhite
##          1.4590885         -0.1107791
##      genreOther          genrePop
##          3.1521627          5.8477239
##      genreRock            energy
##          1.2441293         -0.0129477
##      danceability        happiness
##          0.0177017          0.0116070
##          bpm            loudness_db
##          -0.0041044          0.0309261
##      length_sec        age:genreOther
##          0.0157296         -0.0621346
##      age:genrePop        age:genreRock
##          -0.1589752          NA
##      age:length_sec      raceWhite:energy
##          -0.0004902         -0.0290771
##      raceWhite:bpm      raceWhite:loudness_db
##          0.0097808          0.1737385
##      raceWhite:length_sec  genderNot Male:energy
##          0.0077875          0.0151987
## genderNot Male:danceability
##          -0.0284454
```

```
plot(model_subpop2_2, which = 1)
```



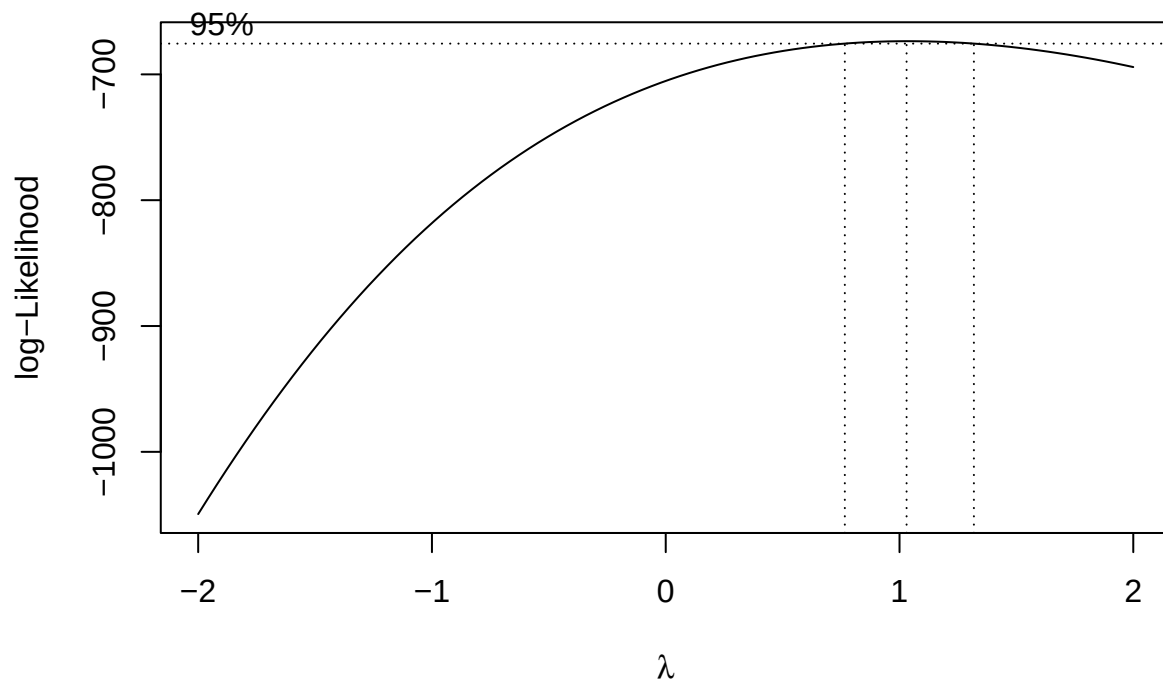
```
plot(model_subpop2_2, which = 2)
```

```
## Warning: not plotting observations with leverage one:
##      89
```



lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...

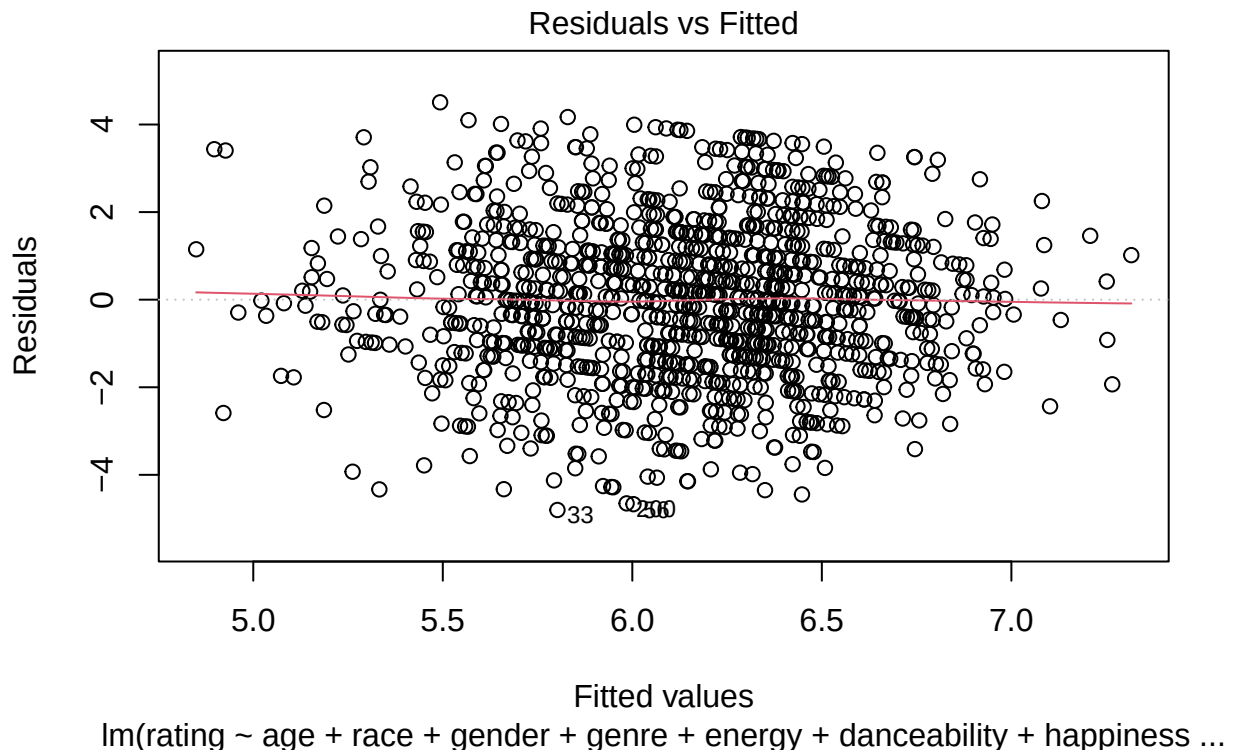
```
boxcox(model_subpop2_2, plotit=T)
```



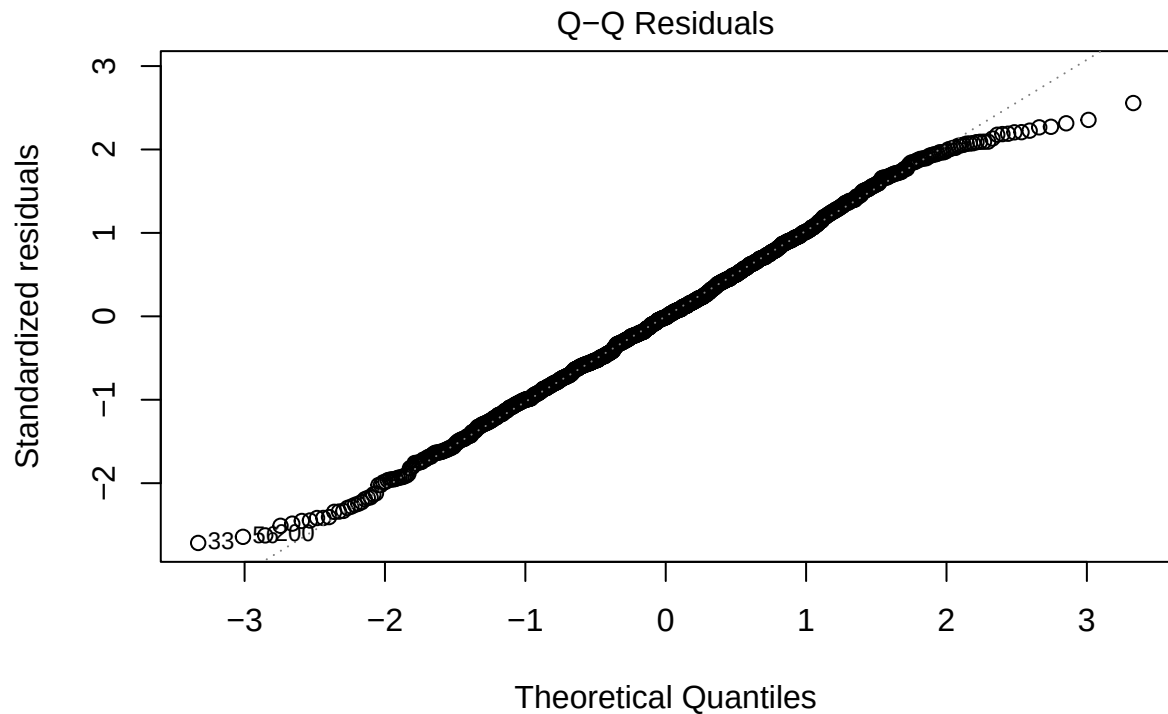
```
model_total_1 <- lm(data = cleaned_data, rating ~ age + race + gender + genre + energy + danceability +
summary(model_total_1)
```

```
##
## Call:
## lm(formula = rating ~ age + race + gender + genre + energy +
##     danceability + happiness + bpm + loudness_db + length_sec,
```

```
##      data = cleaned_data)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -4.8024 -1.2244 -0.0203  1.2248  4.5071
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   7.0722615   0.6747344   10.482 < 2e-16 ***
## age          -0.0235027   0.0080080    -2.935  0.003403 **
## raceWhite    -0.4425431   0.1231832    -3.593  0.000341 ***
## genderNot Male  0.2008111   0.1139569     1.762  0.078309 .
## genreJazz      1.1915765   0.8098788     1.471  0.141485
## genreOther     0.2896938   0.3717489     0.779  0.435981
## genrePop     -0.0309389   0.3602833    -0.086  0.931582
## genreRock      0.5806698   0.3666694     1.584  0.113555
## energy       -0.0086781   0.0045002    -1.928  0.054054 .
## danceability  -0.0098823   0.0043489    -2.272  0.023249 *
## happiness      0.0103522   0.0028992     3.571  0.000371 ***
## bpm          -0.0006552   0.0021639    -0.303  0.762097
## loudness_db    0.0442563   0.0223927     1.976  0.048353 *
## length_sec     0.0027224   0.0011510     2.365  0.018188 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.78 on 1140 degrees of freedom
## Multiple R-squared:  0.05086,    Adjusted R-squared:  0.04003
## F-statistic: 4.699 on 13 and 1140 DF,  p-value: 5.732e-08
plot(model_total_1, which = 1)
```

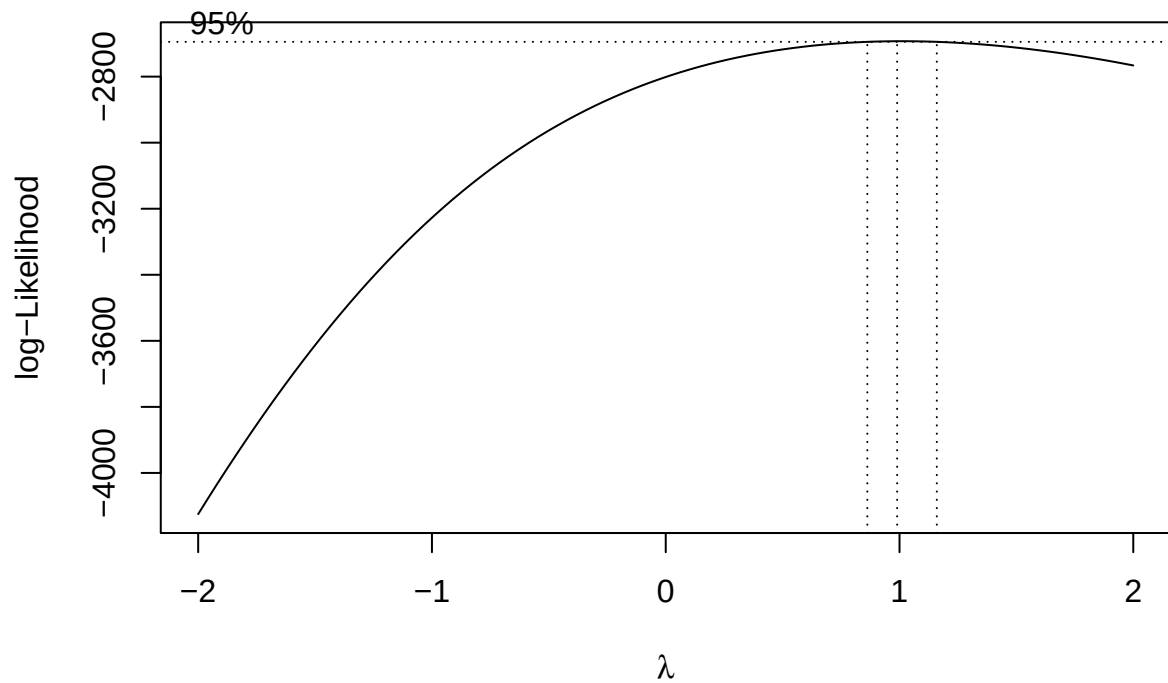


```
plot(model_total_1, which = 2)
```



lm(rating ~ age + race + gender + genre + energy + danceability + happiness ...

```
boxcox(model_total_1, plotit=T)
```



```
model_total_2 <- lm(data = cleaned_data, rating ~ age + gender + race + age*genre + age*energy + age*danceability + age*happiness)
summary(model_total_2)
```

```
##
```

```
## Call:
## lm(formula = rating ~ age + gender + race + age * genre + age *
##     energy + age * danceability + age * happiness + age * bpm +
##     age * loudness_db + age * length_sec + race * genre + race *
##     energy + race * danceability + race * happiness + race *
##     bpm + race * loudness_db + race * length_sec + +gender *
##     genre + gender * energy + gender * danceability + gender *
##     happiness + gender * bpm + gender * loudness_db + gender *
##     length_sec, data = cleaned_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.7534 -1.1493 -0.0184  1.1834  4.4488
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    9.492e+00  3.315e+00   2.863  0.00427 **
## age           -6.391e-02  9.526e-02  -0.671  0.50243
## genderNot Male  3.706e-01  1.367e+00   0.271  0.78633
## raceWhite     -1.378e+00  1.751e+00  -0.787  0.43120
## genreJazz      4.795e-01  2.932e+00   0.164  0.87014
## genreOther    -1.873e+00  2.329e+00  -0.804  0.42133
## genrePop      -1.092e+00  2.295e+00  -0.476  0.63416
## genreRock     -8.983e-01  2.346e+00  -0.383  0.70184
## energy        1.182e-02  2.016e-02   0.586  0.55775
## danceability  -3.883e-02  2.153e-02  -1.803  0.07165
## happiness     1.270e-02  1.342e-02   0.946  0.34425
## bpm          -3.741e-03  1.020e-02  -0.367  0.71388
## loudness_db   4.239e-02  1.071e-01   0.396  0.69223
## length_sec    1.714e-03  5.632e-03   0.304  0.76091
## age:genreJazz  -2.004e-02  7.403e-02  -0.271  0.78671
## age:genreOther  4.622e-02  6.149e-02   0.752  0.45237
## age:genrePop   -1.034e-02  6.027e-02  -0.172  0.86385
## age:genreRock  -1.651e-03  6.219e-02  -0.027  0.97883
## age:energy     -8.336e-04  6.954e-04  -1.199  0.23091
## age:danceability 9.308e-04  6.870e-04   1.355  0.17574
## age:happiness  1.533e-04  4.417e-04   0.347  0.72865
## age:bpm        7.048e-05  3.299e-04   0.214  0.83087
## age:loudness_db -1.306e-03  3.605e-03  -0.362  0.71717
## age:length_sec -4.083e-05  1.896e-04  -0.215  0.82959
## raceWhite:genreJazz 1.658e+00  1.996e+00   0.831  0.40631
## raceWhite:genreOther 9.835e-01  1.363e+00   0.721  0.47084
## raceWhite:genrePop 1.669e+00  1.355e+00   1.232  0.21811
## raceWhite:genreRock 1.991e+00  1.384e+00   1.439  0.15043
## raceWhite:energy  7.483e-04  9.274e-03   0.081  0.93570
## raceWhite:danceability -3.031e-03  8.984e-03  -0.337  0.73587
## raceWhite:happiness -7.637e-03  5.873e-03  -1.300  0.19377
## raceWhite:bpm    -3.436e-04  4.386e-03  -0.078  0.93758
## raceWhite:loudness_db 8.519e-02  4.553e-02   1.871  0.06158
## raceWhite:length_sec 4.248e-03  2.388e-03   1.778  0.07561
## genderNot Male:genreJazz NA         NA         NA         NA
## genderNot Male:genreOther -1.514e+00  8.832e-01  -1.714  0.08683
## genderNot Male:genrePop -1.077e+00  8.687e-01  -1.240  0.21536
## genderNot Male:genreRock -1.739e+00  9.057e-01  -1.920  0.05517
```

```
## genderNot Male:energy      4.152e-03  9.688e-03   0.429  0.66831
## genderNot Male:danceability 1.616e-02  9.138e-03   1.768  0.07734 .
## genderNot Male:happiness    -7.980e-03  6.116e-03  -1.305  0.19225
## genderNot Male:bpm          3.348e-03  4.539e-03   0.738  0.46084
## genderNot Male:loudness_db  -5.966e-03  4.702e-02  -0.127  0.89907
## genderNot Male:length_sec   -3.146e-04  2.412e-03  -0.130  0.89627
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.764 on 1111 degrees of freedom
## Multiple R-squared:  0.09126,    Adjusted R-squared:  0.05691
## F-statistic: 2.657 on 42 and 1111 DF,  p-value: 9.427e-08
```

```
step(model_total_2, direction="backward")
```

```
## Start:  AIC=1351.99
## rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + +gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##
##           Df Sum of Sq  RSS   AIC
## - race:bpm      1      0.019 3456.6 1350.0
## - race:energy    1      0.020 3456.6 1350.0
## - gender:loudness_db 1      0.050 3456.6 1350.0
## - gender:length_sec 1      0.053 3456.6 1350.0
## - age:bpm        1      0.142 3456.7 1350.0
## - age:length_sec  1      0.144 3456.7 1350.0
## - race:danceability 1      0.354 3456.9 1350.1
## - age:happiness   1      0.375 3456.9 1350.1
## - age:loudness_db 1      0.408 3457.0 1350.1
## - gender:energy   1      0.571 3457.1 1350.2
## - gender:bpm      1      1.693 3458.3 1350.5
## - age:energy       1      4.470 3461.0 1351.5
## - race:happiness   1      5.260 3461.8 1351.7
## - gender:happiness 1      5.296 3461.9 1351.8
## - age:danceability 1      5.711 3462.3 1351.9
## <none>                        3456.6 1352.0
## - gender:genre     3     19.903 3476.5 1352.6
## - age:genre         4     26.831 3483.4 1352.9
## - gender:danceability 1      9.725 3466.3 1353.2
## - race:length_sec   1      9.840 3466.4 1353.3
## - race:loudness_db  1     10.894 3467.5 1353.6
## - race:genre        4     33.047 3489.6 1355.0
##
## Step:  AIC=1349.99
## rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec + age:genre +
##   age:energy + age:danceability + age:happiness + age:bpm +
##   age:loudness_db + age:length_sec + race:genre + race:energy +
##   race:danceability + race:happiness + race:loudness_db + race:length_sec +
##   gender:genre + gender:energy + gender:danceability + gender:happiness +
```

```

##      gender:bpm + gender:loudness_db + gender:length_sec
##
##      Df Sum of Sq  RSS   AIC
## - race:energy      1    0.015 3456.6 1348.0
## - gender:loudness_db 1    0.047 3456.6 1348.0
## - gender:length_sec  1    0.051 3456.6 1348.0
## - age:bpm           1    0.139 3456.7 1348.0
## - age:length_sec     1    0.146 3456.7 1348.0
## - race:danceability  1    0.350 3456.9 1348.1
## - age:happiness      1    0.380 3457.0 1348.1
## - age:loudness_db    1    0.408 3457.0 1348.1
## - gender:energy      1    0.561 3457.2 1348.2
## - gender:bpm         1    1.771 3458.4 1348.6
## - age:energy         1    4.462 3461.1 1349.5
## - race:happiness     1    5.242 3461.8 1349.7
## - gender:happiness   1    5.285 3461.9 1349.8
## - age:danceability   1    5.702 3462.3 1349.9
## <none>                3456.6 1350.0
## - gender:genre       3   19.886 3476.5 1350.6
## - age:genre          4   26.813 3483.4 1350.9
## - gender:danceability 1    9.741 3466.3 1351.2
## - race:length_sec    1    9.888 3466.5 1351.3
## - race:loudness_db   1   10.905 3467.5 1351.6
## - race:genre         4   33.031 3489.6 1353.0
##
## Step:  AIC=1348
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:danceability +
##      race:happiness + race:loudness_db + race:length_sec + gender:genre +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:bpm + gender:loudness_db + gender:length_sec
##
##      Df Sum of Sq  RSS   AIC
## - gender:loudness_db  1    0.043 3456.6 1346.0
## - gender:length_sec   1    0.050 3456.7 1346.0
## - age:bpm             1    0.140 3456.7 1346.0
## - age:length_sec      1    0.151 3456.8 1346.0
## - race:danceability   1    0.361 3457.0 1346.1
## - age:happiness       1    0.372 3457.0 1346.1
## - age:loudness_db     1    0.430 3457.0 1346.1
## - gender:energy       1    0.547 3457.2 1346.2
## - gender:bpm          1    1.793 3458.4 1346.6
## - age:energy          1    4.460 3461.1 1347.5
## - gender:happiness    1    5.270 3461.9 1347.8
## - age:danceability    1    5.730 3462.3 1347.9
## <none>                3456.6 1348.0
## - race:happiness     1    6.582 3463.2 1348.2
## - gender:genre       3   19.874 3476.5 1348.6
## - age:genre          4   26.797 3483.4 1348.9
## - gender:danceability 1    9.732 3466.3 1349.2
## - race:length_sec    1   10.232 3466.8 1349.4
## - race:genre         4   33.016 3489.6 1351.0

```



```

## - race:loudness_db      1      21.451 3478.1 1353.1
##
## Step: AIC=1346.01
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:danceability +
##      race:happiness + race:loudness_db + race:length_sec + gender:genre +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:bpm + gender:length_sec
##
##
##      Df Sum of Sq    RSS    AIC
## - gender:length_sec      1      0.040 3456.7 1344.0
## - age:bpm                 1      0.135 3456.8 1344.1
## - age:length_sec          1      0.147 3456.8 1344.1
## - age:happiness           1      0.367 3457.0 1344.1
## - race:danceability        1      0.385 3457.0 1344.1
## - age:loudness_db         1      0.426 3457.1 1344.2
## - gender:energy           1      0.665 3457.3 1344.2
## - gender:bpm              1      1.819 3458.5 1344.6
## - age:energy              1      4.440 3461.1 1345.5
## - gender:happiness         1      5.591 3462.2 1345.9
## - age:danceability         1      5.716 3462.4 1345.9
## <none>                     3456.6 1346.0
## - race:happiness          1      6.565 3463.2 1346.2
## - gender:genre            3     19.858 3476.5 1346.6
## - age:genre               4     26.857 3483.5 1346.9
## - gender:danceability      1      9.746 3466.4 1347.3
## - race:length_sec          1     10.206 3466.9 1347.4
## - race:genre              4     33.010 3489.7 1349.0
## - race:loudness_db         1     21.859 3478.5 1351.3
##
## Step: AIC=1344.03
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:bpm +
##      age:loudness_db + age:length_sec + race:genre + race:danceability +
##      race:happiness + race:loudness_db + race:length_sec + gender:genre +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:bpm
##
##
##      Df Sum of Sq    RSS    AIC
## - age:bpm                 1      0.141 3456.8 1342.1
## - age:length_sec          1      0.159 3456.8 1342.1
## - age:happiness           1      0.365 3457.1 1342.2
## - race:danceability        1      0.397 3457.1 1342.2
## - age:loudness_db         1      0.430 3457.1 1342.2
## - gender:energy           1      0.634 3457.3 1342.2
## - gender:bpm              1      1.855 3458.5 1342.7
## - age:energy              1      4.419 3461.1 1343.5
## - age:danceability         1      5.722 3462.4 1343.9
## - gender:happiness         1      5.797 3462.5 1344.0
## <none>                     3456.7 1344.0
## - race:happiness          1      6.575 3463.3 1344.2

```

```

## - gender:genre          3      20.243 3476.9 1344.8
## - age:genre             4      26.971 3483.7 1345.0
## - gender:danceability   1       9.751 3466.4 1345.3
## - race:length_sec       1     10.488 3467.2 1345.5
## - race:genre            4     33.041 3489.7 1347.0
## - race:loudness_db      1     21.844 3478.5 1349.3
##
## Step:  AIC=1342.07
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:loudness_db +
##      age:length_sec + race:genre + race:danceability + race:happiness +
##      race:loudness_db + race:length_sec + gender:genre + gender:energy +
##      gender:danceability + gender:happiness + gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - age:length_sec      1      0.129 3457.0 1340.1
## - age:happiness        1      0.309 3457.1 1340.2
## - race:danceability     1      0.389 3457.2 1340.2
## - age:loudness_db      1      0.455 3457.3 1340.2
## - gender:energy         1      0.612 3457.4 1340.3
## - gender:bpm            1      1.810 3458.6 1340.7
## - age:energy            1      4.280 3461.1 1341.5
## - age:danceability      1      5.768 3462.6 1342.0
## - gender:happiness      1      5.773 3462.6 1342.0
## <none>                  3456.8 1342.1
## - race:happiness        1      6.705 3463.5 1342.3
## - gender:genre          3     20.180 3477.0 1342.8
## - age:genre             4     27.117 3483.9 1343.1
## - gender:danceability   1      9.691 3466.5 1343.3
## - race:length_sec       1     10.558 3467.4 1343.6
## - race:genre            4     32.932 3489.8 1345.0
## - race:loudness_db      1     21.783 3478.6 1347.3
##
## Step:  AIC=1340.12
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + age:loudness_db +
##      race:genre + race:danceability + race:happiness + race:loudness_db +
##      race:length_sec + gender:genre + gender:energy + gender:danceability +
##      gender:happiness + gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - age:loudness_db      1      0.399 3457.4 1338.2
## - race:danceability     1      0.427 3457.4 1338.3
## - gender:energy         1      0.631 3457.6 1338.3
## - age:happiness         1      0.745 3457.7 1338.4
## - gender:bpm            1      1.825 3458.8 1338.7
## - age:energy            1      5.119 3462.1 1339.8
## - gender:happiness      1      5.801 3462.8 1340.0
## - age:danceability      1      5.846 3462.8 1340.1
## <none>                  3457.0 1340.1
## - race:happiness        1      6.634 3463.6 1340.3
## - gender:genre          3     20.156 3477.1 1340.8

```

```

## - age:genre          4      26.990 3483.9 1341.1
## - gender:danceability 1       9.688 3466.6 1341.3
## - race:length_sec    1      10.489 3467.4 1341.6
## - race:genre         4      32.817 3489.8 1343.0
## - race:loudness_db   1      21.729 3478.7 1345.3
##
## Step:  AIC=1338.25
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + race:genre +
##      race:danceability + race:happiness + race:loudness_db + race:length_sec +
##      gender:genre + gender:energy + gender:danceability + gender:happiness +
##      gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - race:danceability  1      0.391 3457.7 1336.4
## - gender:energy      1      0.687 3458.0 1336.5
## - age:happiness      1      1.160 3458.5 1336.6
## - gender:bpm         1      1.777 3459.1 1336.8
## - age:danceability   1      5.894 3463.3 1338.2
## - gender:happiness   1      5.905 3463.3 1338.2
## <none>                3457.4 1338.2
## - race:happiness     1      6.681 3464.0 1338.5
## - gender:genre       3     20.644 3478.0 1339.1
## - age:genre          4     26.895 3484.3 1339.2
## - gender:danceability 1      9.669 3467.0 1339.5
## - race:length_sec    1     10.443 3467.8 1339.7
## - age:energy         1     13.584 3470.9 1340.8
## - race:genre         4     32.662 3490.0 1341.1
## - race:loudness_db   1     21.920 3479.3 1343.5
##
## Step:  AIC=1336.38
## rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec + age:genre +
##      age:energy + age:danceability + age:happiness + race:genre +
##      race:happiness + race:loudness_db + race:length_sec + gender:genre +
##      gender:energy + gender:danceability + gender:happiness +
##      gender:bpm
##
##              Df Sum of Sq    RSS    AIC
## - gender:energy      1      0.608 3458.4 1334.6
## - age:happiness      1      1.118 3458.9 1334.8
## - gender:bpm         1      1.817 3459.6 1335.0
## - gender:happiness   1      5.781 3463.5 1336.3
## <none>                3457.7 1336.4
## - age:danceability   1      6.089 3463.8 1336.4
## - age:genre          4     26.770 3484.5 1337.3
## - gender:genre       3     20.857 3478.6 1337.3
## - race:length_sec    1     10.060 3467.8 1337.7
## - race:happiness     1     10.259 3468.0 1337.8
## - gender:danceability 1     10.419 3468.2 1337.8
## - age:energy         1     13.745 3471.5 1339.0
## - race:genre         4     36.911 3494.7 1340.6
## - race:loudness_db   1     21.534 3479.3 1341.5

```

```

##
## Step: AIC=1334.58
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:danceability + age:happiness + race:genre +
## race:happiness + race:loudness_db + race:length_sec + gender:genre +
## gender:danceability + gender:happiness + gender:bpm
##
##
## Df Sum of Sq RSS AIC
## - age:happiness 1 1.139 3459.5 1333.0
## - gender:bpm 1 2.219 3460.6 1333.3
## - gender:happiness 1 5.205 3463.6 1334.3
## <none> 3458.4 1334.6
## - age:danceability 1 6.050 3464.4 1334.6
## - gender:genre 3 20.559 3478.9 1335.4
## - age:genre 4 27.056 3485.4 1335.6
## - race:length_sec 1 10.061 3468.4 1335.9
## - race:happiness 1 10.123 3468.5 1336.0
## - gender:danceability 1 10.987 3469.3 1336.2
## - age:energy 1 13.985 3472.3 1337.2
## - race:genre 4 36.921 3495.3 1338.8
## - race:loudness_db 1 21.254 3479.6 1339.7
##
## Step: AIC=1332.96
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:danceability + race:genre + race:happiness +
## race:loudness_db + race:length_sec + gender:genre + gender:danceability +
## gender:happiness + gender:bpm
##
##
## Df Sum of Sq RSS AIC
## - gender:bpm 1 2.336 3461.8 1331.7
## - gender:happiness 1 5.618 3465.1 1332.8
## <none> 3459.5 1333.0
## - age:genre 4 26.019 3485.5 1333.6
## - gender:genre 3 20.285 3479.8 1333.7
## - age:danceability 1 9.624 3469.1 1334.2
## - race:length_sec 1 9.798 3469.3 1334.2
## - race:happiness 1 9.828 3469.3 1334.2
## - gender:danceability 1 11.054 3470.5 1334.6
## - age:energy 1 12.846 3472.3 1335.2
## - race:genre 4 36.800 3496.3 1337.2
## - race:loudness_db 1 21.219 3480.7 1338.0
##
## Step: AIC=1331.74
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec + age:genre +
## age:energy + age:danceability + race:genre + race:happiness +
## race:loudness_db + race:length_sec + gender:genre + gender:danceability +
## gender:happiness
##
##
## Df Sum of Sq RSS AIC
## - bpm 1 0.277 3462.1 1329.8
## - gender:happiness 1 5.275 3467.1 1331.5

```

```

## <none> 3461.8 1331.7
## - gender:genre 3 20.237 3482.1 1332.5
## - age:genre 4 26.561 3488.4 1332.6
## - age:danceability 1 9.508 3471.3 1332.9
## - race:length_sec 1 9.748 3471.6 1333.0
## - race:happiness 1 9.754 3471.6 1333.0
## - gender:danceability 1 10.687 3472.5 1333.3
## - age:energy 1 13.542 3475.4 1334.2
## - race:genre 4 36.268 3498.1 1335.8
## - race:loudness_db 1 20.614 3482.4 1336.6
##
## Step: AIC=1329.83
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + loudness_db + length_sec + age:genre + age:energy +
## age:danceability + race:genre + race:happiness + race:loudness_db +
## race:length_sec + gender:genre + gender:danceability + gender:happiness
##
## Df Sum of Sq RSS AIC
## - gender:happiness 1 5.393 3467.5 1329.6
## <none> 3462.1 1329.8
## - gender:genre 3 20.130 3482.2 1330.5
## - age:genre 4 26.501 3488.6 1330.6
## - age:danceability 1 9.581 3471.7 1331.0
## - race:length_sec 1 9.695 3471.8 1331.1
## - race:happiness 1 9.796 3471.9 1331.1
## - gender:danceability 1 10.760 3472.9 1331.4
## - age:energy 1 13.719 3475.8 1332.4
## - race:genre 4 36.213 3498.3 1333.8
## - race:loudness_db 1 20.707 3482.8 1334.7
##
## Step: AIC=1329.63
## rating ~ age + gender + race + genre + energy + danceability +
## happiness + loudness_db + length_sec + age:genre + age:energy +
## age:danceability + race:genre + race:happiness + race:loudness_db +
## race:length_sec + gender:genre + gender:danceability
##
## Df Sum of Sq RSS AIC
## <none> 3467.5 1329.6
## - gender:danceability 1 6.283 3473.8 1329.7
## - gender:genre 3 19.968 3487.5 1330.3
## - age:genre 4 26.600 3494.1 1330.5
## - race:happiness 1 8.978 3476.5 1330.6
## - race:length_sec 1 9.145 3476.6 1330.7
## - age:danceability 1 10.356 3477.9 1331.1
## - age:energy 1 13.092 3480.6 1332.0
## - race:genre 4 34.348 3501.8 1333.0
## - race:loudness_db 1 21.430 3488.9 1334.7
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
## danceability + happiness + loudness_db + length_sec + age:genre +
## age:energy + age:danceability + race:genre + race:happiness +
## race:loudness_db + race:length_sec + gender:genre + gender:danceability,

```

```

##      data = cleaned_data)
##
## Coefficients:
##              (Intercept)                age
##              8.6220178                -0.0408272
##      genderNot Male                raceWhite
##              0.7280638                -1.4809661
##              genreJazz                genreOther
##              0.5285699                -1.6051711
##              genrePop                genreRock
##              -0.9281648                -0.6090139
##              energy                danceability
##              0.0159972                -0.0429294
##              happiness                loudness_db
##              0.0141879                0.0021139
##              length_sec                age:genreJazz
##              0.0006249                -0.0211649
##              age:genreOther                age:genrePop
##              0.0377222                -0.0147169
##              age:genreRock                age:energy
##              -0.0100985                -0.0009260
##              age:danceability                raceWhite:genreJazz
##              0.0010893                1.6839226
##              raceWhite:genreOther                raceWhite:genrePop
##              0.9821741                1.6598432
##              raceWhite:genreRock                raceWhite:happiness
##              1.9609966                -0.0076927
##              raceWhite:loudness_db                raceWhite:length_sec
##              0.0856656                0.0039525
##      genderNot Male:genreJazz                genderNot Male:genreOther
##              NA                -1.3988850
##      genderNot Male:genrePop                genderNot Male:genreRock
##              -1.0005892                -1.6516031
##      genderNot Male:danceability
##              0.0112198
anova(model_subpop1_1, model_subpop1_2)

## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##      danceability + age * happiness + age * bpm + age * loudness_db +
##      age * length_sec + race * genre + race * energy + race *
##      danceability + race * happiness + race * bpm + race * loudness_db +
##      race * length_sec + +gender * genre + gender * energy + gender *
##      danceability + gender * happiness + gender * bpm + gender *
##      loudness_db + gender * length_sec
##      Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      729 2510.6
## 2      700 2364.4 29    146.23 1.4928 0.0474 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

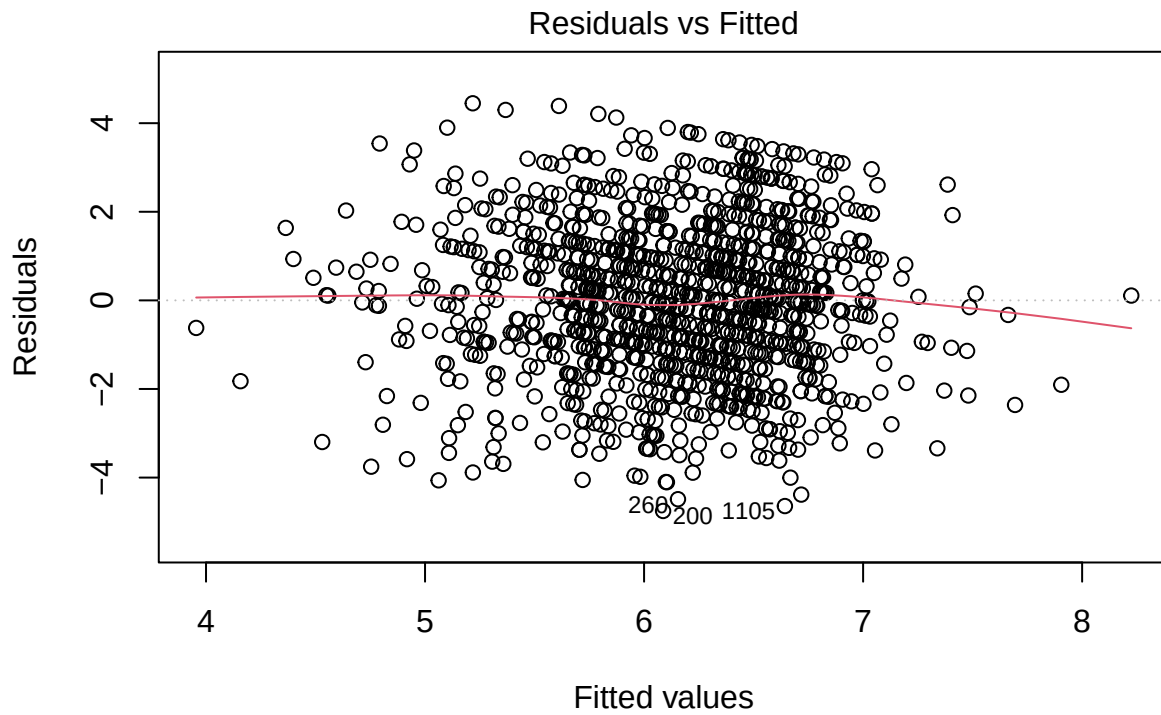
```
anova(model_subpop2_1, model_subpop2_2)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + +gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      398 1012.58
## 2      374  919.86 24    92.717 1.5707 0.04418 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model_total_1, model_total_2)
```

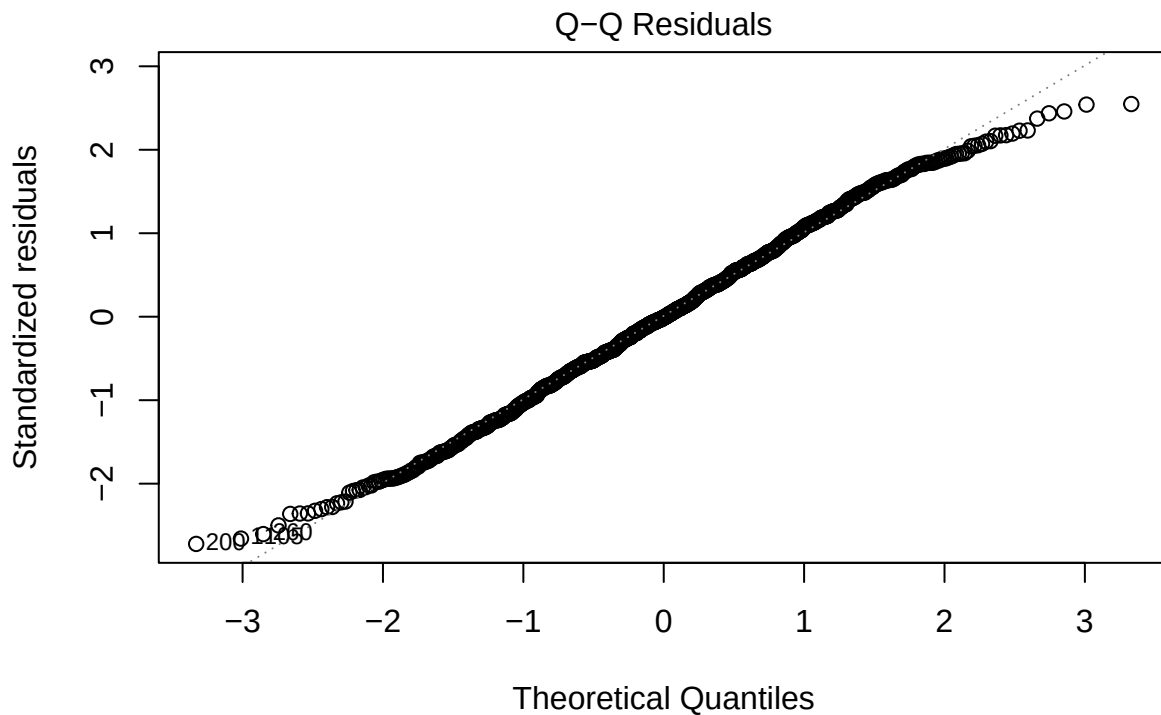
```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + +gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      1140 3610.3
## 2      1111 3456.6 29    153.69 1.7034 0.01179 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
plot(model_total_2, which = 1)
```



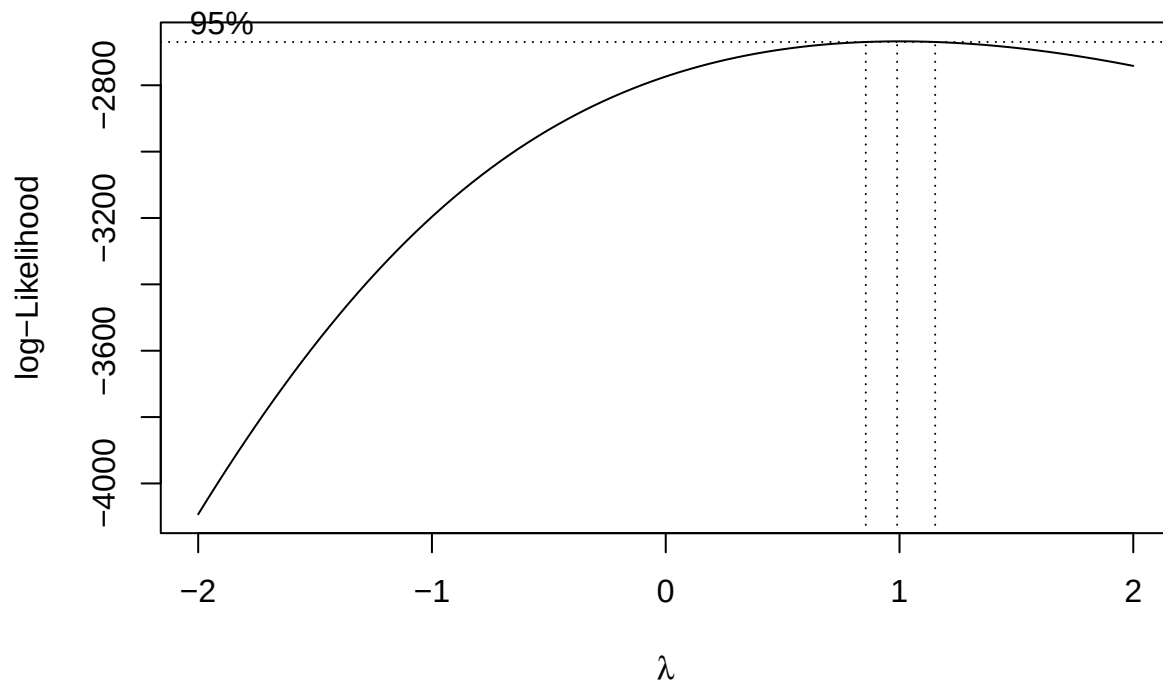
`lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...`

```
plot(model_total_2, which = 2)
```



`lm(rating ~ age + gender + race + age * genre + age * energy + age * dancea ...`

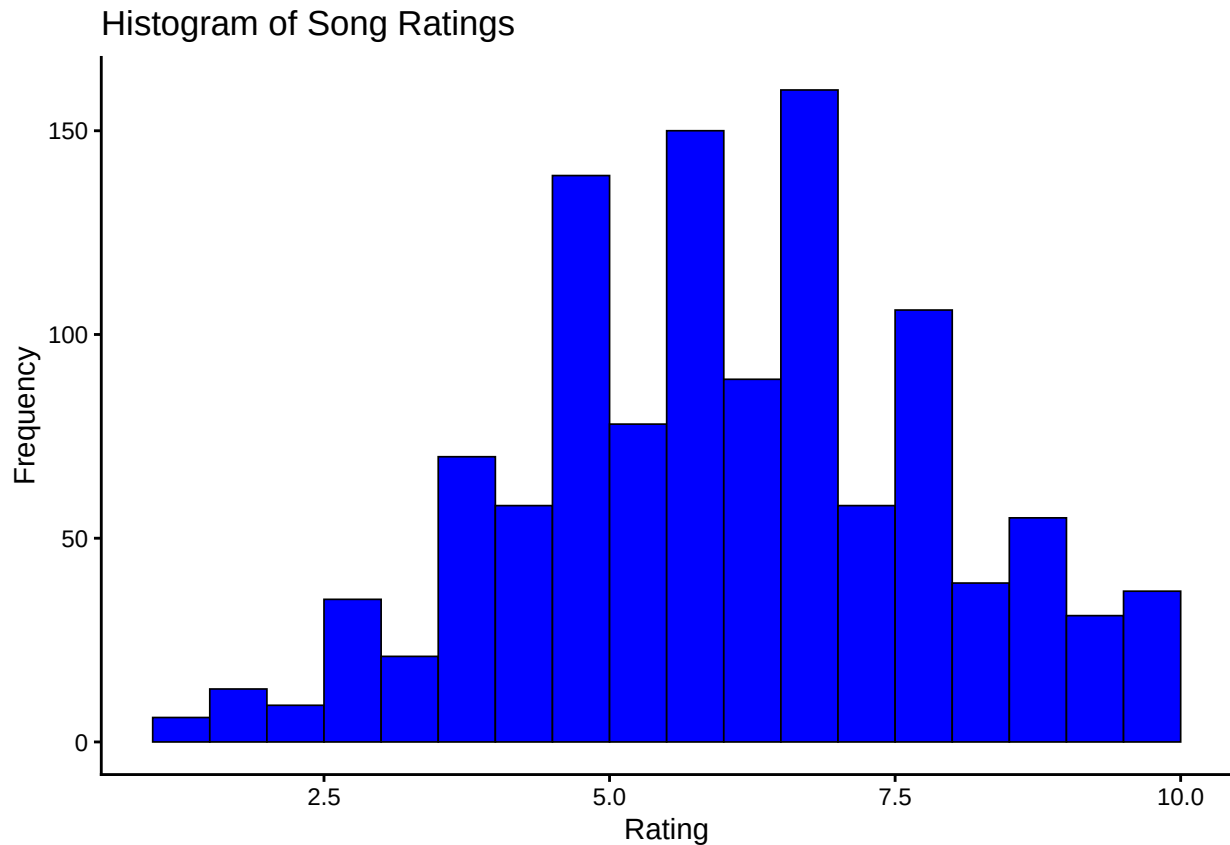
```
boxcox(model_total_2, plotit=T)
```

Preliminary exploration of data through visual analysis

```
library(ggplot2)

ggplot(cleaned_data, aes(x = rating)) +
  geom_histogram(
    binwidth = 0.5,
    boundary = 0.5,
    fill = "blue",
    color = "black",
    linewidth = 0.3
  ) +
  labs(
    title = "Histogram of Song Ratings",
    x = "Rating",
    y = "Frequency"
  ) +
  theme_classic()
```



The histogram of ratings appears approximately symmetric and unimodal, resembling a normal distribution. Although the ratings are discrete (integer-valued), the distribution spans multiple levels, creating a shape that approximates continuity. This supports the treating of rating as a continuous response variable in the regression analysis.

```
library(patchwork)
```

```
##  
## Attaching package: 'patchwork'  
## The following object is masked from 'package:MASS':  
##  
## area  
  
p1 <- ggplot(first_subpop, aes(x = age, y = rating)) +  
  geom_point(color = "black", size = 1.5) +  
  geom_smooth(method = "lm", color = "red", se = F) +  
  theme_classic() +  
  ggtitle("Rating vs Age")  
  
p2 <- ggplot(first_subpop, aes(x = energy, y = rating)) +  
  geom_point(color = "black", size = 1.5) +  
  geom_smooth(method = "lm", color = "red", se = F) +  
  theme_classic() +  
  ggtitle("Rating vs Energy")  
  
p3 <- ggplot(first_subpop, aes(x = danceability, y = rating)) +  
  geom_point(color = "black", size = 1.5) +  
  geom_smooth(method = "lm", color = "red", se = F) +
```

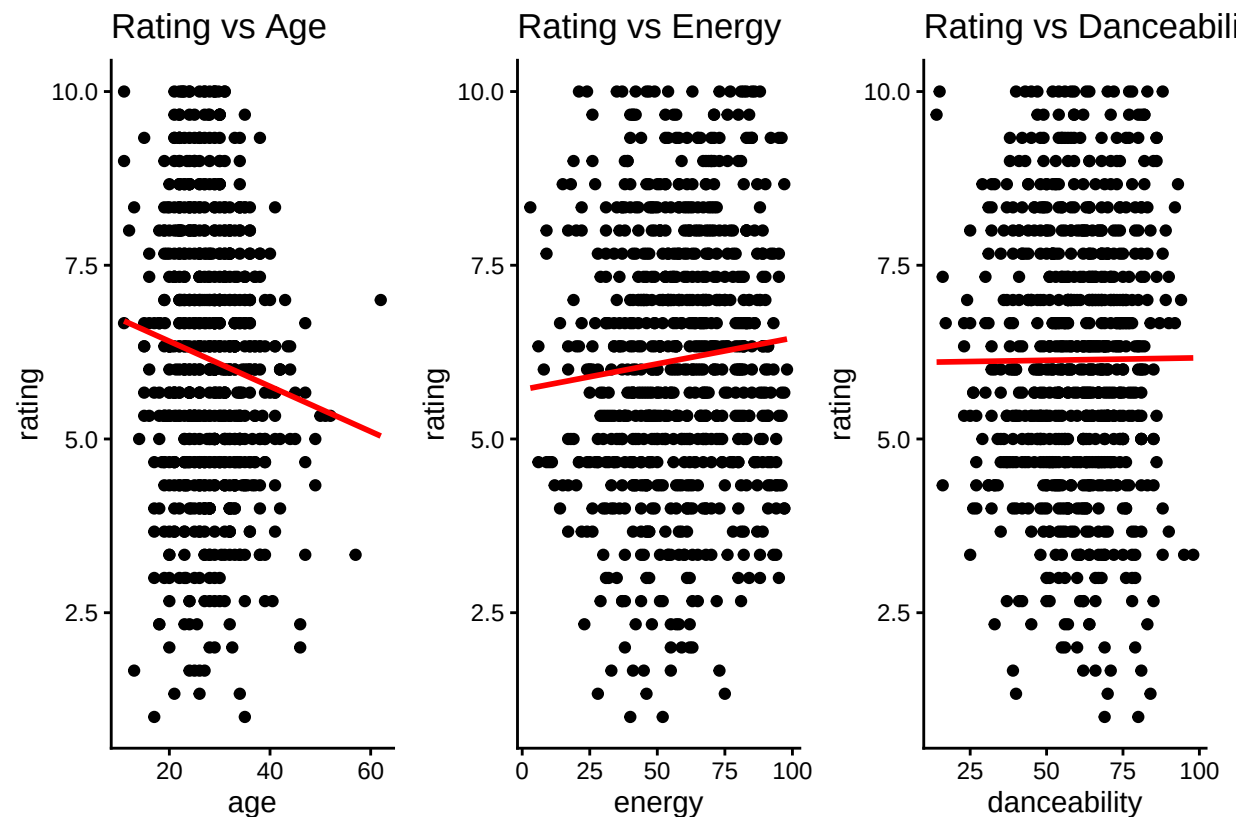
```
theme_classic() +
ggtitle("Rating vs Danceability")
```

```
p1 + p2 + p3
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



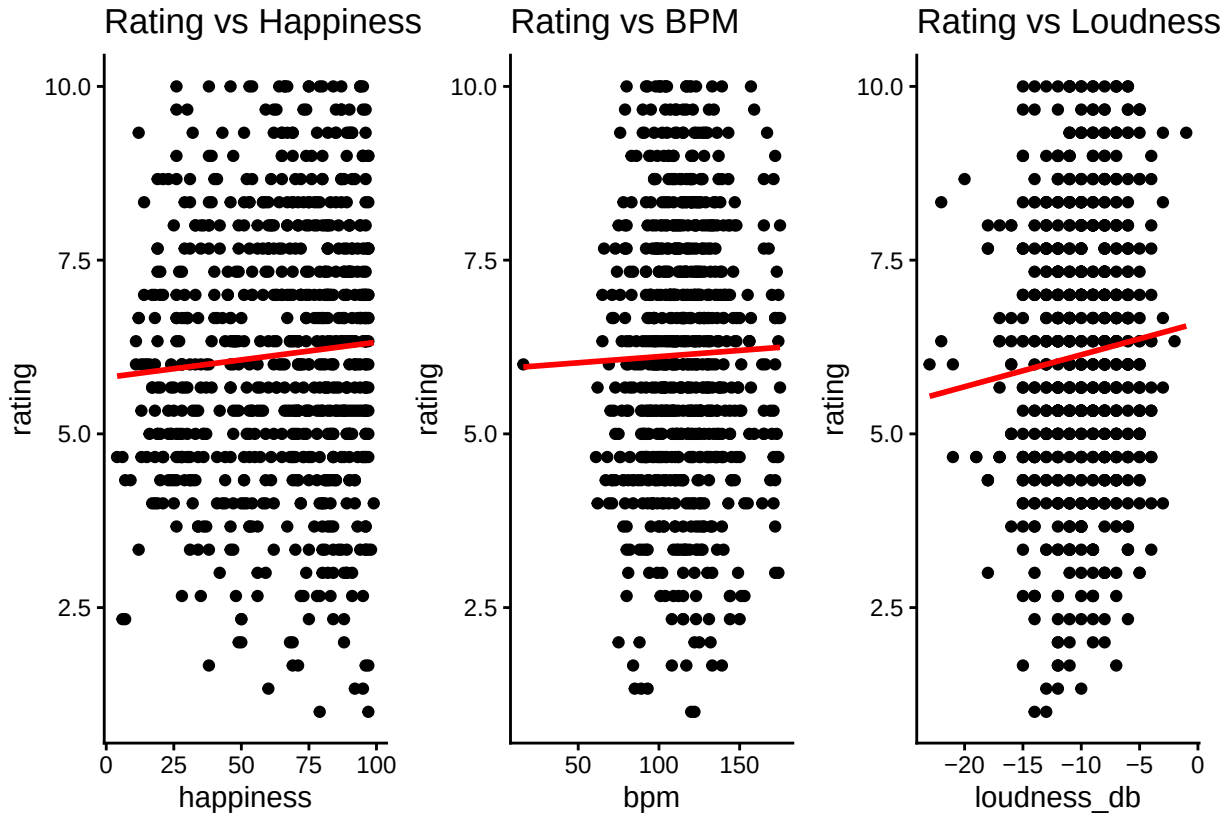
```
p4 <- ggplot(first_subpop, aes(x = happiness, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Happiness")
```

```
p5 <- ggplot(first_subpop, aes(x = bpm, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs BPM")
```

```
p6 <- ggplot(first_subpop, aes(x = loudness_db, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Loudness")
```

```
p4 + p5 + p6
```

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```



```
p7 <- ggplot(first_subpop, aes(x = length_sec, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Song Length")
```

```
p7
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

